

Information Retrieval

2. Boolean retrieval

Ghislain Fourny
Spring 2022



2



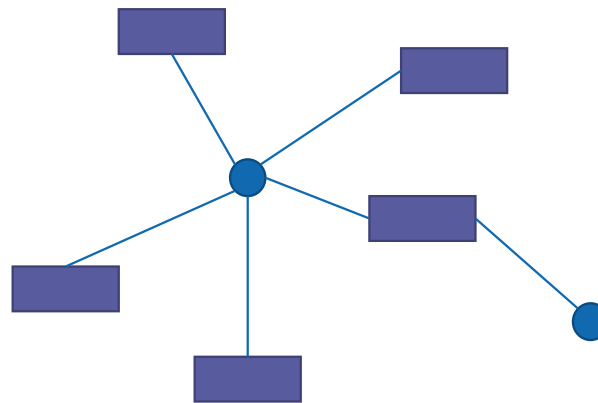
Data Shapes

Data Shapes: Tables

Data Shapes: Trees



Data Shapes: Graphs



Data Shapes: Cubes



Data Shapes: Unstructured text

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam vel erat nec dui aliquet vulputate sed quis nulla. Donec eget ultricies magna, eu dignissim elit. Nullam sed urna nec nisl rhoncus ullamcorper placerat et enim. Integer varius ornare libero quis consequat. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aenean eu efficitur orci. Aenean ac posuere tellus. Ut id commodo turpis.

Praesent nec libero metus. Praesent at turpis placerat, congue ipsum eget, scelerisque justo. Ut volutpat, massa ac lacinia cursus, nisl dui volutpat arcu, quis interdum sapien turpis in tellus. Suspendisse potenti. Vestibulum pharetra justo massa, ac venenatis mi condimentum nec. Proin viverra tortor non orci suscipit rutrum. Phasellus sit amet euismod diam. Nullam convallis nunc sit amet diam suscipit dapibus. Integer porta hendrerit nunc. Quisque pharetra congue porta. Suspendisse vestibulum sed mi in euismod. Etiam a purus suscipit, accumsan nibh vel, posuere ipsum. Nulla nec tempor nibh, id venenatis lectus. Duis lobortis id urna eget tincidunt.

Finding **unstructured** data

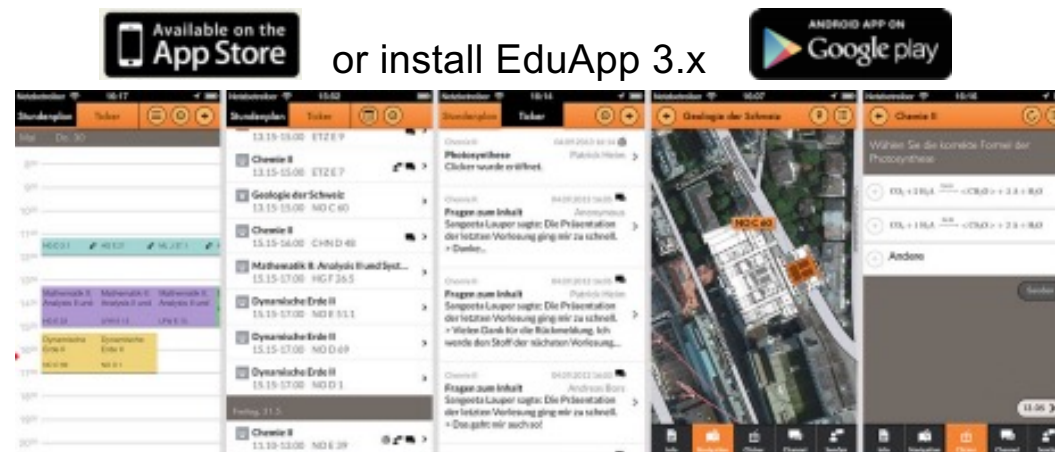
**Finding unstructured data
that satisfies an information need**

Finding **unstructured data**
that satisfies an **information need**
from within **large collections**

Poll

Go *now* to:

<https://eduapp-app1.ethz.ch/>





First, naive approach

Example of corpus

CHAPTER I

holmes.txt (3 MB)

Mr. Sherlock Holmes

Mr. Sherlock Holmes, who was usually very late in the mornings, save upon those not infrequent occasions when he was up all night, was seated at the breakfast table. I stood upon the hearth-rug and picked up the stick which our visitor had left behind him the night before.

It was a fine, thick piece of wood, bulbous-headed, of the sort which is known as a "Penang lawyer." Just under the head was a broad silver band nearly an inch across. "To James Mortimer, M.R.C.S., from his friends of the C.C.H.," was engraved upon it, with the date "1884."

It was just such a stick as the old-fashioned family practitioner used to carry--dignified, solid, and reassuring.

"Well, Watson, what do you make of it?"

Holmes was sitting with his back to me, and I had given him no sign of my occupation.

"How did you know what I was doing? I believe you have eyes in the back of your head."

Naive approach

Q: lawyer

Example of corpus

CHAPTER I

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Result

Q: lawyer

CHAPTER I

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Result

Q: lawyer

CHAPTER I

Mr. Sherlock Holmes

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Naive approach

```
$ grep lawyer corpus.txt
```

Naive approach

```
$ grep lawyer corpus.txt
```

William Whyte was. Some pragmatistical seventeenth century lawyer, I
He was a lawyer. That sounded ominous. What was the relation between
"Her banker or her lawyer. There is that double possibility. But I am
"To an English lawyer named Norton."
and send down to Fordham, the Horsham lawyer.'
"I did as he ordered, and when the lawyer arrived I was asked to step
"I signed the paper as directed, and the lawyer took it away with
violin-player, boxer, swordsman, lawyer, and self-poisoner by cocaine
which was a Penang-lawyer weighted with lead, was just such a weapon
a lawyer with a good practice. They had one child, but the yellow
has some claim on half Cunningham's estate, and the lawyers have been
lawyer whose name was given in the paper. There we met two gentlemen,
as he followed the argument of the lawyer. At least, I thought, when
is known as a "Penang lawyer." Just under the head was a broad silver
Holmes to the porter. "A lawyer, is he not, gray-headed, and walks
Reilly the lawyer and take the defense upon myself. Take my word for
"Exactly," said I. "A plausible lawyer could make it out as an act of
daughter of Augusto Barelli, who was the chief lawyer and once the
I showed it to Mr. Sutro, my lawyer, who lives in Harrow. He said to
lawyer of yours a capable man?"
This Sutro, of course, is her lawyer. I made a mistake, I fear, in
gentleman, who introduced himself as the lawyer, together with a
"I only heard of it this morning," the lawyer explained.
I have been recommended to you by my lawyers, but indeed the matter
to the American lawyer. ... Very good. Good-bye!"
lawyer burst excitedly into the room.
"I can only suppose that this American lawyer put it in himself. What
say nothing, as her lawyer had advised her to reserve her defence. We

Naive approach

Q: lawyer AND Penang

Naive approach

```
$ grep lawyer corpus.txt | grep Penang
```

Naive approach

```
$ grep lawyer corpus.txt | grep Penang
```

```
which was a Penang-lawyer weighted with lead, was just such a weapon  
is known as a "Penang lawyer." Just under the head was a broad silver
```

Naive approach

Q: lawyer AND
Penang AND
NOT silver

Naive approach

```
$ grep lawyer corpus.txt | grep Penang | grep -v silver
```


Naive approach

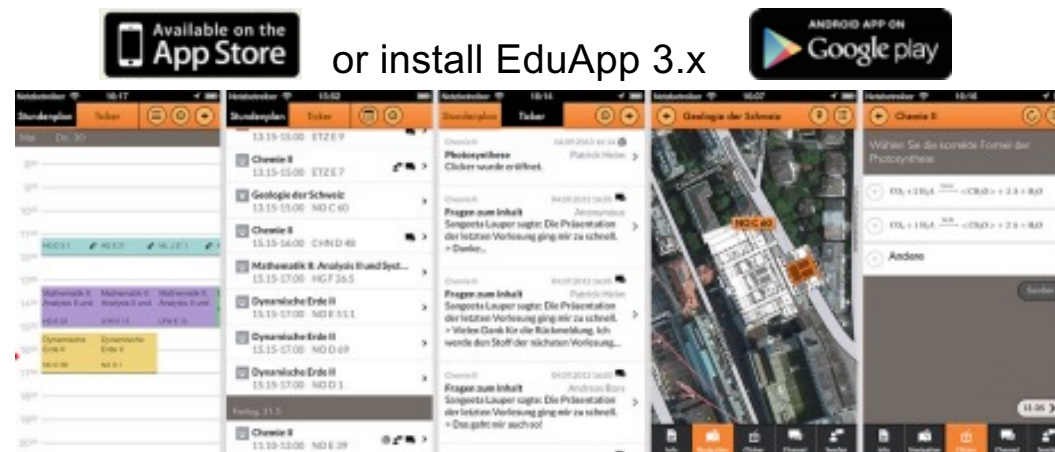
```
$ grep lawyer corpus.txt | grep Penang | grep -v silver
```

```
which was a Penang-lawyer weighted with lead, was just such a weapon
```

Poll

Go *now* to:

<https://eduapp-app1.ethz.ch/>



Grepping

Regular expressions

`foo|bar[a-z]+`

Wildcards

`.*`

Grepping: shortcomings (1)

Grepping: shortcomings (1)

What if we want
to search
an **extremely large**
collection:

The Web

?



Grepping: shortcomings

To do it right, he told Page, you'd really have to capture a significant chunk of the World Wide Web's link structure.

Steven Levy
In The Plex: How Google Thinks, Works, and Shapes Our Lives

Grepping: shortcomings

To do it right, he told Page, you'd really have to capture a significant chunk of the World Wide Web's link structure.

Page said, sure, he'd go and **download the web** and get the structure.

Steven Levy
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Grepping: shortcomings

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He figured it would take a week or something.

Steven Levy

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Grepping: shortcomings

To do it right, he told Page, you'd really have to capture a significant chunk of the World Wide Web's link structure.

Page said, sure, he'd go and **download the web** and get the structure.

He figured it would take a week or something.

“And of course,” he later recalled, “**it took, like, years.**”

Steven Levy

In The Plex: How Google Thinks, Works, and Shapes Our Lives

Grepping: shortcomings (2)

Proximity search?

Zurich NEAR Switzerland

Grepping: shortcomings (3)

How do we rank?

The image shows a Google search interface for the term "röstigraben". The search bar at the top contains the text "röstigraben". Below the search bar, there are tabs for "All", "Images", "Maps", "Shopping", "News", and "More". The "All" tab is selected. Below the tabs, it says "About 72.800 results (0,41 seconds)".

The search results are listed below. The first result is "Röstigraben - Wikipedia" with the URL <https://en.wikipedia.org/wiki/Röstigraben>. The snippet describes it as a humorous term for the cultural boundary between German-speaking and French-speaking parts of Switzerland. The second result is "Röstigraben – Wikipedia" with the URL <https://de.wikipedia.org/wiki/Röstigraben>. The snippet describes it as a term used to refer to the cultural boundary between German-speaking and French-speaking parts of Switzerland. The third result is "Röstigraben — Wikipédia" with the URL <https://fr.wikipedia.org/wiki/Röstigraben>. The snippet describes it as a term used to refer to the cultural boundary between German-speaking and French-speaking parts of Switzerland.

On the right side of the search results, there is a "More images" link. Below the search results, there is a "Feedback" link.

The featured snippet on the right shows a collage of images related to Röstigraben, including a map of Switzerland, a map of the Röstigraben region, and a photo of a Röstigraben sign. Below the images, the title "Röstigraben" is displayed. The snippet text reads: "Röstigraben is a humorous term used to refer to the cultural boundary between German-speaking and French-speaking parts of Switzerland. There is also a term Polentagraben which refers to the Italian-speaking canton of Ticino. Wikipedia".

Usefulness of indices

web searches:
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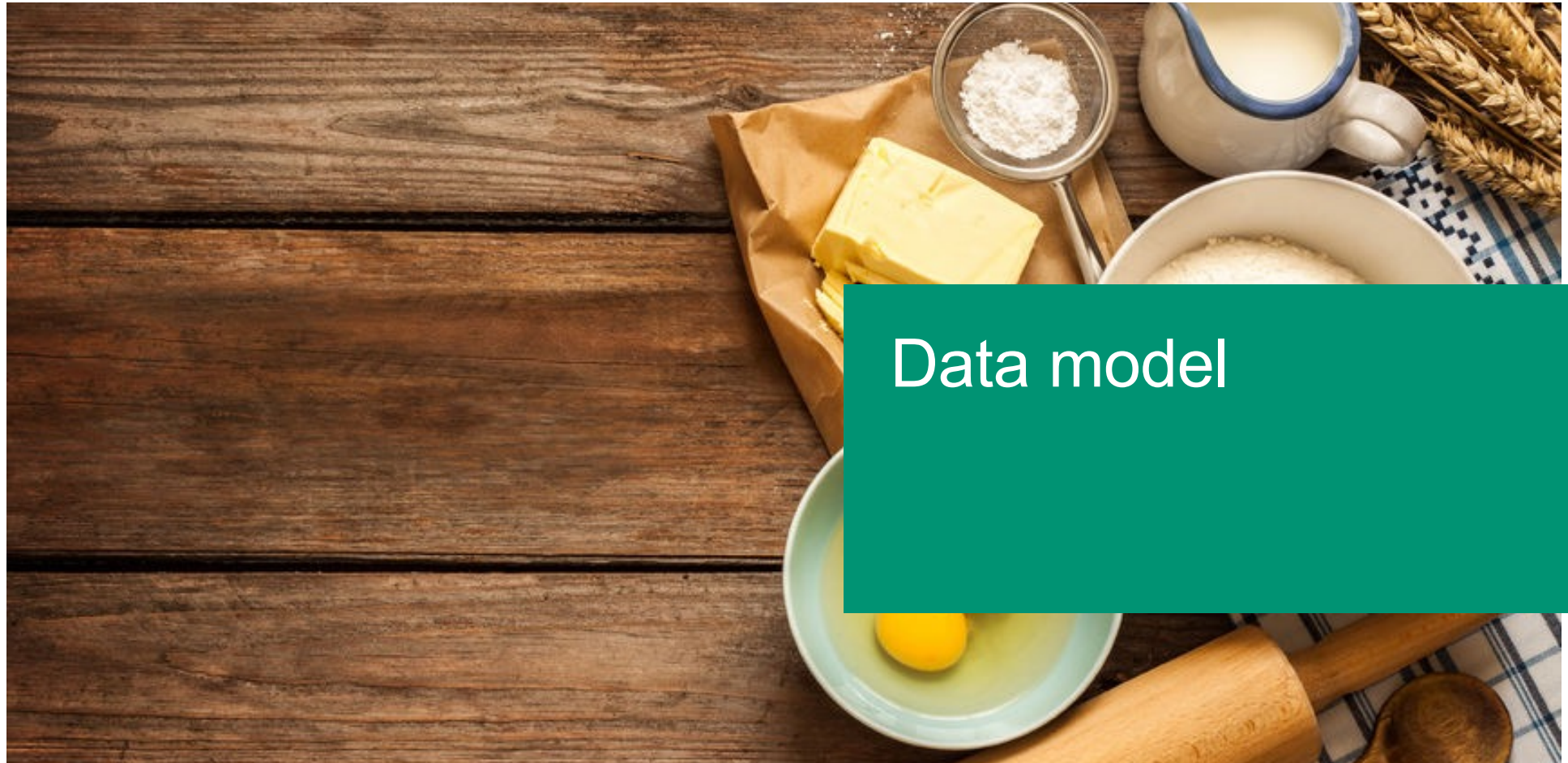
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Indices are everywhere

Look at the **last pages of books**

Steven Levy
In The Plex: How Google Thinks, Works,
and Shapes Our Lives



Data model

Document



Document

Suchergebnisse für «information retrieval»

information retrieval 

Alle Ergebnisse Webseiten News **Dokumente** Personen

Suchergebnisse 1-10 von 175000

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Dateiformat: PDF/Adobe Acrobat

Information Retrieval
Sep 26, 2012 ... Today's Overview. 1. Introduction. 2. Dictionaries. 3. Index Construction. 4. Distributed Indexing. 5. Multiple Query Terms. 6. Advanced Posting List Intersection. 7. Web-scale Index Serving. Class from 9:15-10:45 (no break), 11-12: Exercise. **Information Retrieval**, ETHZ 2012. 2 ...
http://www.systems.ethz.ch/sites/default/files/file/IR_Fall2012/02-indexing.pdf ↓

Dateiformat: PDF/Adobe Acrobat

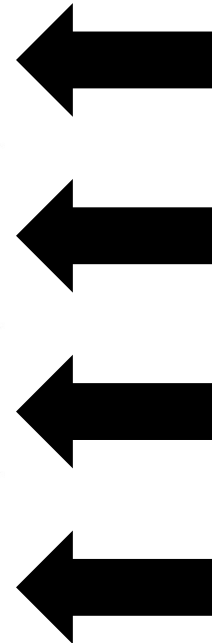
On Robust Combiners for Private Information Retrieval and Other ...
In this work we study robust combiners for private **information retrieval**. (PIR), oblivious transfer (OT), and bit commitment (BC). We propose a (1,2)-robust PIR-to-PIR combiner, and describe various optimizations based on properties of existing PIR protocols. The existence of simple. PIR-to-PIR combiners is somewhat ...
<https://www.tik.ee.ethz.ch/file/1e6e8033267d4abdd79a930fe397524e/crypto06.pdf> ↓

Dateiformat: PDF/Adobe Acrobat

Information Retrieval
Oct 10, 2012 ... Overview. 1. Dictionary Compression. 2. Zipf's Law. 3. Posting List Compression. 4. Gamma Codes. 5. Golomb Code. 6. Index Compression in Practice. **Information Retrieval**, ETHZ 2012. 2 ...
http://www.systems.ethz.ch/sites/default/files/file/IR_Fall2012/04-Index-Compression.pdf ↓

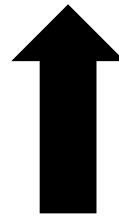
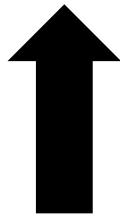
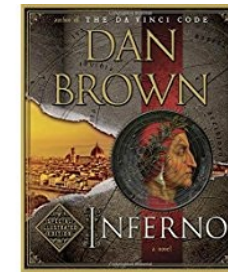
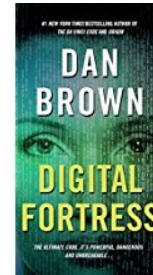
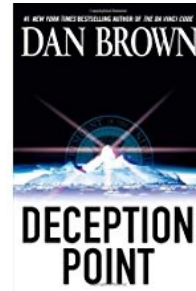
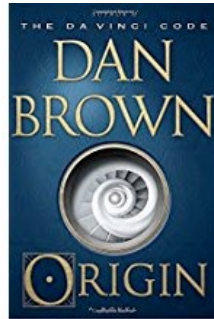
Dateiformat: PDF/Adobe Acrobat

Spatial Information Retrieval From Remote-sensing Images-Part I ...
Abstract— Automatic interpretation of remote-sensing (RS) images and the growing interest for query by image content from large remote-sensing image archives rely on the ability and robustness of **information** extraction from observed data. In Parts I and II of this article, we turn the attention to the modern



Documents

Document



Documents

Term

Sherlock
lawyer
Switzerland
Unterwalden nid dem Wald
ETH Zürich
person
watch
run
paper
book
...

Relationship between documents and terms?

I han es Zündhölzli azündt
Und das het e Flamme gäh
Und i ha für d'Zigarette
Welle Fүү vom Hölzli näh
Aber ds Hölzli isch dervo-
Gspickt und uf e Teppich cho
Und es hätt no fasch es Loch i Teppich gäh dervo

D

Inclusion

I han es Zündhölzli azündt

Und das het e Flamme gäh

Und i ha für d'Zigarette

Welle Fүү vom **Hölzli** näh

Aber ds Hölzli isch dervo-

Gspickt und uf e Teppich cho

Und es hätt no fasch es Loch i Teppich gäh dervo

A term is
included
in a document

"Hölzli" ∈ D

Occurrence

I han es Zündhölzli azündt
Und das het e Flamme gäh

Und i ha für d'Zigarette

Welle Fүүr vom **Hölzli** näh

Aber ds **Hölzli** isch dervo-

Gspickt und uf e Teppich cho

Und es hätt no fasch es Loch i Teppich gäh dervo

A term
can occur
x times
in a document

$\#(\text{"Hölzli"}, D) = 2$

Order

I han es Zündhölzli azündt
Und das het e **Flamme** gäh
Und i ha für a Zigarette
Welle **Füür** vom Hölzli näh
Aber ds Hölzli isch dervo-
Gspickt und uf e Teppich cho
Und es hätt no fasch es Loch i Teppich gäh dervo

A term
can occur
before
another

"Flamme" < "Füür"

Document as list of terms

- ✓ Inclusion
- ✓ Occurrence
- ✓ Order

I han es Zündhölzli azündt
Und das het e Flamme gäh
Und i ha für d'Zigarette
Welle Fүү vom Hölzli näh
Aber ds Hölzli isch dervo-
Gspickt und uf e Teppich cho
Und es hätt no fasch es Loch i Teppich gäh dervo

Document as bag of terms

- ✓ Inclusion
- ✓ Occurrence
- ✗ Order

Und vom es uf
e no Welle es e ha ds
azündt i gäh l cho
Flamme es het Fűr han i fasch
d'Zigarette Hölzli näh
Zündhölzli das Und Aber hätt
Loch dervo- für Und isch Teppich Teppich
Gspickt Hölzli und dervo
gäh

Document as set of terms

- ✓ Inclusion
- ✗ Occurrence
- ✗ Order





Incidence matrix

Set-of-terms model

Documents

contain (or not)

Terms

Incidence Matrix

Documents

1

contain (or not)

0

Terms

Incidence Matrix

$$\text{contains}_{i,j} = \begin{cases} 1 & \text{if document } j \text{ contains term } i \\ 0 & \text{if document } j \text{ does not contain term } i \end{cases}$$

Incidence Matrix

		Documents									
		1	2	3	4	5	6	7	8	9	10
Terms	t	1	0	1	1	1	0	0	0	1	1
	u	0	0	1	0	1	1	1	1	0	0
	v	0	1	1	1	0	1	0	1	0	1
	w	0	0	0	1	1	0	0	0	0	0
	x	1	0	1	1	1	0	1	0	0	1
	y	0	0	0	0	1	0	0	1	0	1

Incidence Matrix

		Documents									
		1	2	3	4	5	6	7	8	9	10
Terms	t	1	0	1	1	1	0	0	0	1	1
	u	0	0	1	0	1	1	1	1	0	0
	v	0	1	1	1	0	1	0	1	0	1
	w	0	0	0	1	1	0	0	0	0	0
	x	1	0	1	1	1	0	1	0	0	1
	y	0	0	0	0	1	0	0	1	0	1

↑
 $\text{contains}_{y,5} = 1$

Incidence Matrix

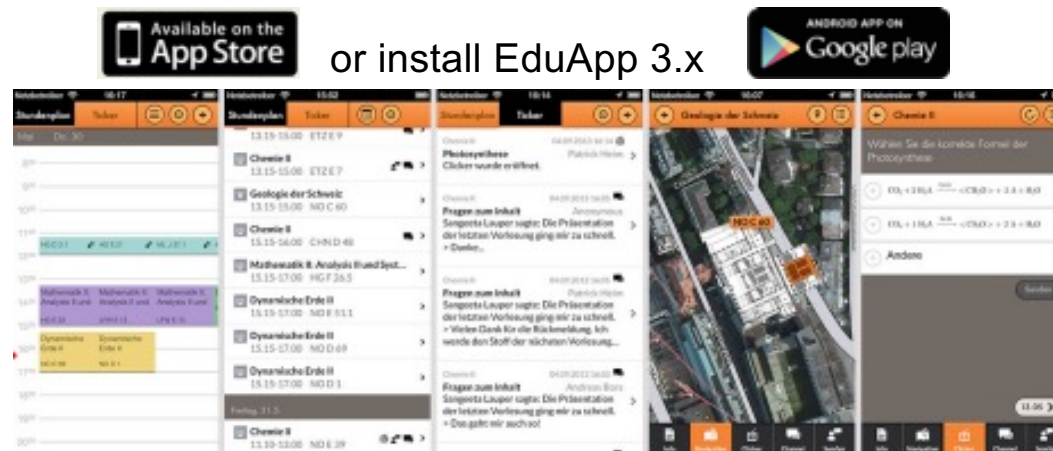
		Documents									
		1	2	3	4	5	6	7	8	9	10
Terms	t										
	u										
	v										
	w										
	x										
	y										

contains $\in \mathbb{B}^{\text{Terms} \times \text{Documents}}$

Poll

Go *now* to:

<https://eduapp-app1.ethz.ch/>



Boolean Query

u

Incidence Matrix

		Documents									
		1	2	3	4	5	6	7	8	9	10
Terms	t	1	0	1	1	1	0	0	0	1	1
	u	0	0	1	0	1	1	1	1	0	0
	v	0	1	1	1	0	1	0	1	0	1
	w	0	0	0	1	1	0	0	0	0	0
	x	1	0	1	1	1	0	1	0	0	1
	y	0	0	0	0	1	0	0	1	0	1

Incidence Matrix

		Documents									
		1	2	3	4	5	6	7	8	9	10
Terms	t	1	0	1	1	1	0	0	0	1	1
	u	0	0	1	0	1	1	1	1	0	0
	v	0	1	1	1	0	1	0	1	0	1
	w	0	0	0	1	1	0	0	0	0	0
	x	1	0	1	1	1	0	1	0	0	1
	y	0	0	0	0	1	0	0	1	0	1

Incidence Matrix

		Documents									
		1	2	3	4	5	6	7	8	9	10
Terms	t	1	0	↑	1	↑	↑	↑	↑	1	1
	u	0	0	1	0	1	1	1	1	0	0
	v	0	1	1	1	0	1	0	1	0	1
	w	0	0	0	1	1	0	0	0	0	0
	x	1	0	1	1	1	0	1	0	0	1
	y	0	0	0	0	1	0	0	1	0	1

Boolean Query

NOT u

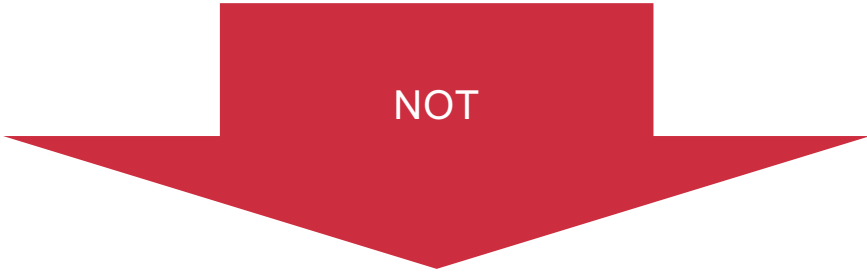
Incidence Matrix

		Documents									
		1	2	3	4	5	6	7	8	9	10
Terms	t	1	0	1	1	1	0	0	0	1	1
	u	0	0	1	0	1	1	1	1	0	0
	v	0	1	1	1	0	1	0	1	0	1
	w	0	0	0	1	1	0	0	0	0	0
	x	1	0	1	1	1	0	1	0	0	1
	y	0	0	0	0	1	0	0	1	0	1

Incidence Matrix

	Documents									
	1	2	3	4	5	6	7	8	9	10
u	0	0	1	0	1	1	1	1	0	0

Incidence Matrix

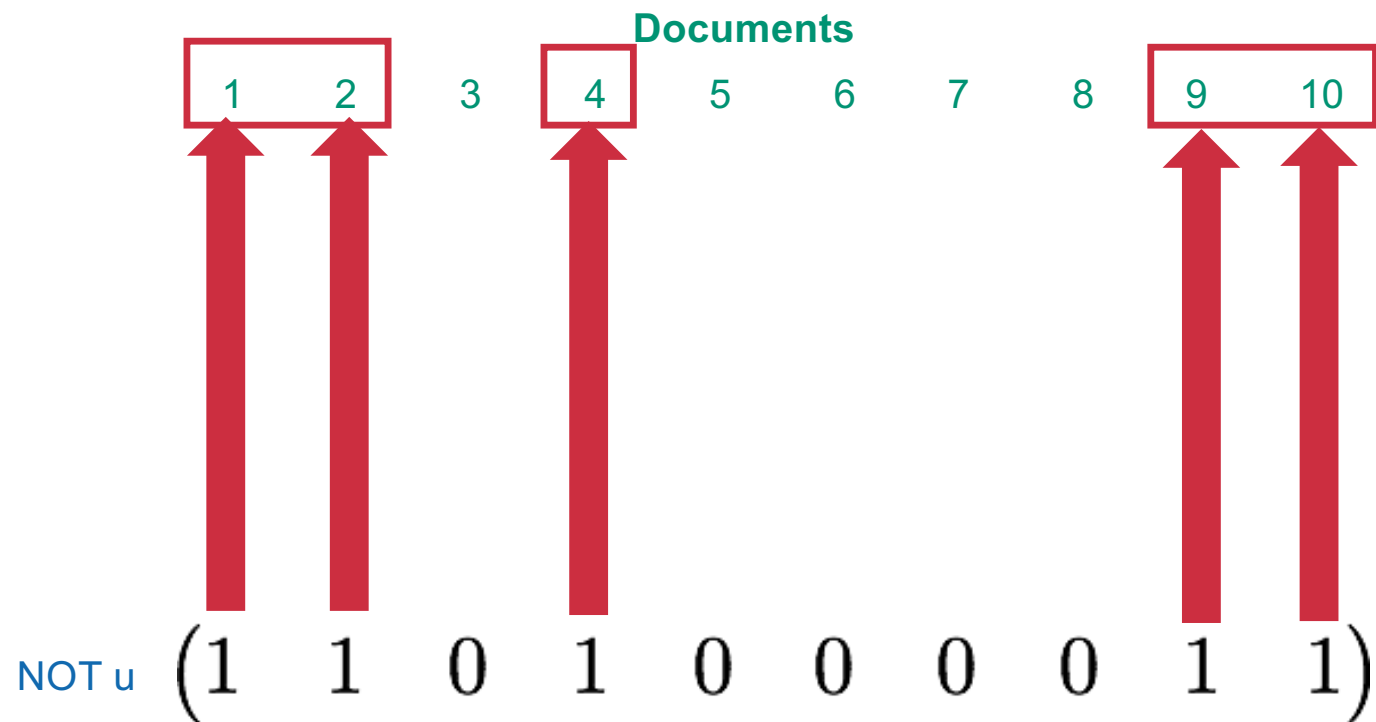
	Documents									
	1	2	3	4	5	6	7	8	9	10
u	0	0	1	0	1	1	1	1	0	0
										
$\text{NOT } u$	1	1	0	1	0	0	0	0	1	1

Incidence Matrix

Documents
1 2 3 4 5 6 7 8 9 10

$$\text{NOT } u \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix}$$

Incidence Matrix



Boolean Query

u AND x

Incidence Matrix

		Documents									
		1	2	3	4	5	6	7	8	9	10
Terms	t	1	0	1	1	1	0	0	0	1	1
	u	0	0	1	0	1	1	1	1	0	0
	v	0	1	1	1	0	1	0	1	0	1
	w	0	0	0	1	1	0	0	0	0	0
	x	1	0	1	1	1	0	1	0	0	1
	y	0	0	0	0	1	0	0	1	0	1

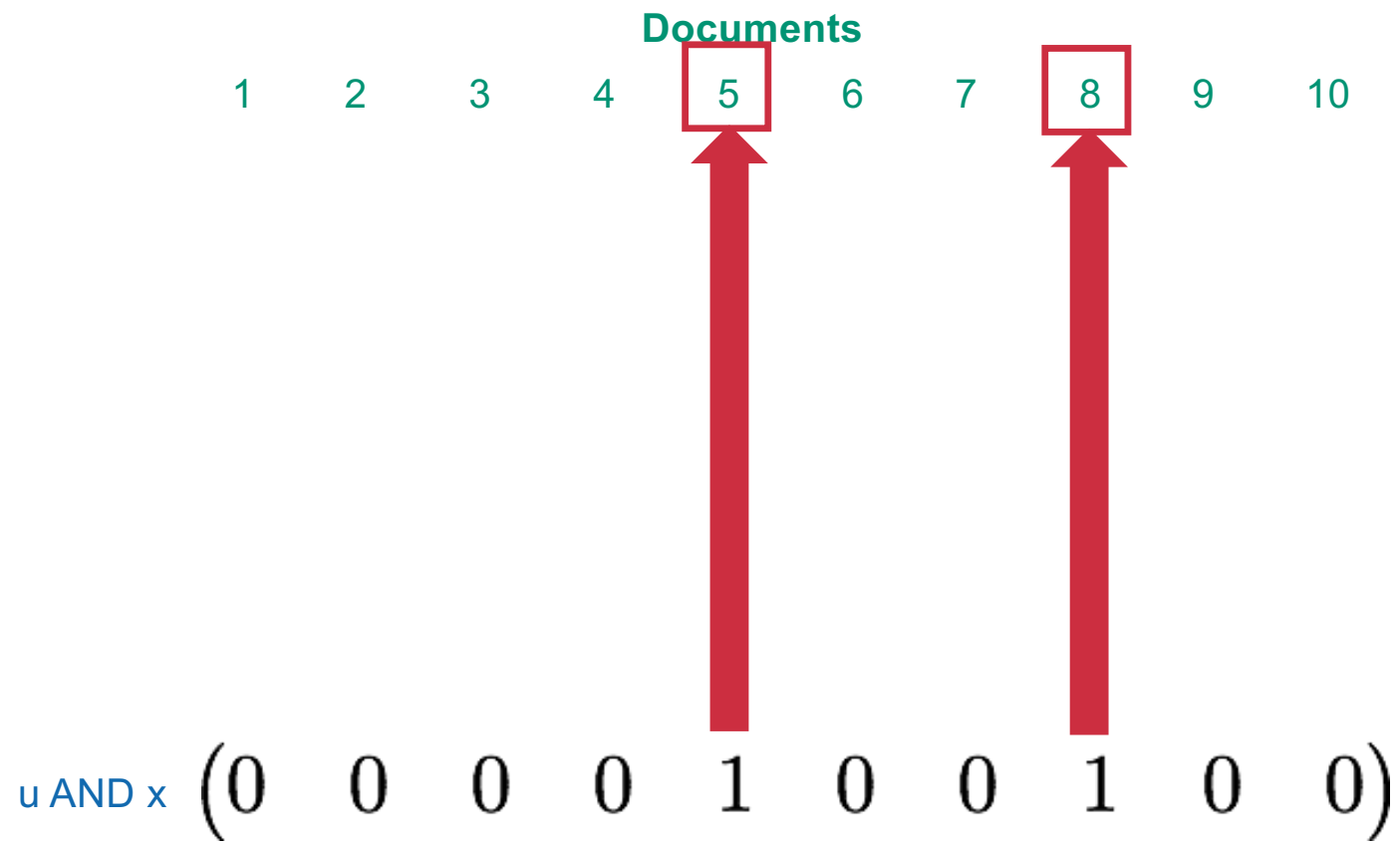
Incidence Matrix

		Documents									
		1	2	3	4	5	6	7	8	9	10
Terms	u	0	0	1	0	1	1	1	1	0	0
	x	0	0	0	0	1	0	0	1	0	1

Incidence Matrix

		Documents									
		1	2	3	4	5	6	7	8	9	10
Terms	u	(0	0	1	0	1	1	1	1	0	0)
	x	(0	0	0	0	1	0	0	1	0	1)
		AND									
u AND x		(0	0	0	0	1	0	0	1	0	0)

Incidence Matrix



Boolean Query

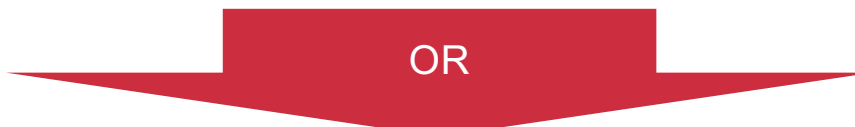
u OR x

Incidence Matrix

		Documents									
		1	2	3	4	5	6	7	8	9	10
Terms	t	1	0	1	1	1	0	0	0	1	1
	u	0	0	1	0	1	1	1	1	0	0
	v	0	1	1	1	0	1	0	1	0	1
	w	0	0	0	1	1	0	0	0	0	0
	x	1	0	1	1	1	0	1	0	0	1
	y	0	0	0	0	1	0	0	1	0	1

Incidence Matrix

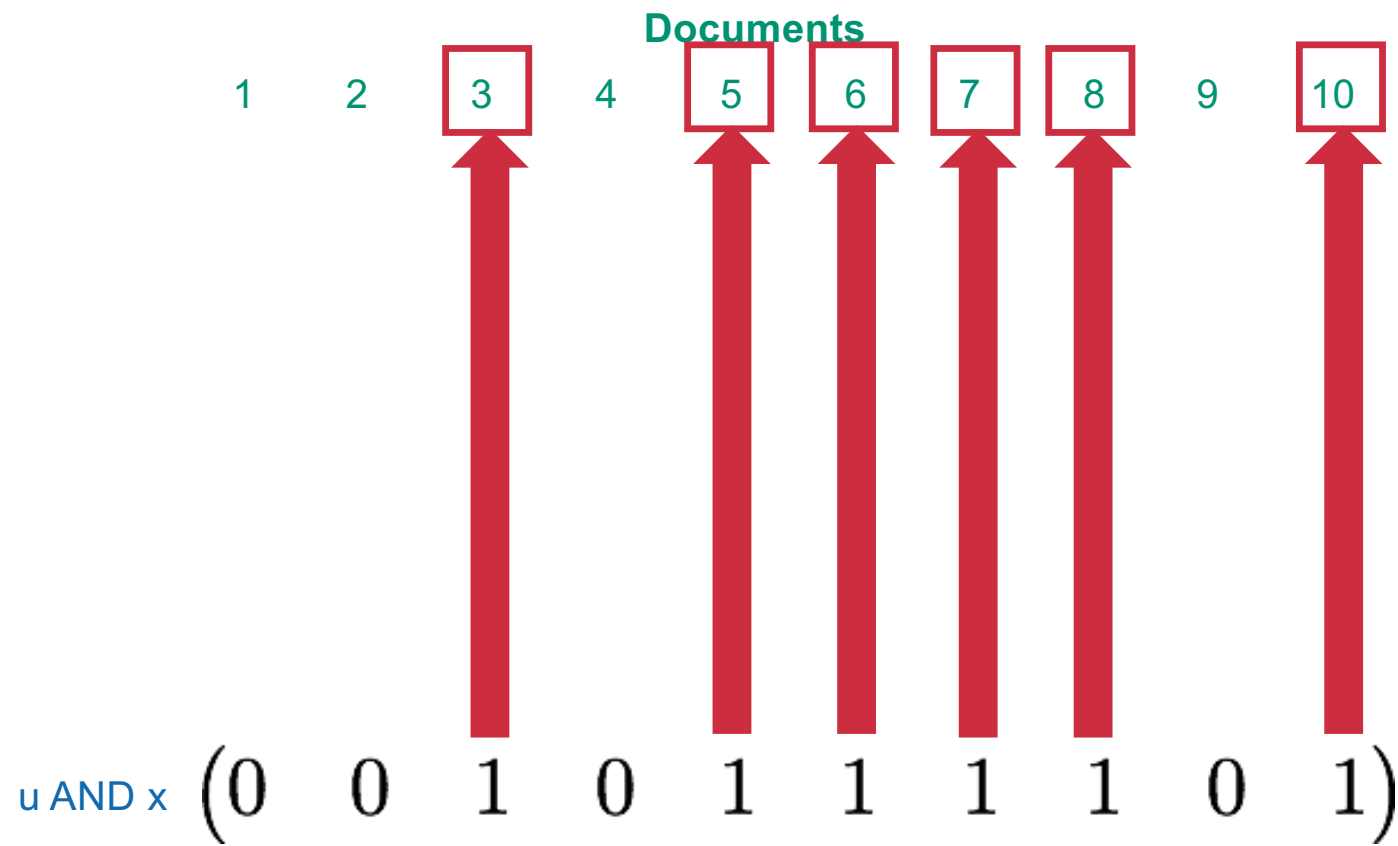
		Documents									
		1	2	3	4	5	6	7	8	9	10
Terms	u	(0	0	1	0	1	1	1	1	0	0)
	x	(0	0	0	0	1	0	0	1	0	1)
	u OR x	(0	0	1	0	1	1	1	1	0	1)



Incidence Matrix

		Documents									
		1	2	3	4	5	6	7	8	9	10
Terms	u	0	0	1	0	1	1	1	1	0	0
	x	0	0	0	0	1	0	0	1	0	1

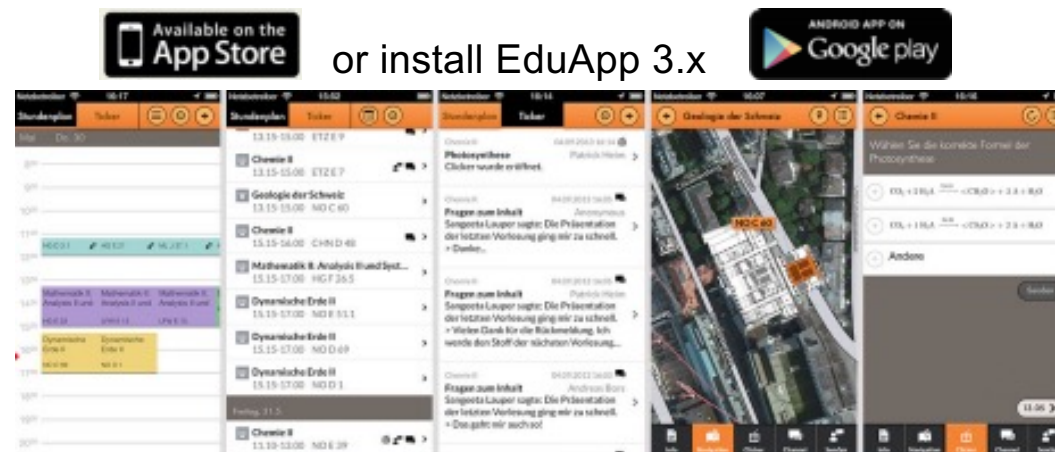
Incidence Matrix



Poll

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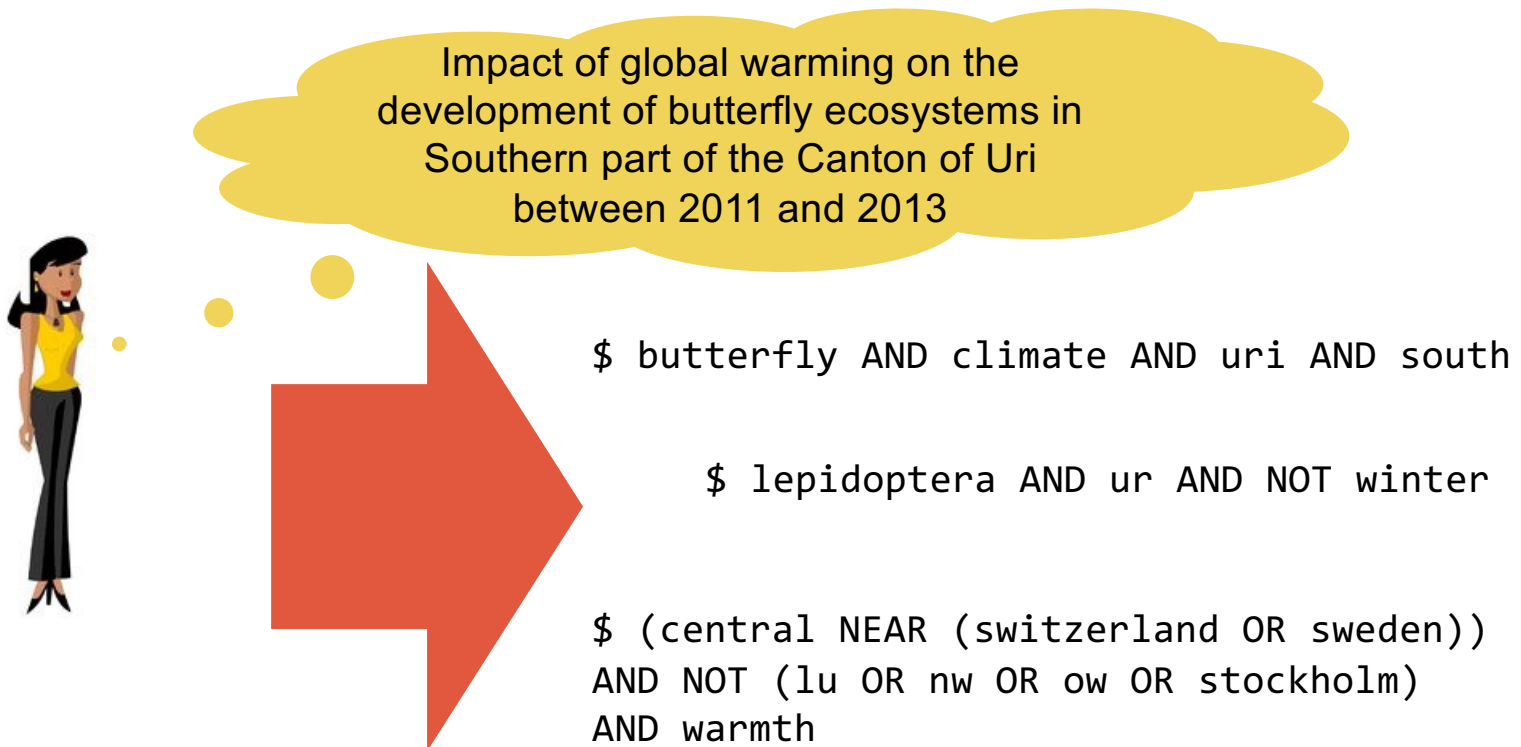
Effectiveness of an IR system

Information need

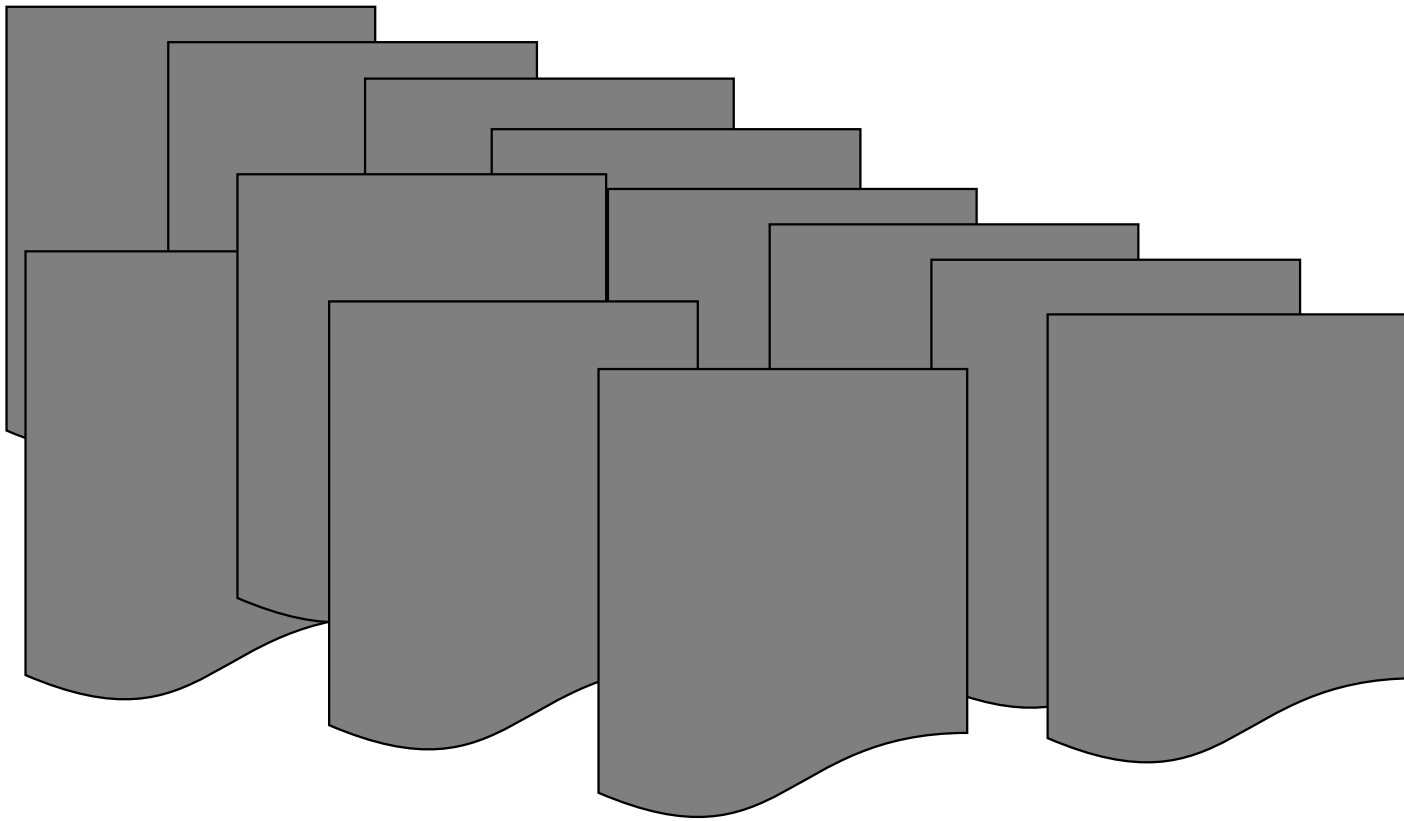


Impact of global warming on the
development of butterfly ecosystems in
Southern part of the Canton of Uri
between 2011 and 2013

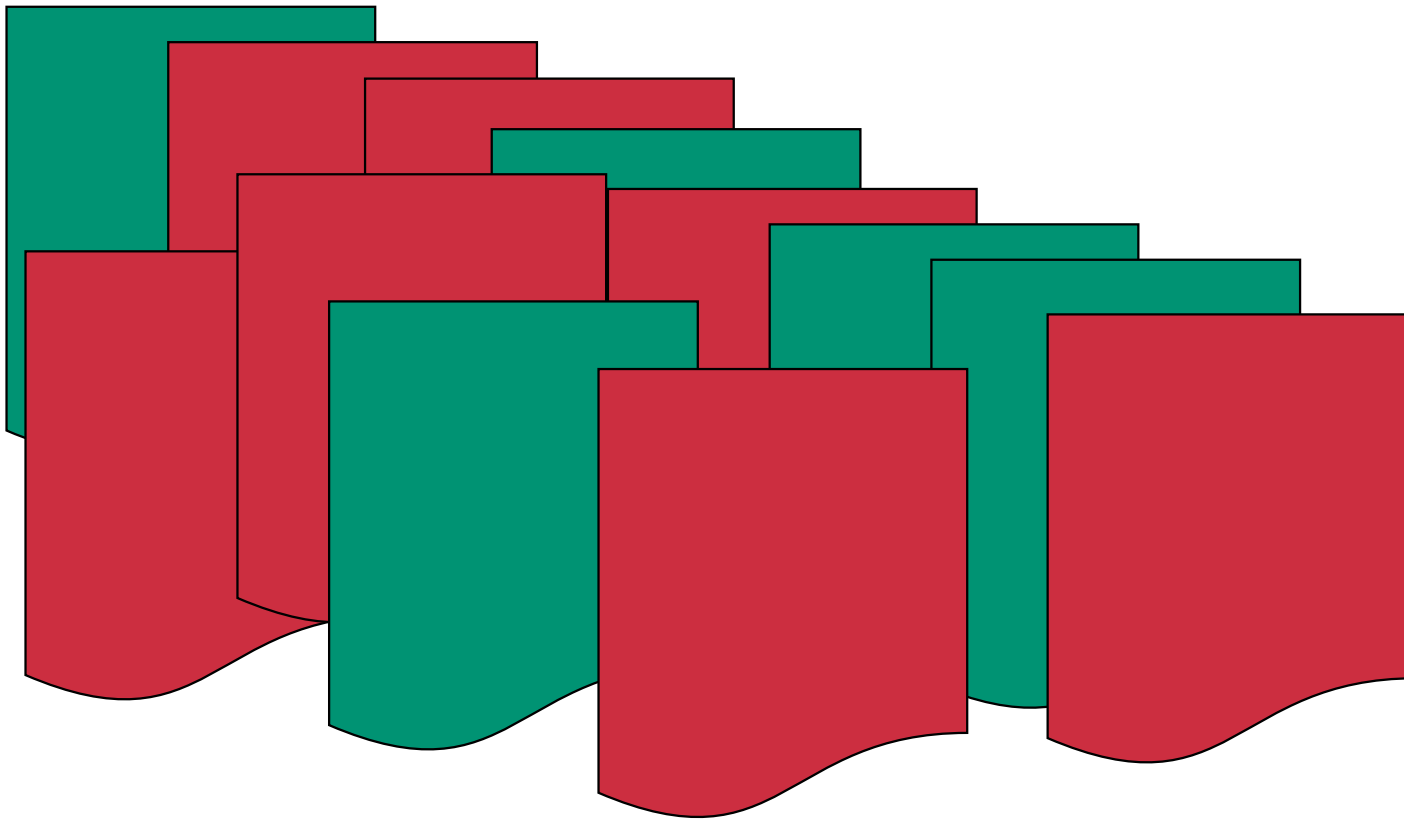
Information need



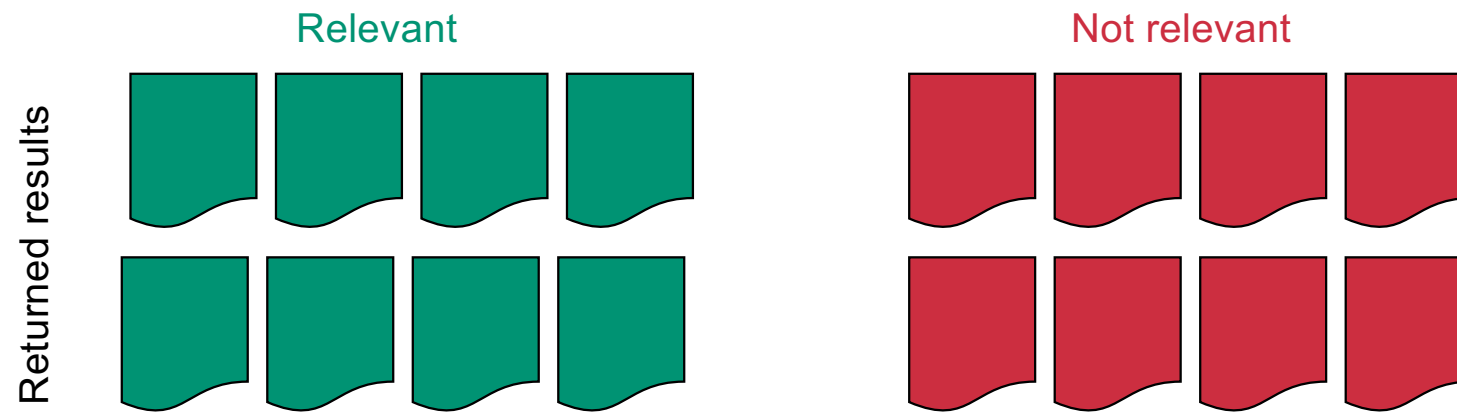
Results



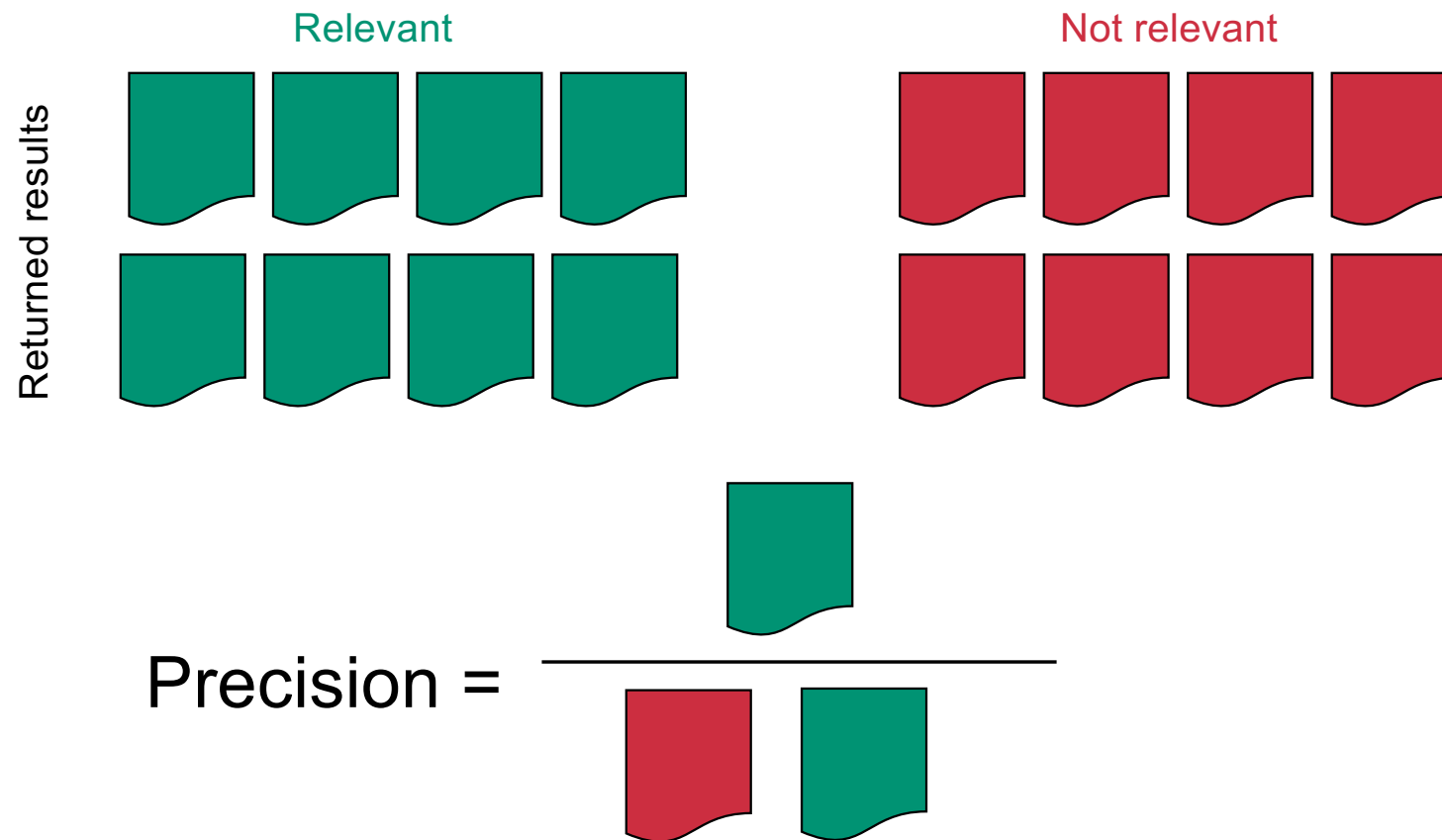
Results



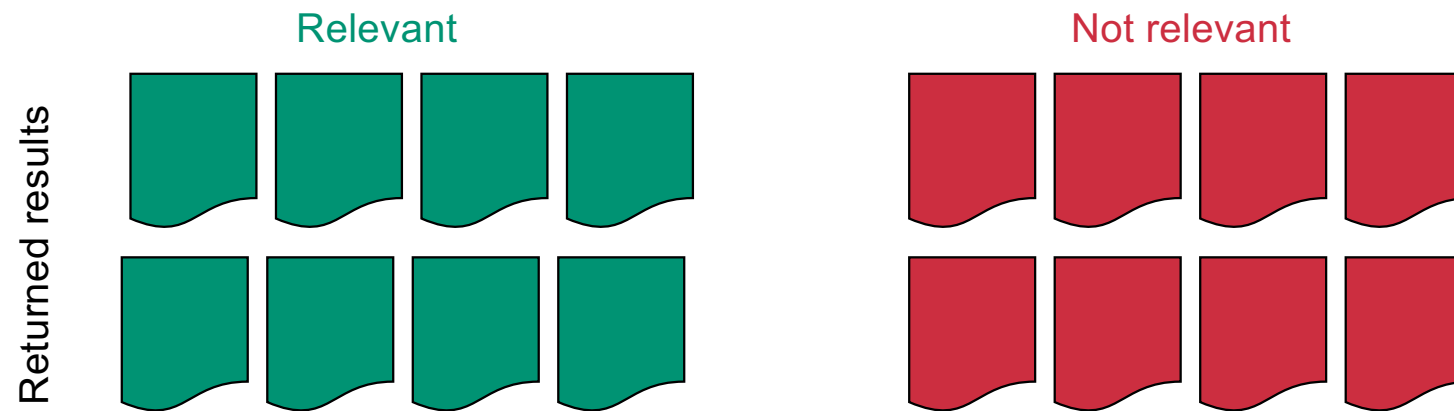
Results



Results

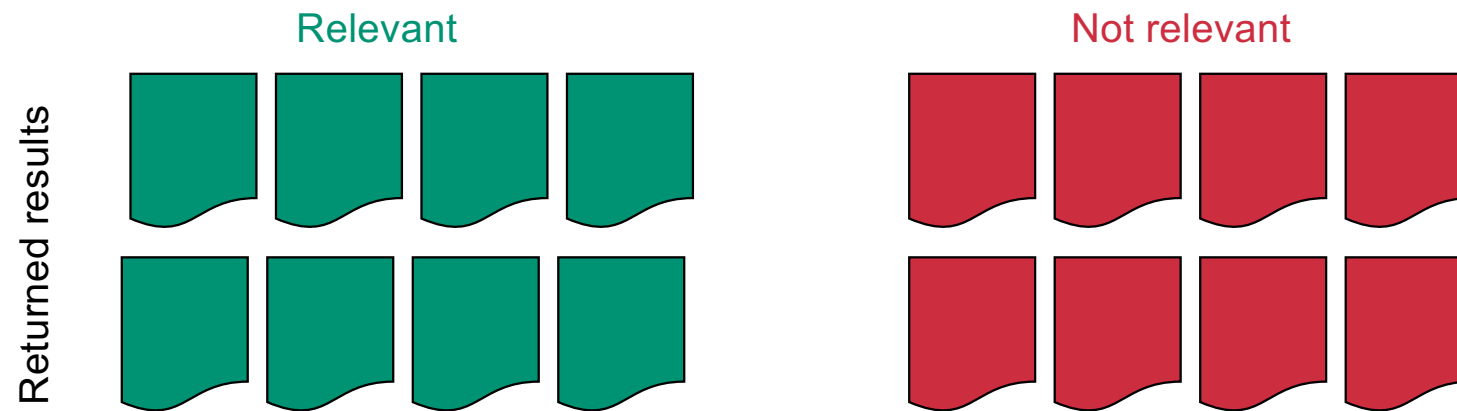


Results



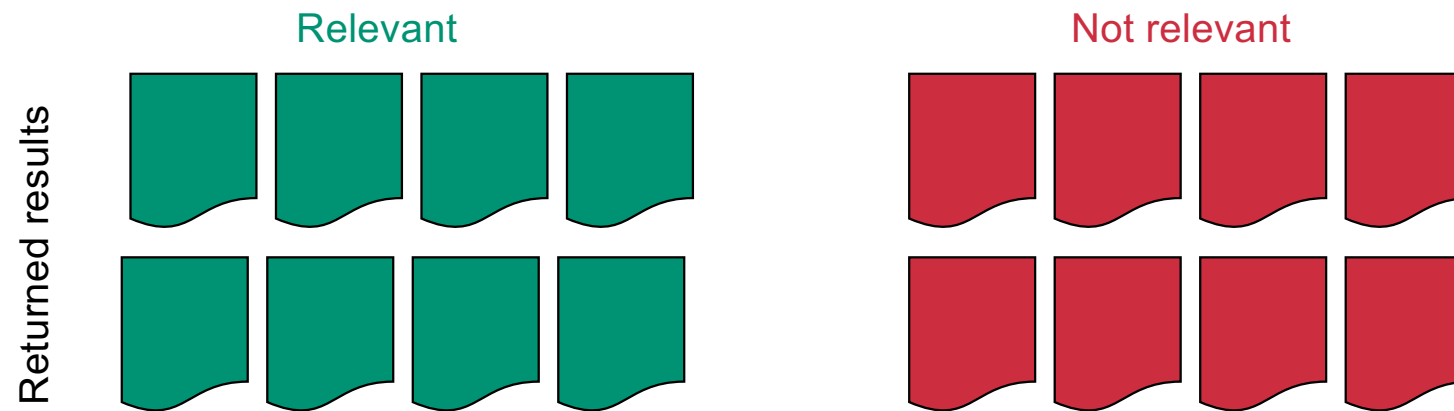
$$\text{Precision} = \frac{\text{true positives}}{\text{returned}}$$

Results



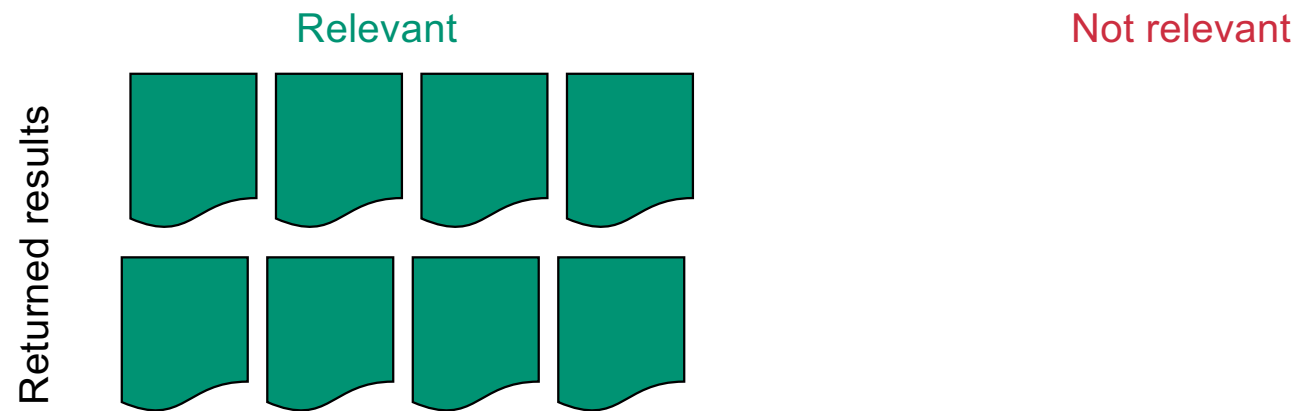
Precision =

Results



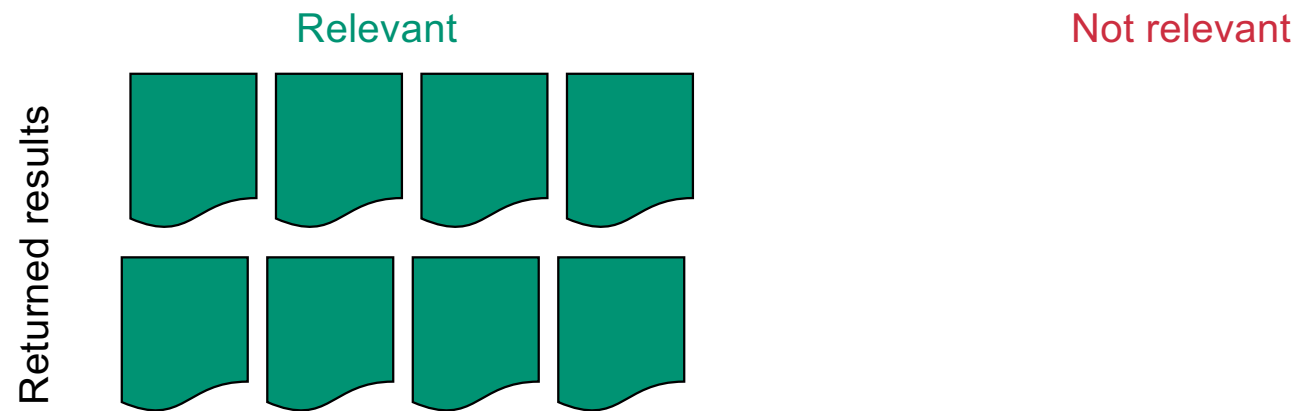
Precision = 50%

Results



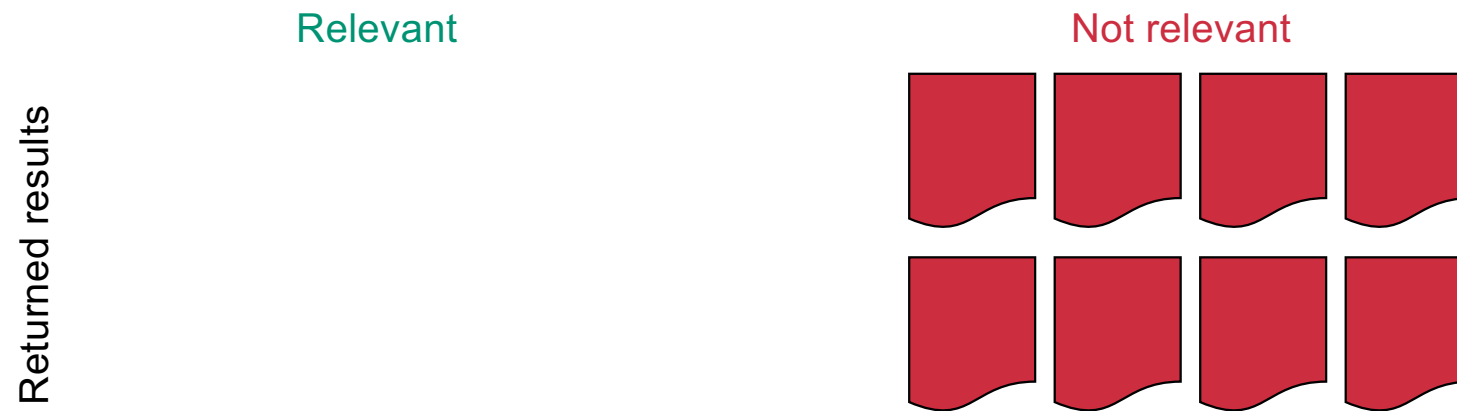
Precision =

Results



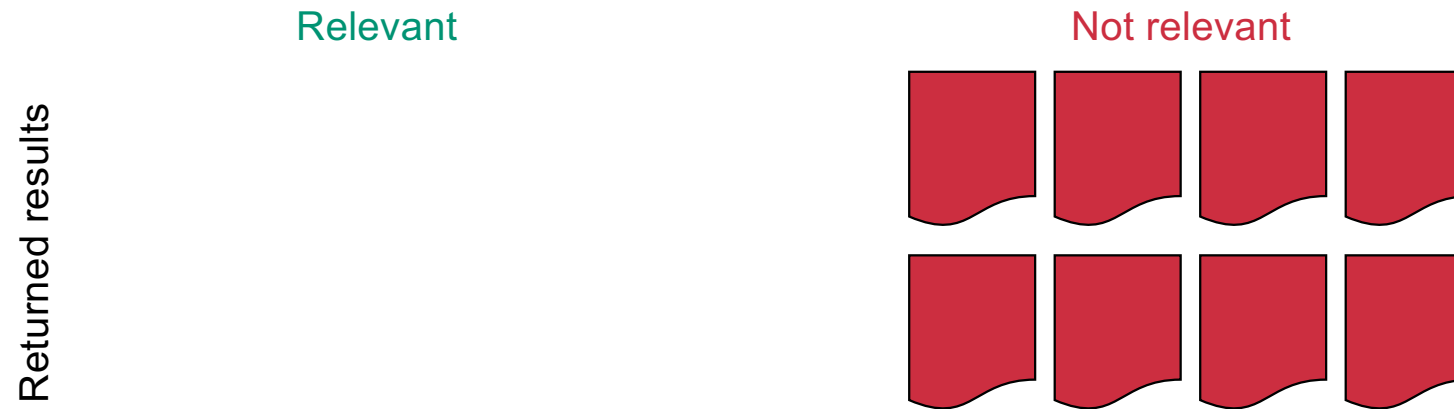
Precision = 100%

Results



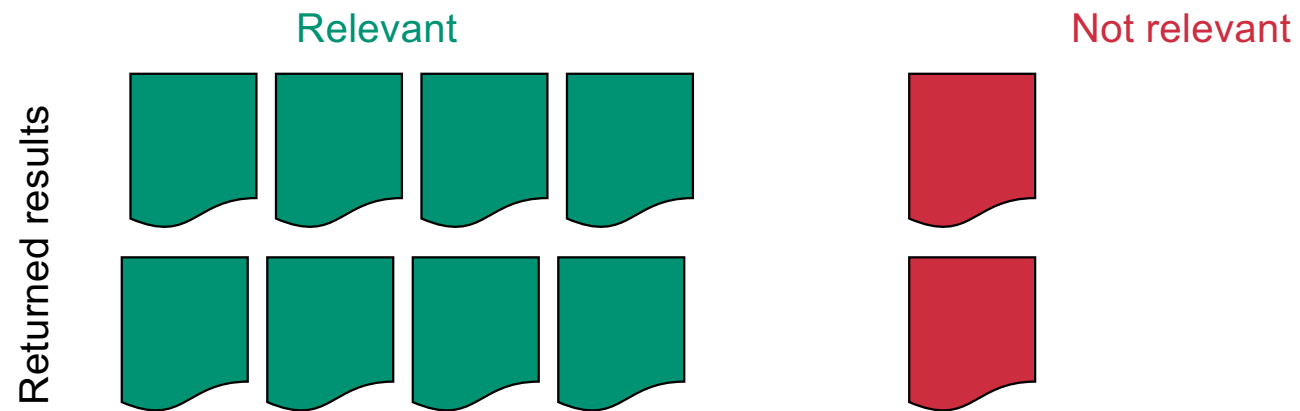
Precision =

Results



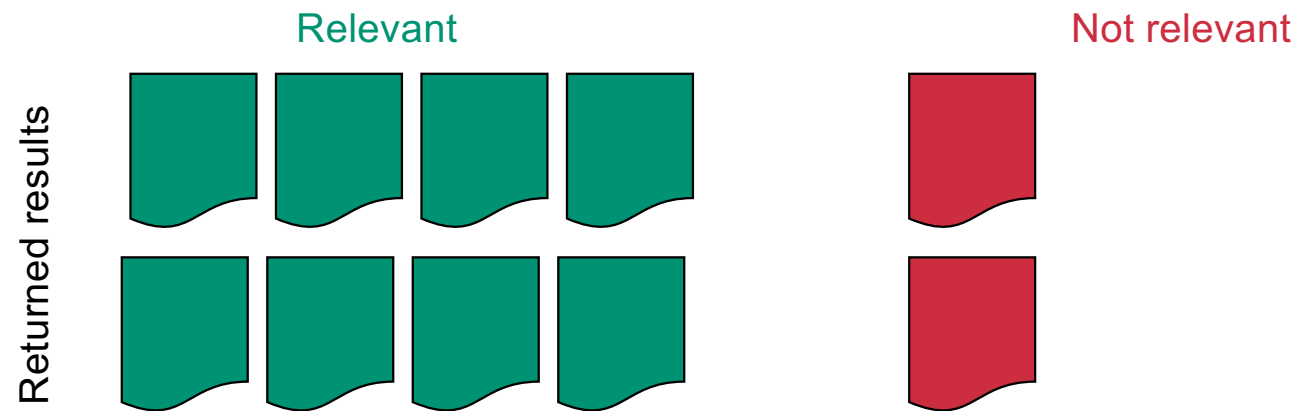
Precision = 0%

Results



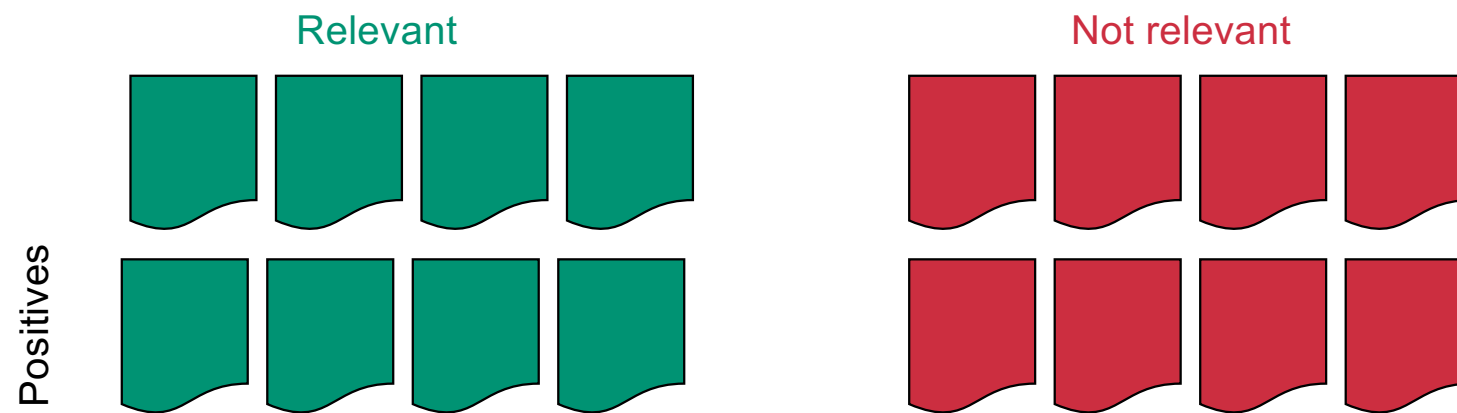
Precision =

Results

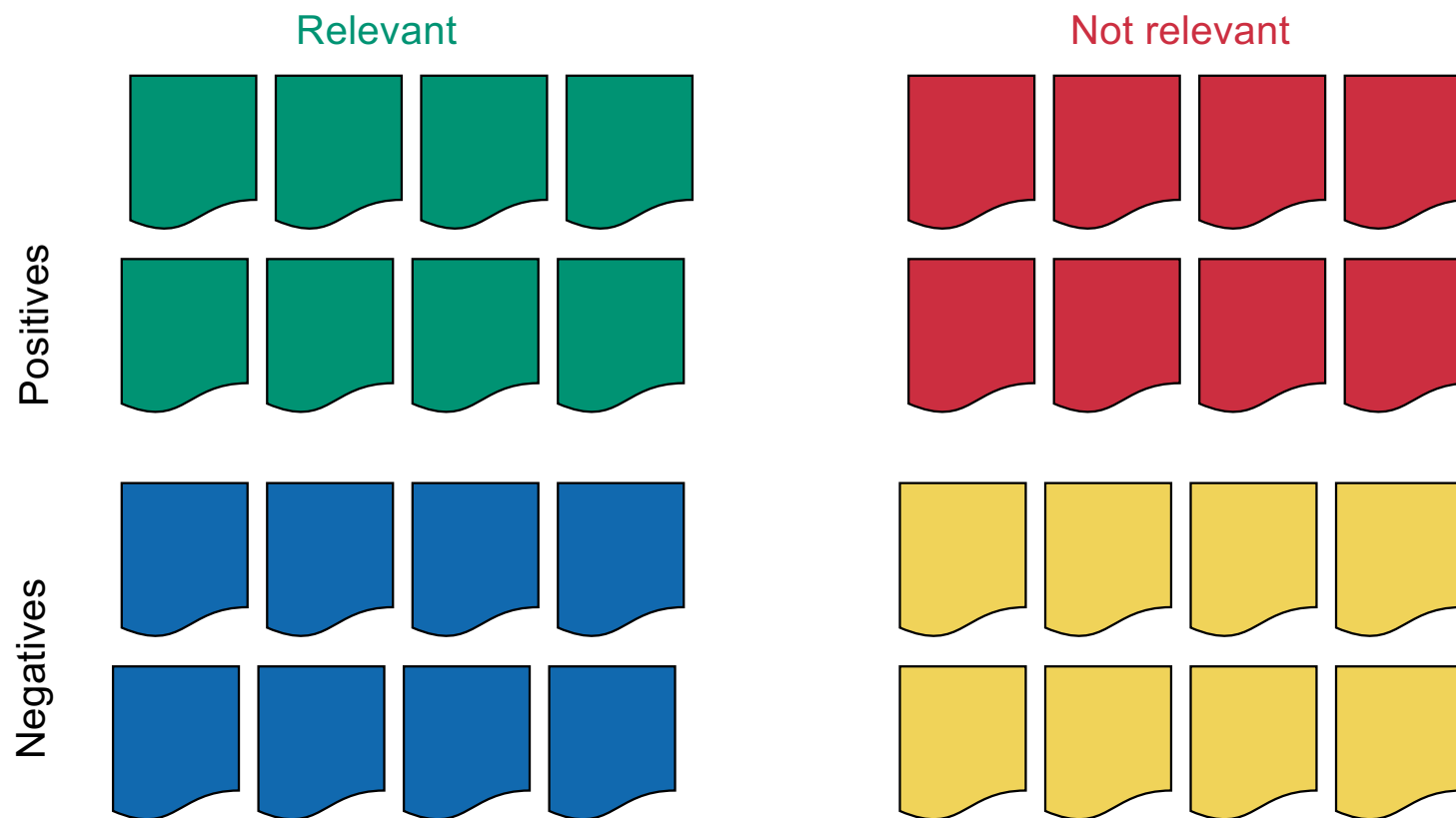


Precision = 80%

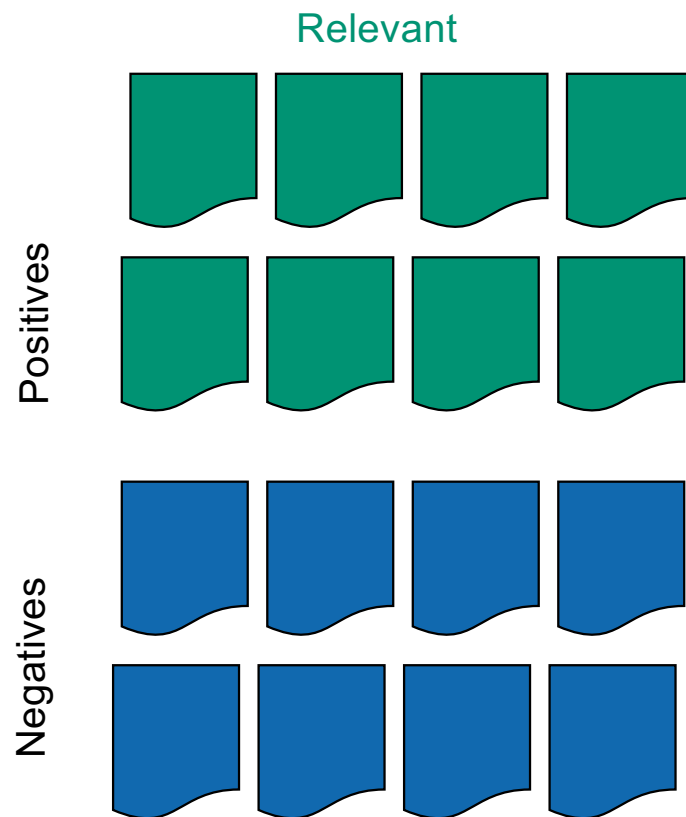
Results



Results



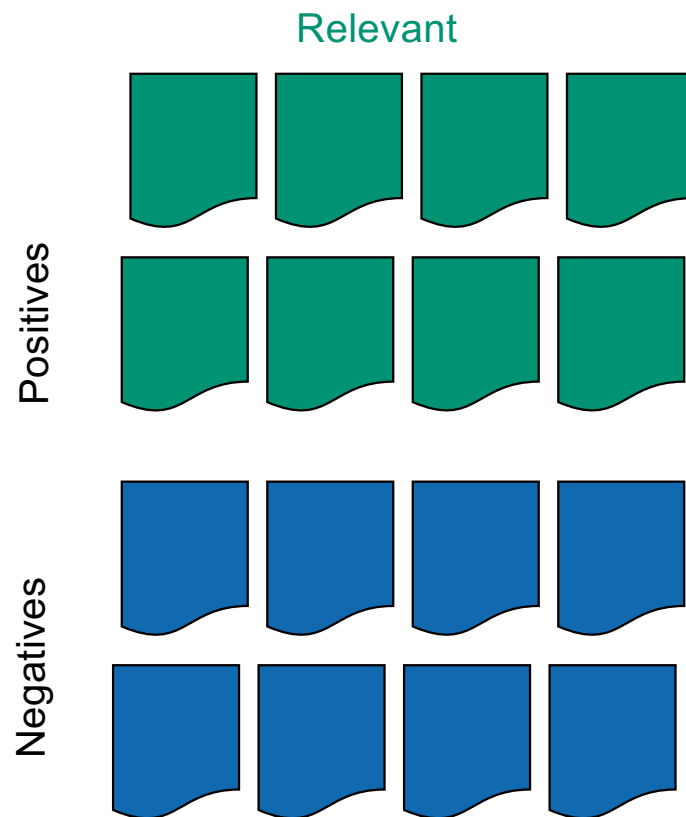
Results



$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

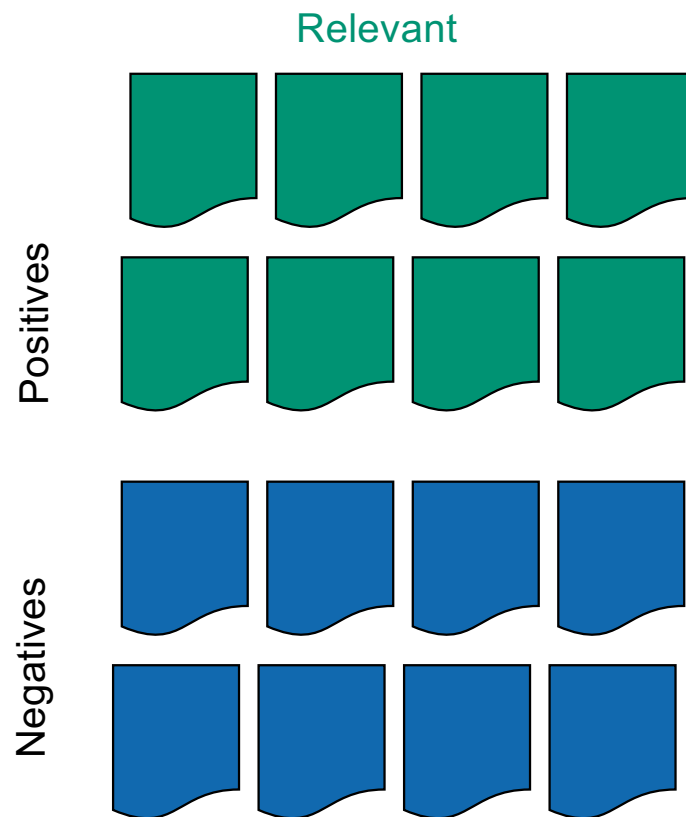
Category	Count
True Positives	1
False Negatives	1

Results



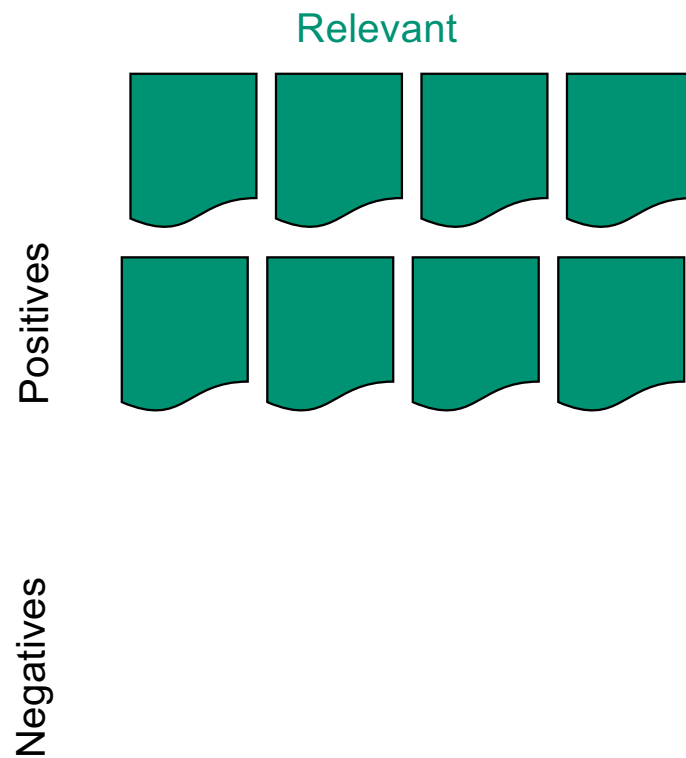
$$\text{Recall} = \frac{\text{true positives}}{\text{relevant}}$$

Results



Recall = 50%

Results



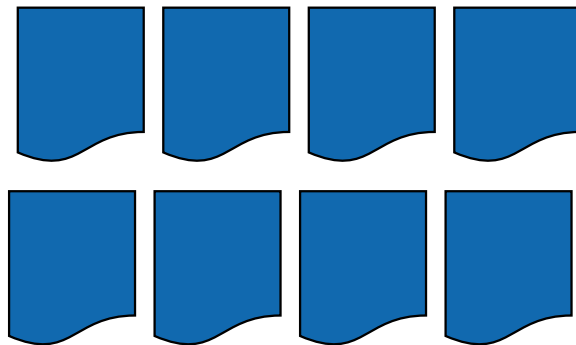
Recall = 100%

Results

Relevant

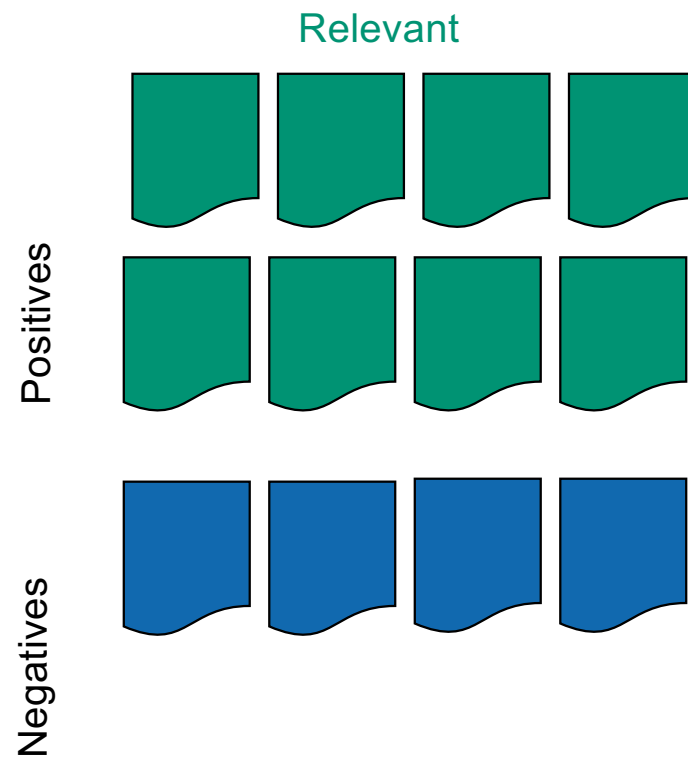
Positives

Negatives



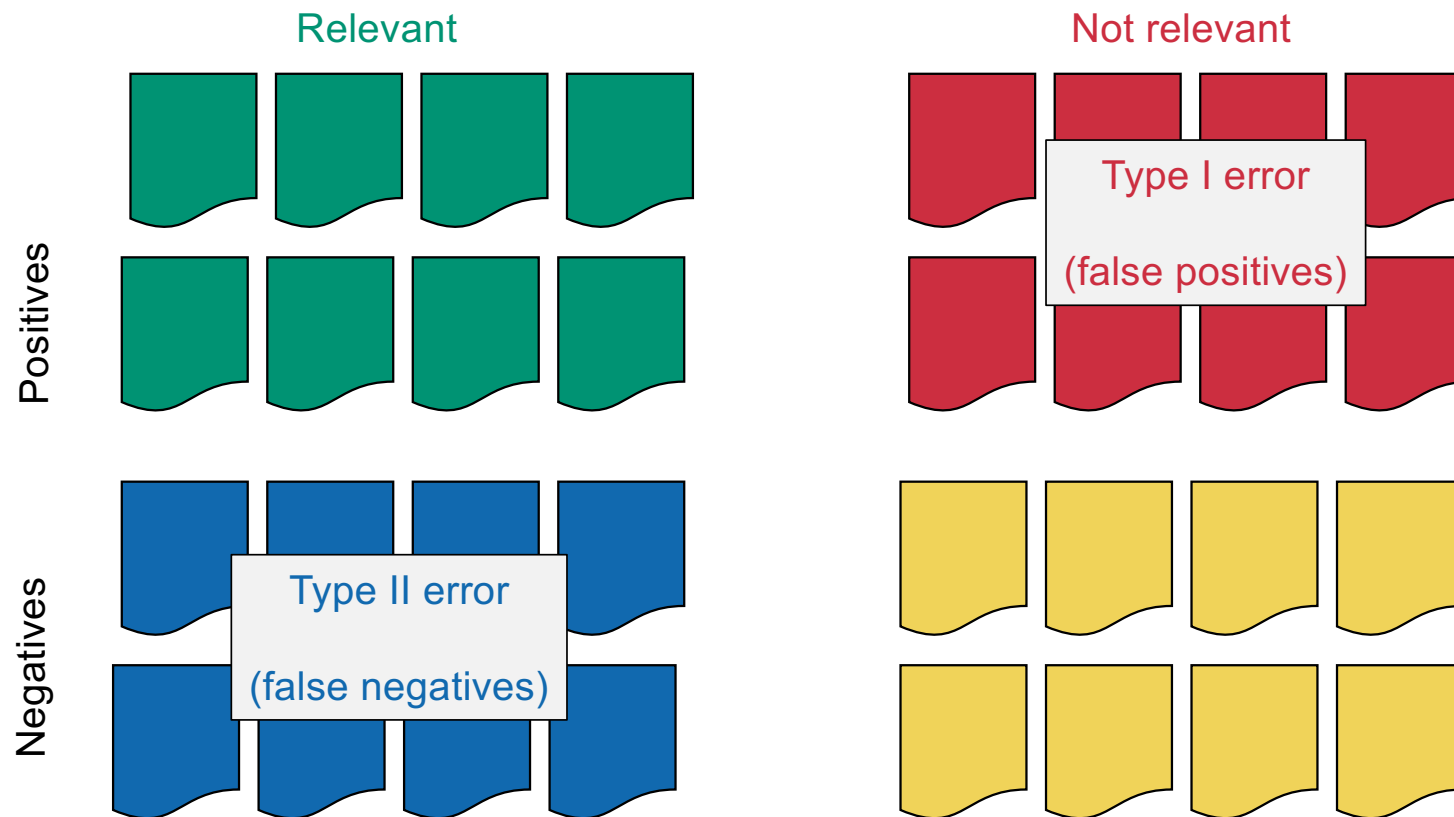
Recall = 0%

Results

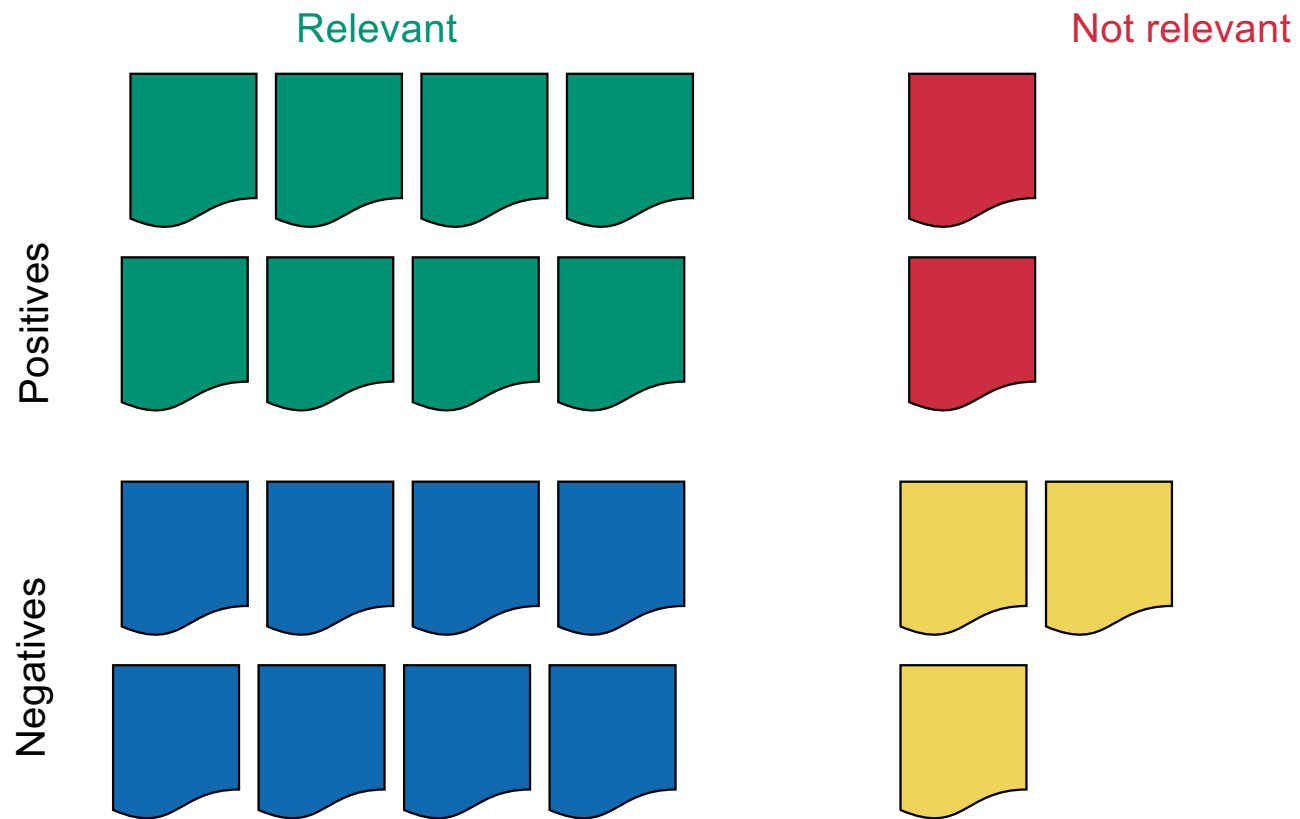


Recall = 67%

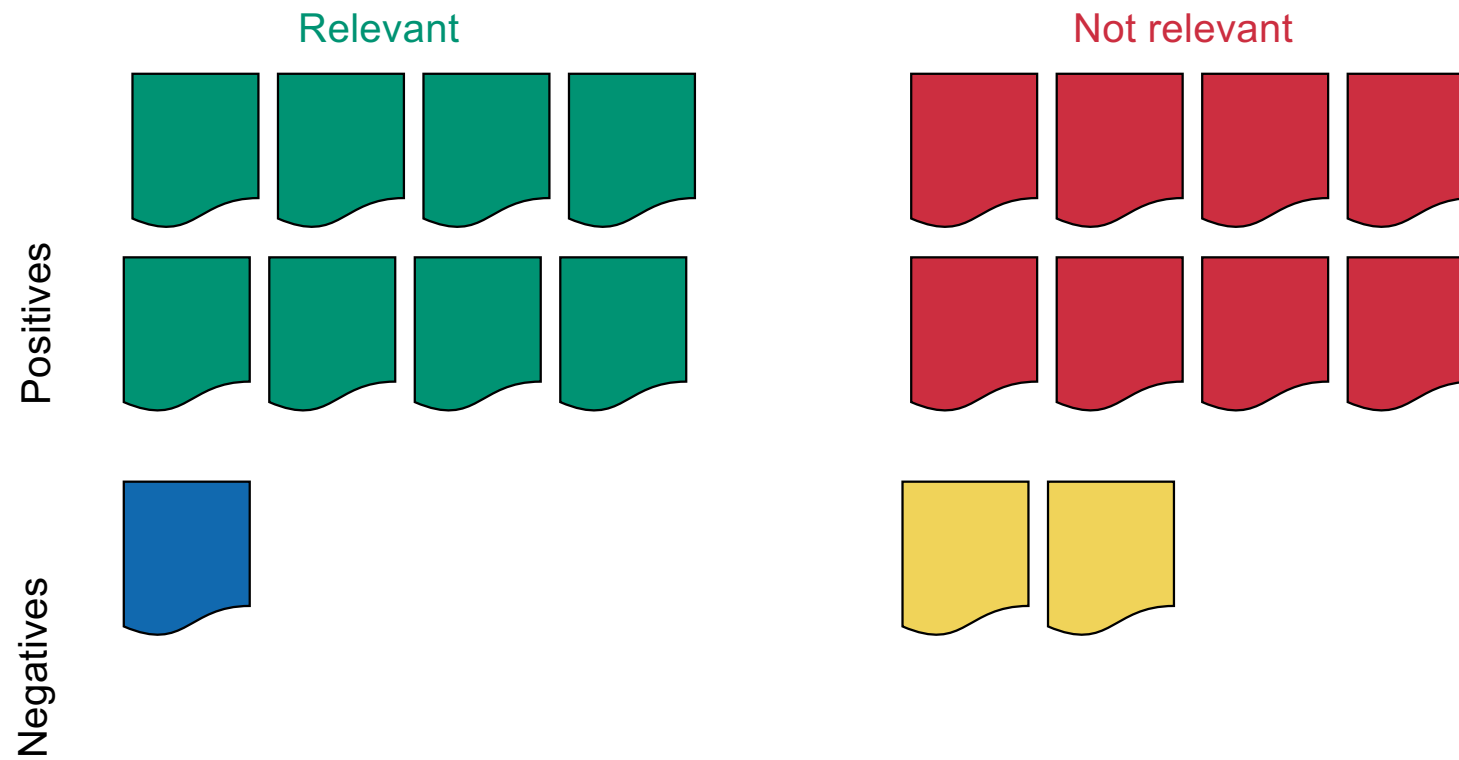
Type I vs. Type II error



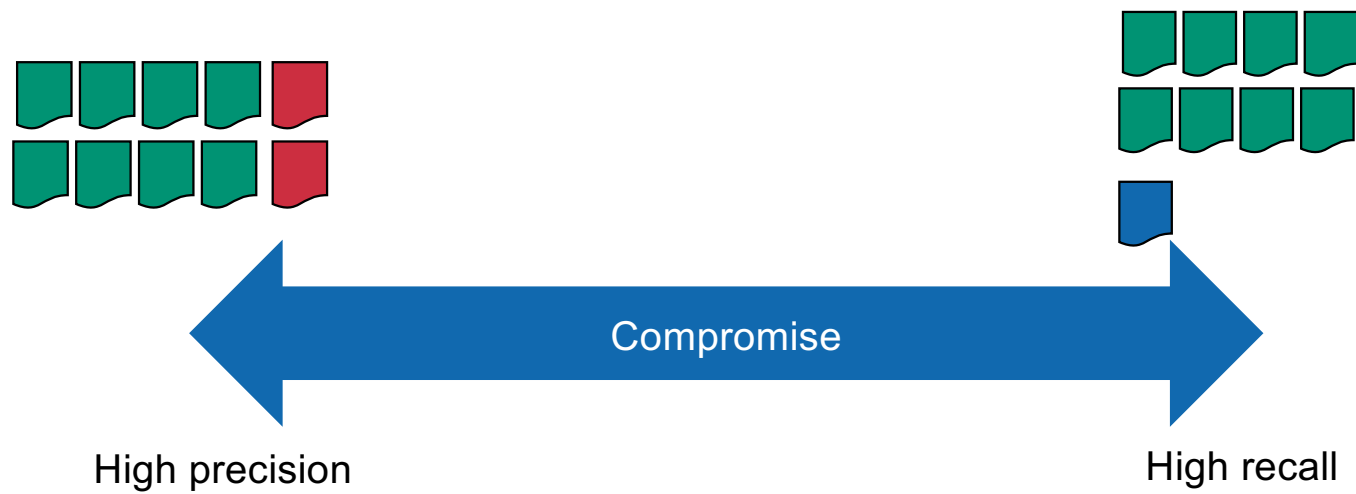
High precision

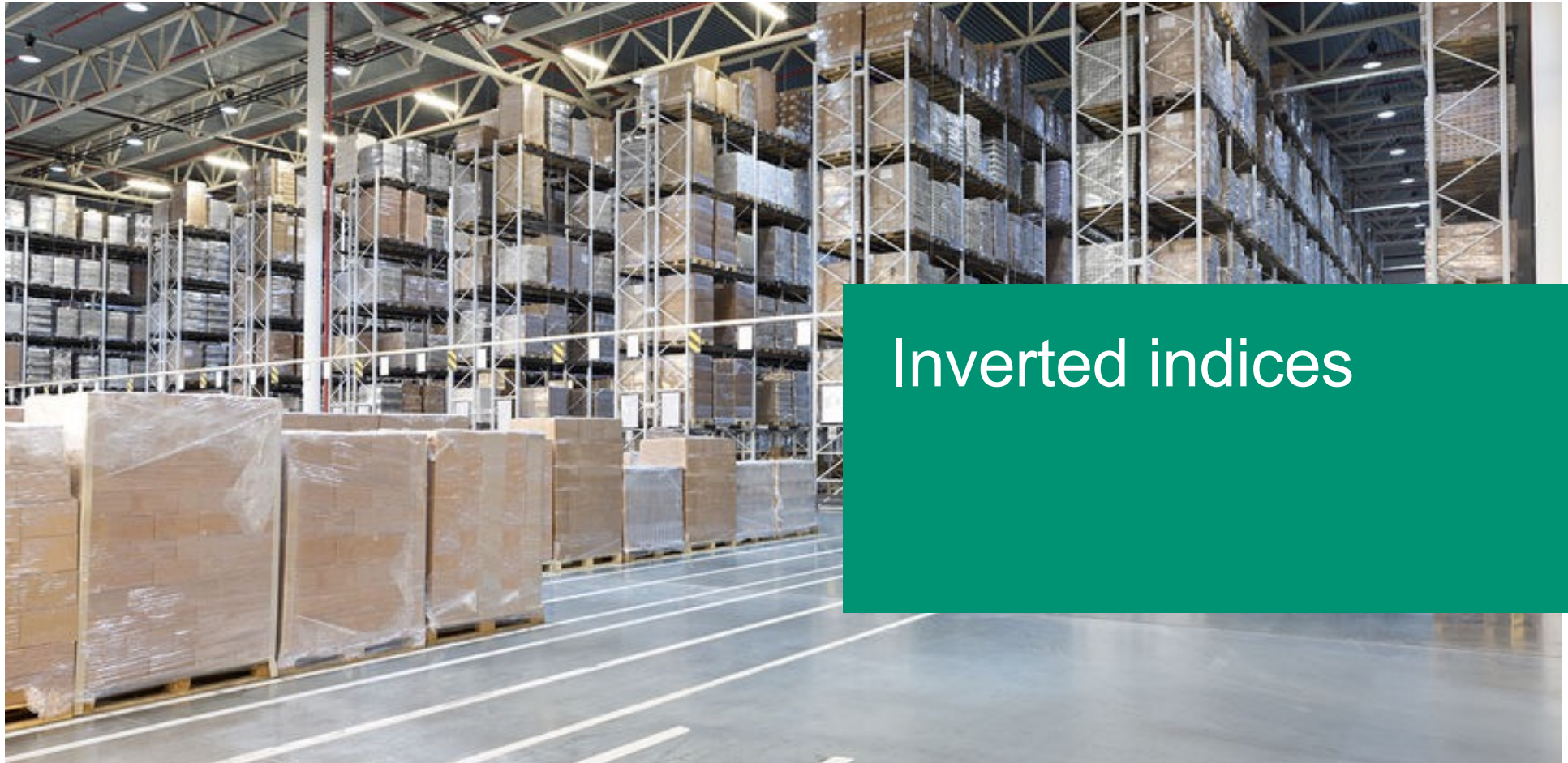


High recall



Compromise



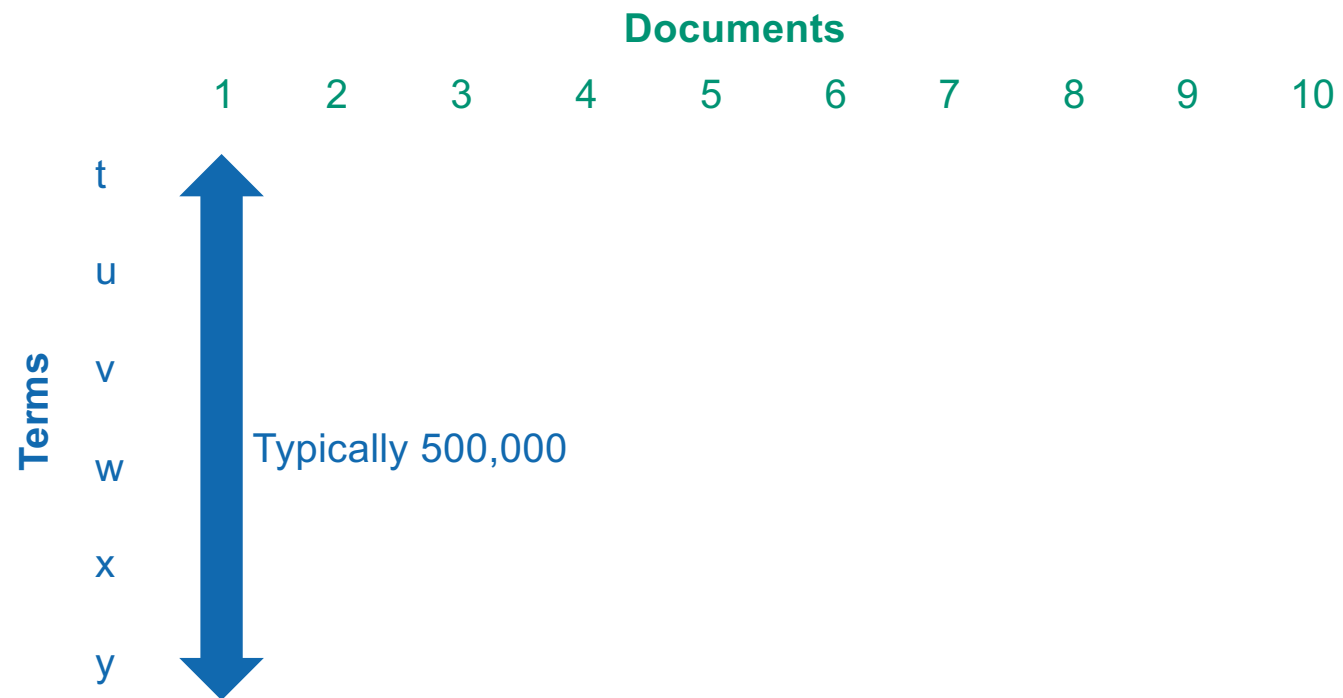


Inverted indices

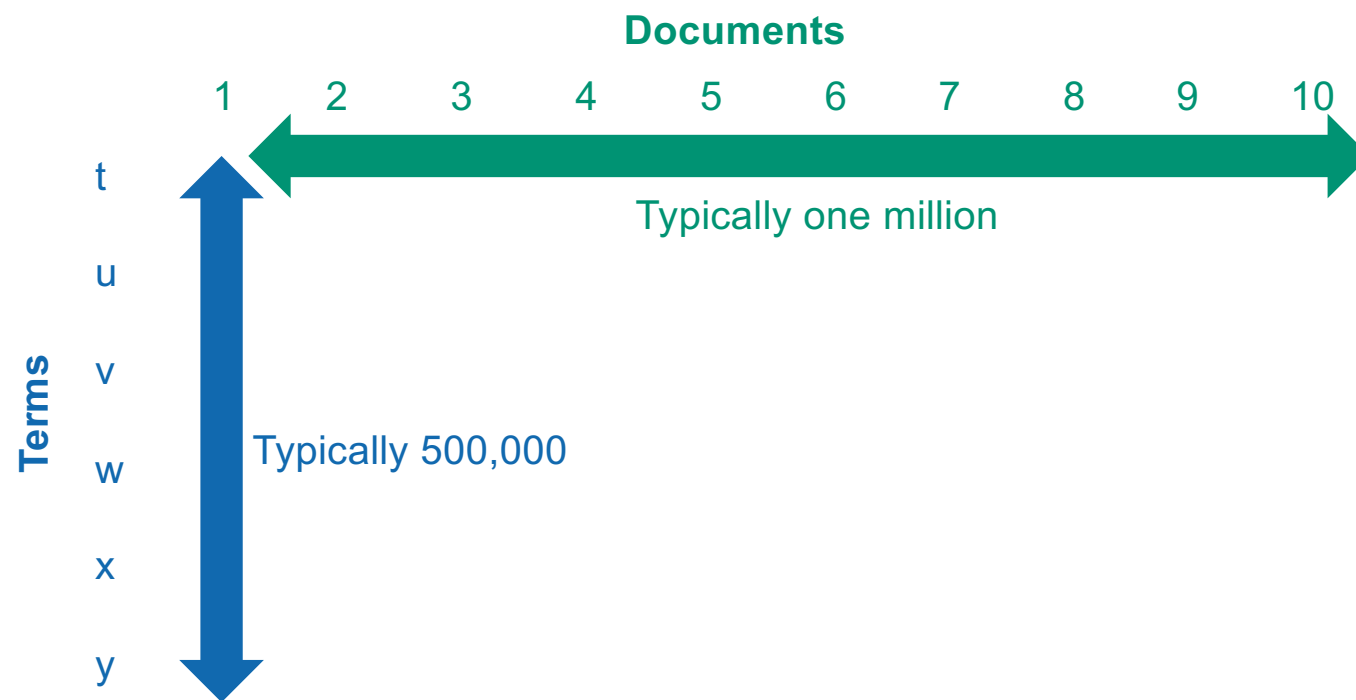
Incidence Matrix

		Documents									
		1	2	3	4	5	6	7	8	9	10
Terms	t	1	0	1	1	1	0	0	0	1	1
	u	0	0	1	0	1	1	1	1	0	0
	v	0	1	1	1	0	1	0	1	0	1
	w	0	0	0	1	1	0	0	0	0	0
	x	1	0	1	1	1	0	1	0	0	1
	y	0	0	0	0	1	0	0	1	0	1

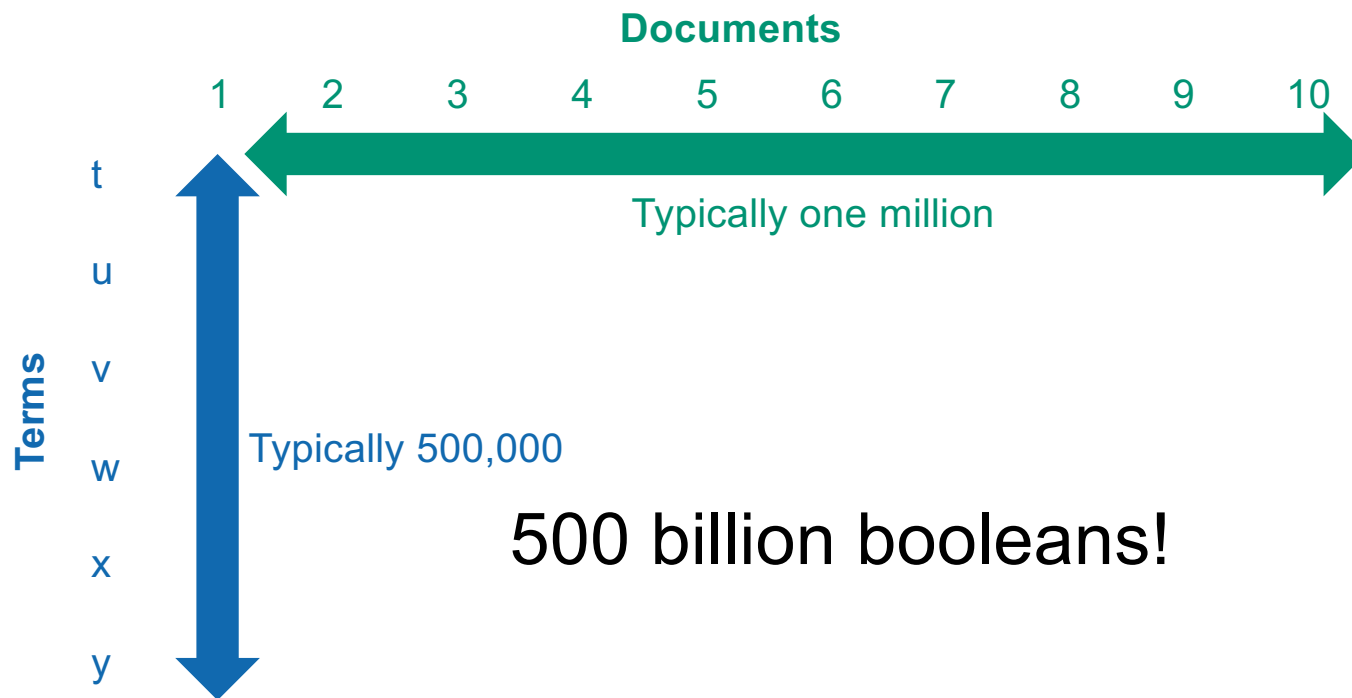
Incidence Matrix



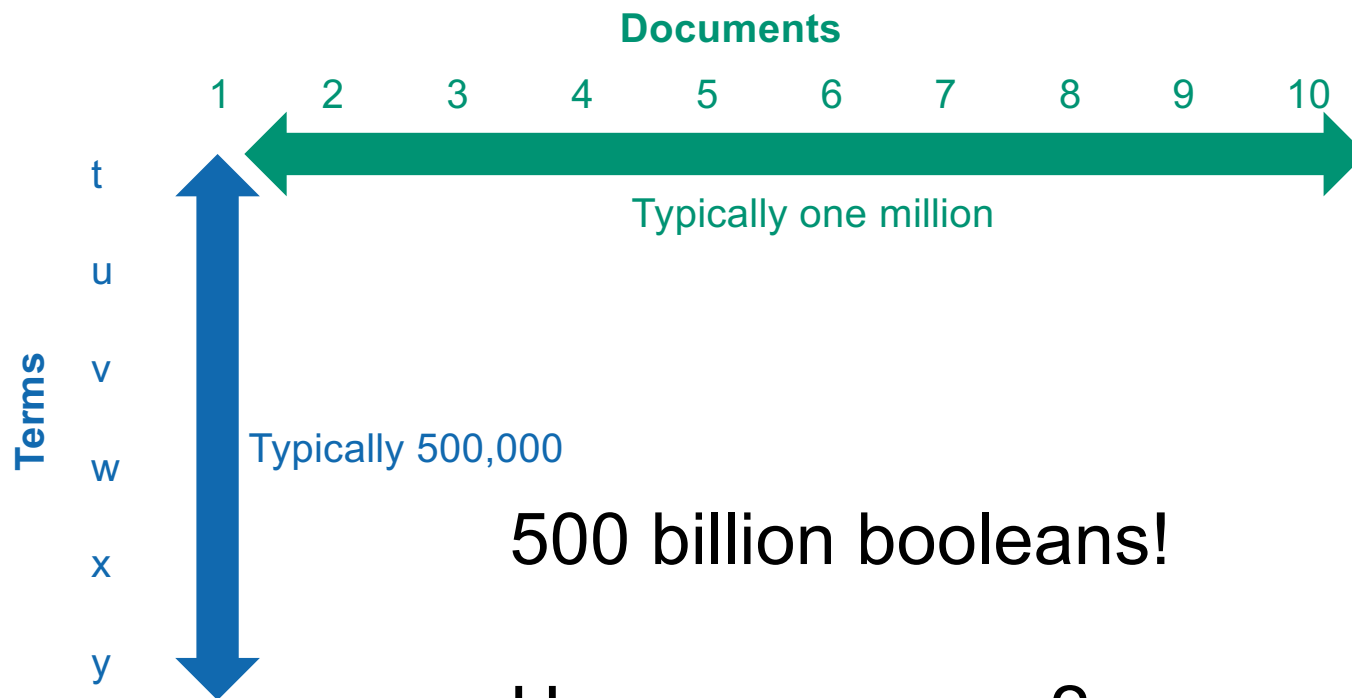
Incidence Matrix



Incidence Matrix



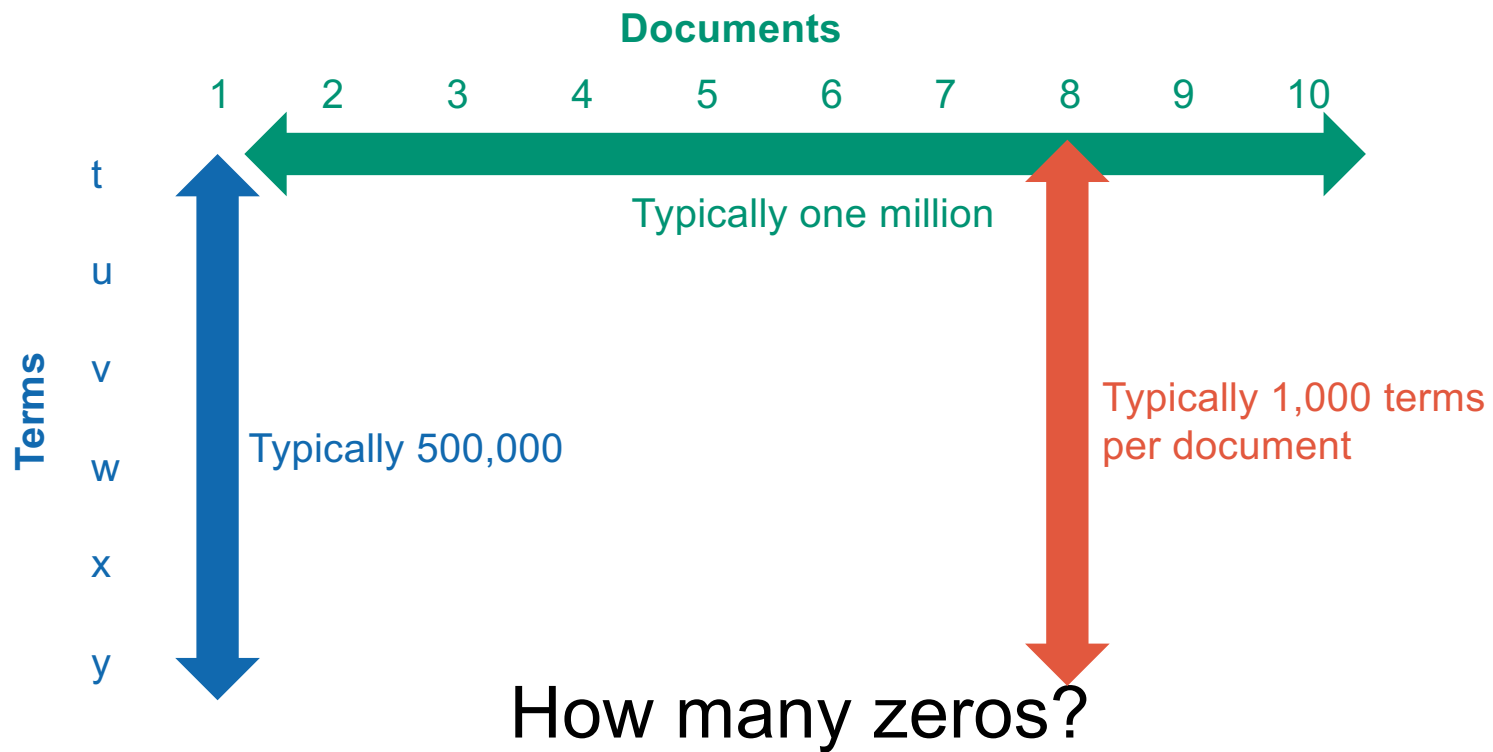
Incidence Matrix



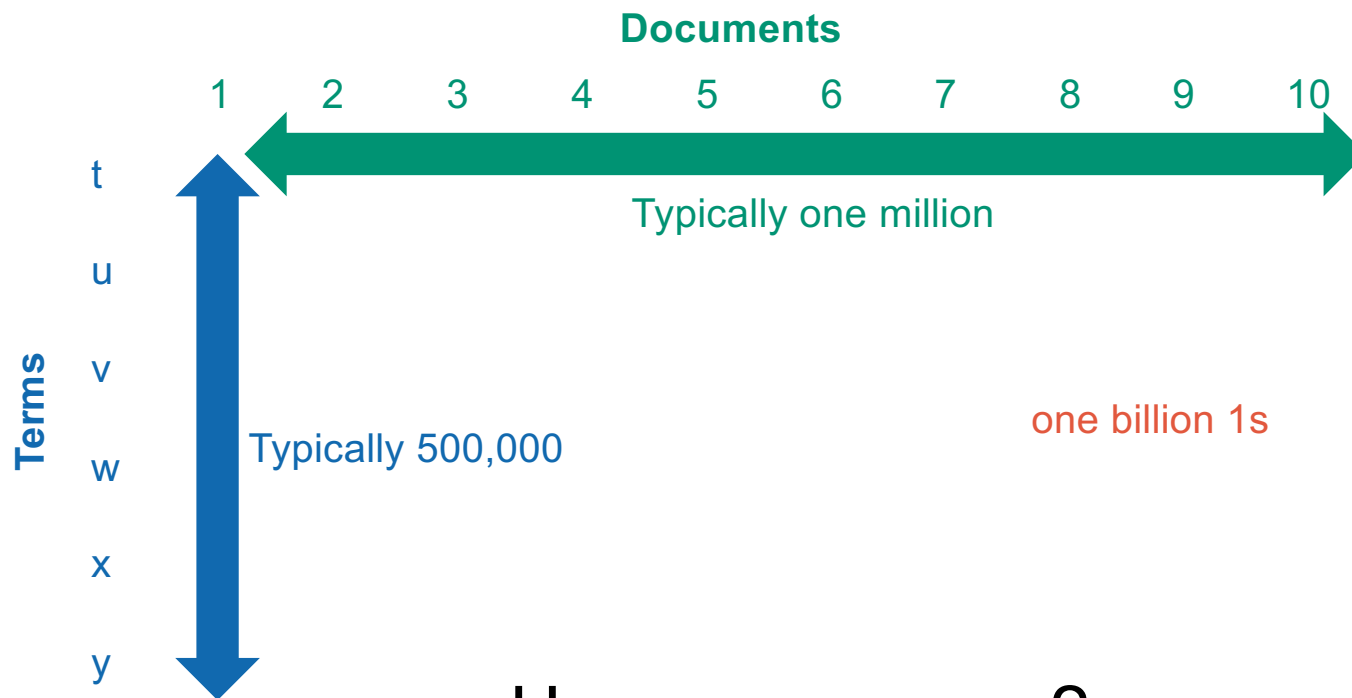
500 billion booleans!

How many zeros?

Incidence Matrix

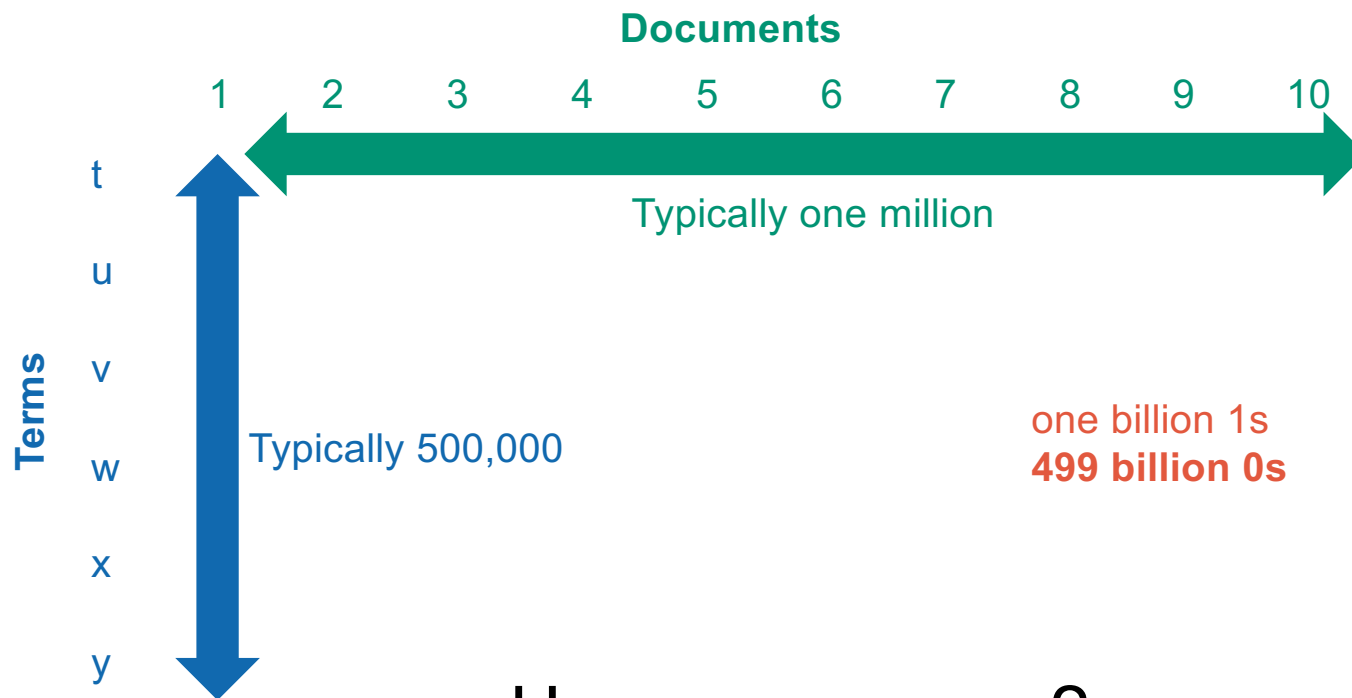


Incidence Matrix



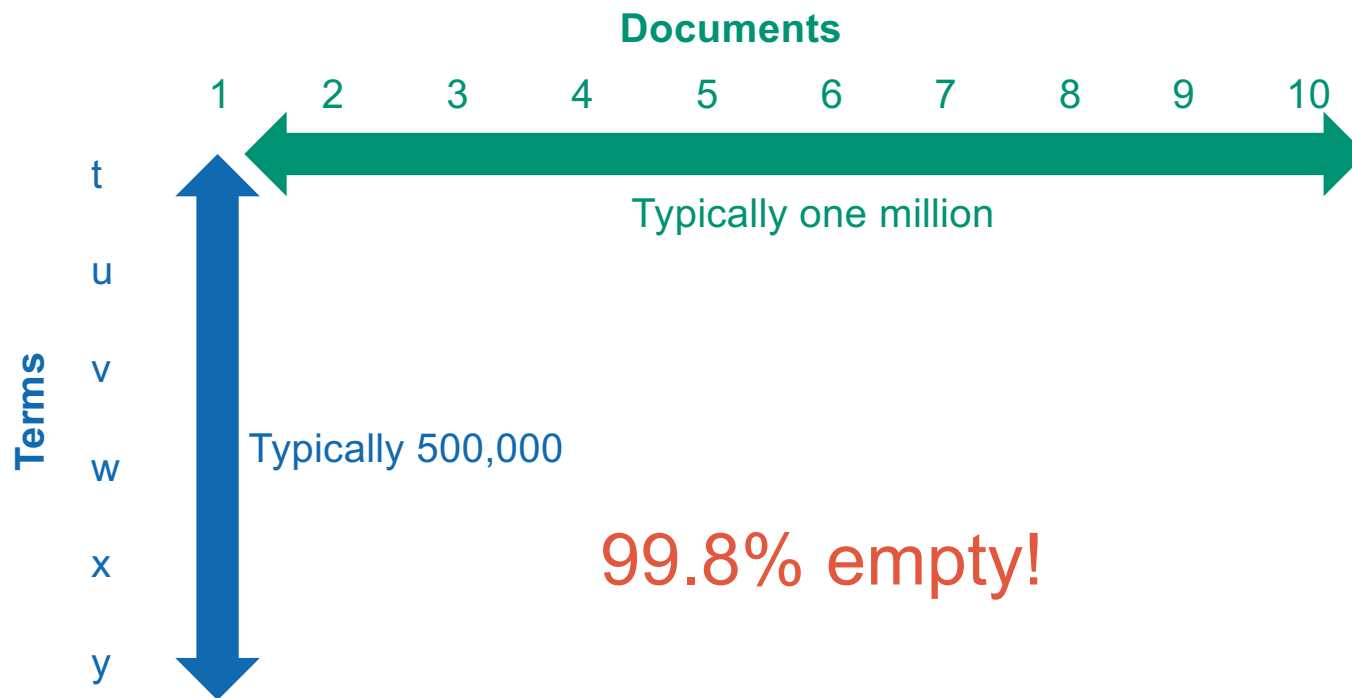
How many zeros?

Incidence Matrix

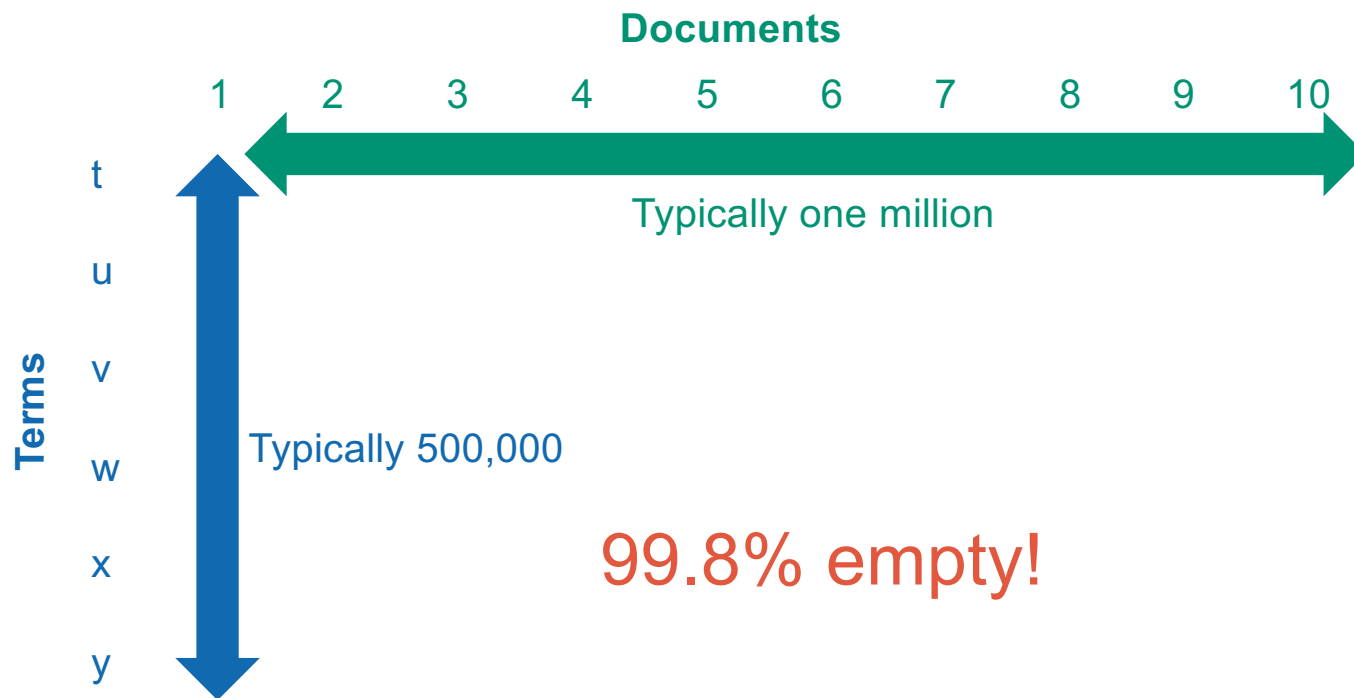


How many zeros?

Incidence Matrix



Incidence Matrix



99.8% empty!

This is space-inefficient.

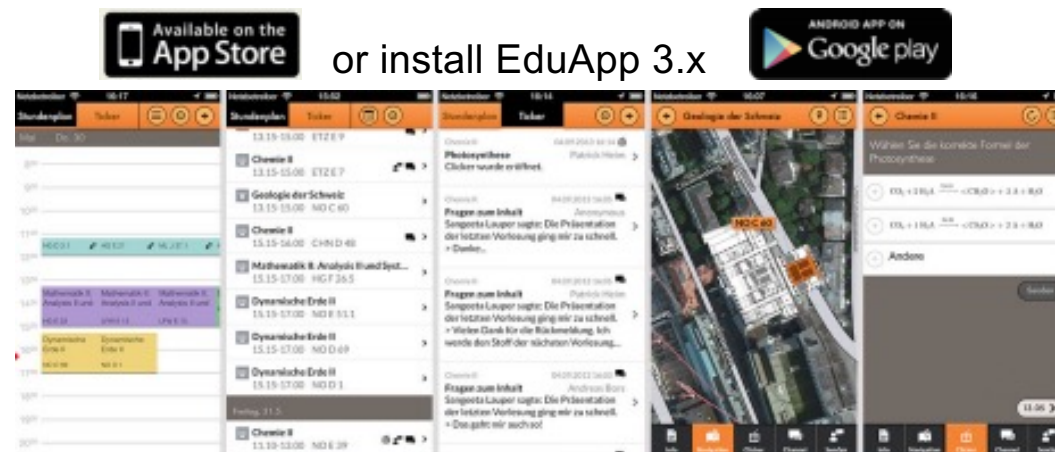
Can we store this in a better way?

		Documents									
		1	2	3	4	5	6	7	8	9	10
Terms	t	1	0	0	1	0	0	0	0	1	1
	u	0	0	0	0	1	1	1	1	0	0
	v	0	1	0	1	0	1	0	1	0	1
	w	0	0	0	0	1	0	0	0	0	0
	x	1	0	1	1	0	0	1	0	0	0
	y	0	0	0	0	1	0	0	1	0	1

Poll

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Can we store this in a better way?

		Documents									
		1	2	3	4	5	6	7	8	9	10
Terms	t	✓			✓					✓	✓
	u					✓	✓	✓			
	v		✓		✓		✓		✓		✓
	w					✓					
	x	✓		✓	✓			✓			
	y					✓			✓		✓

Can we store this in a better way?

		Documents									
		1	2	3	4	5	6	7	8	9	10
Terms	t		1, 4, 9, 10								
	u		5, 6, 7								
	v			2, 4, 6, 8, 10							
	w	5									
	x		1, 3, 4, 7								
	y	5, 8, 10									

Inverted index (inverted file)

Terms	t	1, 4, 9, 10
	u	5, 6, 7
	v	2, 4, 6, 8, 10
	w	5
	x	1, 3, 4, 7
	y	5, 8, 10

Inverted index (inverted file)

Terms	t	1, 4, 9, 10
	u	5, 6, 7
	v	2, 4, 6, 8, 10
	w	5
	x	1, 3, 4, 7
	y	5, 8, 10

Dictionary

Inverted index (inverted file)

Terms	t	1, 4, 9, 10
	u	5, 6, 7
	v	2, 4, 6, 8, 10
	w	5
	x	1, 3, 4, 7
	y	5, 8, 10

Posting

Inverted index (inverted file)

Terms	t	1, 4, 9, 10
	u	5, 6, 7
	v	2, 4, 6, 8, 10
	w	5
	x	1, 3, 4, 7
	y	5, 8, 10

Posting

Assumption:

Each document has a **Document ID (docID)**

Inverted index (inverted file)

Terms	t	1, 4, 9, 10
	u	5, 6, 7
	v	2, 4, 6, 8, 10
	w	5
	x	1, 3, 4, 7
	y	5, 8, 10

Postings list

Inverted index (inverted file)

Terms	t	1, 4, 9, 10
	u	5, 6, 7
	v	2, 4, 6, 8, 10
	w	5
	x	1, 3, 4, 7
	y	5, 8, 10

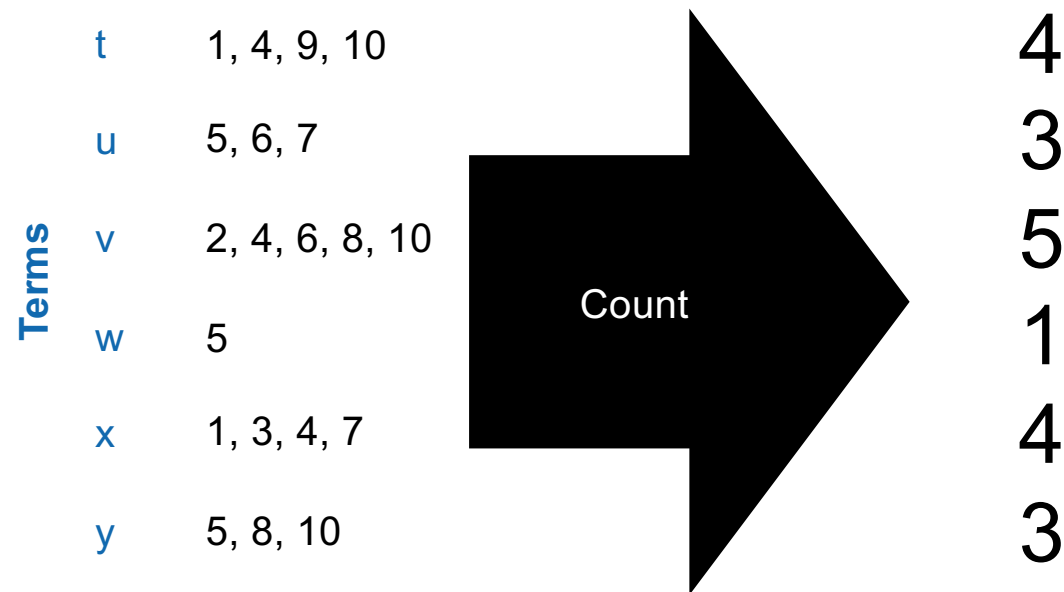
Postings

Inverted index (inverted file)

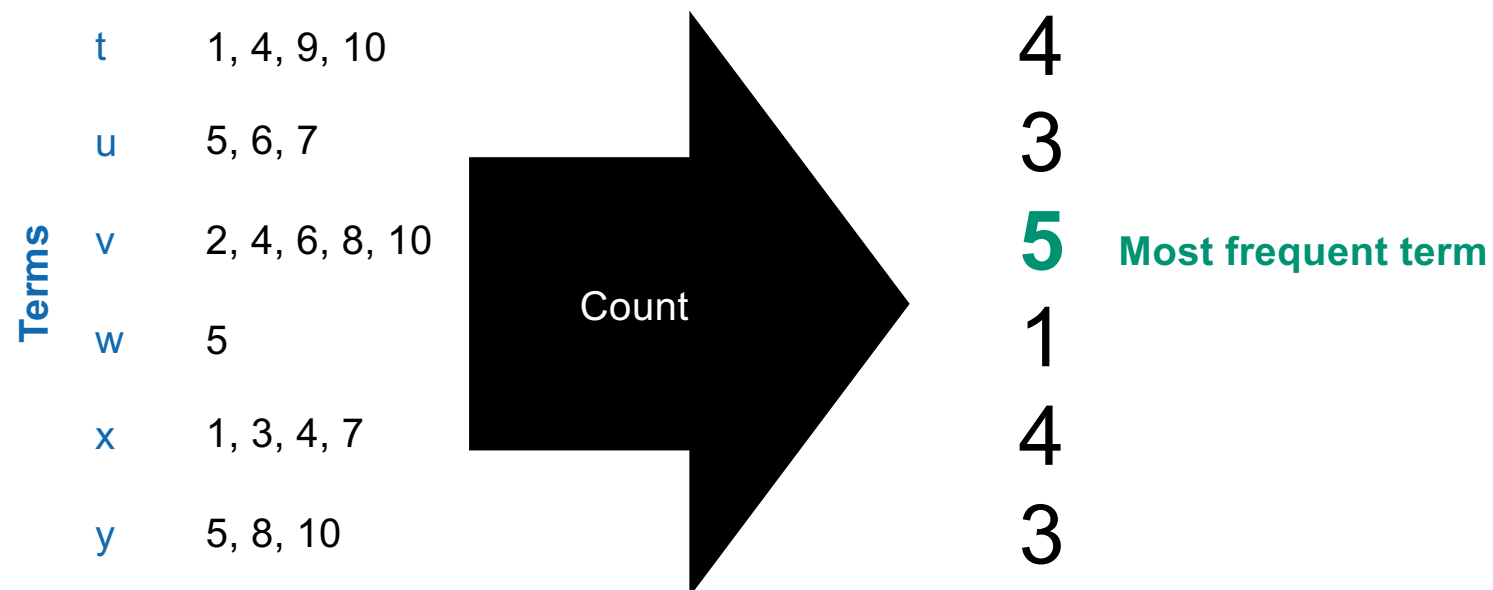
Terms	t	1, 4, 9, 10
	u	5, 6, 7
	v	2, 4, 6, 8, 10
	w	5
	x	1, 3, 4, 7
	y	5, 8, 10

Index

Document frequency



Document frequency



Building an inverted index

Document 1

You come most carefully upon your hour.

Document 2

Take thy fair hour, Laertes; time be thine,

Document 3

My hour is almost come,

Document 4

Possess it merely. That it should come to this!

Tokenize

You come most carefully upon your hour

Take fair thy hour Laertes time be thine

My hour is almost come

Possess it merely That it should come to this

Linguistic pre-processing

you come most carefully upon your hour

take fair thy hour Laertes time be thine

my hour is almost come

Possess it merely that it should come to this

Assign document IDs

you	1	take	2	my	3	possess	4
come	1	fair	2	hour	3	it	4
most	1	thy	2	is	3	merely	4
carefully	1	hour	2	almost	3	that	4
upon	1	Laertes	2	come	3	it	4
your	1	time	2			should	4
hour	1	be	2			come	4
		thine	2			to	4
						this	4

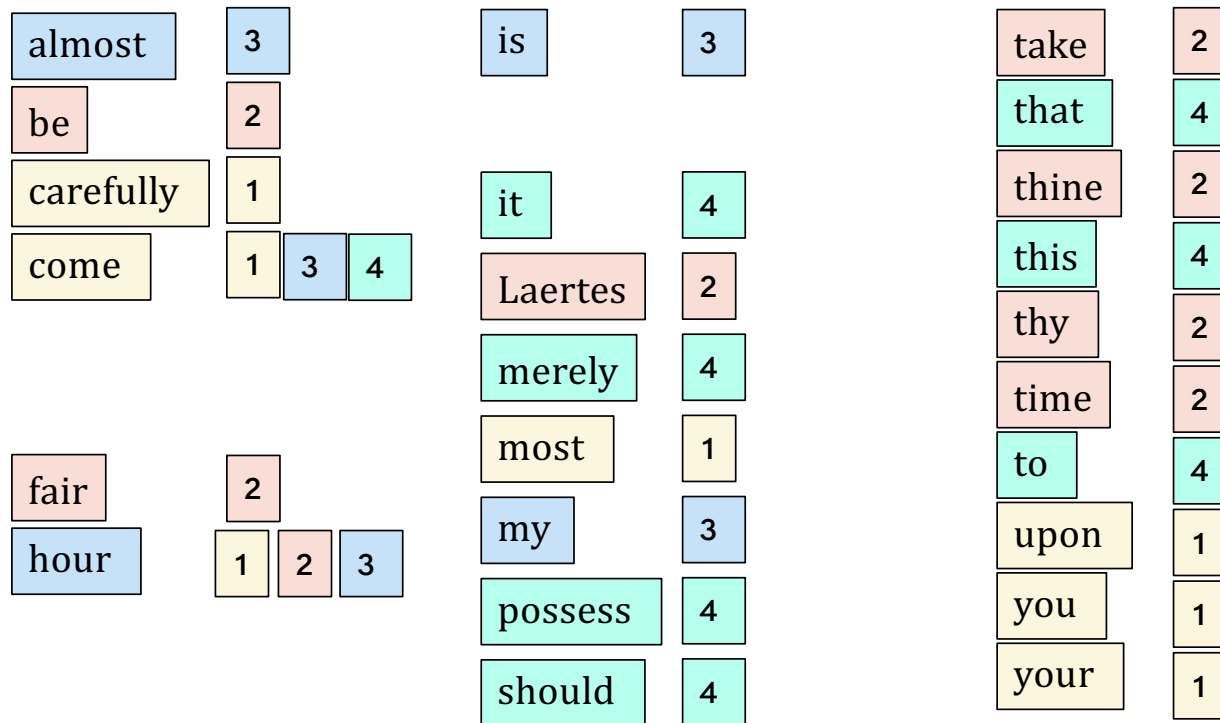
Sort

almost	3
be	2
carefully	1
come	1
come	4
come	3
fair	2
hour	3
hour	1
hour	2

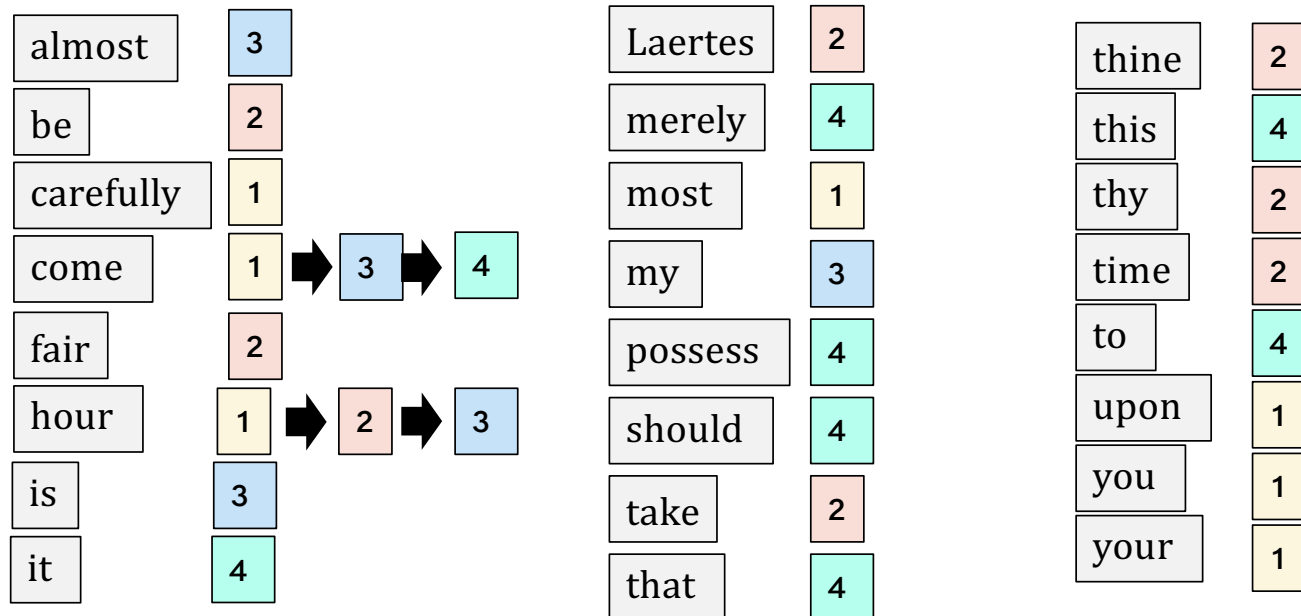
is	3
it	4
it	4
Laertes	2
merely	4
most	1
my	3
possess	4
should	4

take	2
that	4
thine	2
this	4
thy	2
time	2
to	4
upon	1
you	1
your	1

Merge



Merge



Add document frequency

almost	1	3		Laertes	1	2		thine	1	2
be	1	2		merely	1	4		this	1	4
carefully	1	1		most	1	1		thy	1	2
come	3	1	→ 3 → 4	my	1	3		time	1	2
fair	1	2		possess	1	4		to	1	4
hour	3	1	→ 2 → 3	should	1	4		upon	1	1
is	1	3		take	1	2		you	1	1
it	1	4		that	1	4		your	1	1

Single term query

hour

Single term query

Figure 1 displays a visualization of the word co-occurrence matrix for the sentence "almost be carefully come fair hour is it Laertes merely most my possess should take that thine this thy time to upon you your". The matrix is structured as a grid where rows represent words and columns represent their neighbors. The grid is divided into three sections. The first section (left) shows the words "almost", "be", "carefully", "come", "fair", "hour", "is", and "it". The second section (middle) shows the words "Laertes", "merely", "most", "my", "possess", "should", "take", and "that". The third section (right) shows the words "thine", "this", "thy", "time", "to", "upon", "you", and "your". Each cell in the grid contains a numerical value representing the co-occurrence count, and the cells are color-coded: dark blue for 1, light blue for 3, yellow for 1, orange for 2, and green for 4. Arrows indicate transitions between words in the sequence.

Single term query

almost	1	3		Laertes	1	2		thine	1	2
be	1	2		merely	1	4		this	1	4
carefully	1	1		most	1	1		thy	1	2
come	3	1	→ 3 → 4	my	1	3		time	1	2
fair	1	2		possess	1	4		to	1	4
hour	3	1	→ 2 → 3	should	1	4		upon	1	1
is	1	3		take	1	2		you	1	1
it	1	4		that	1	4		your	1	1

Single term query

almost	1	3		Laertes	1	2		thine	1	2
be	1	2		merely	1	4		this	1	4
carefully	1	1		most	1	1		thy	1	2
come	3	1	→ 3 → 4	my	1	3		time	1	2
fair	1	2		possess	1	4		to	1	4
hour	3	1	→ 2 → 3	should	1	4		upon	1	1
is	1	3		take	1	2		you	1	1
it	1	4		that	1	4		your	1	1

Results

Document 1
You come most carefully upon your **hour**.

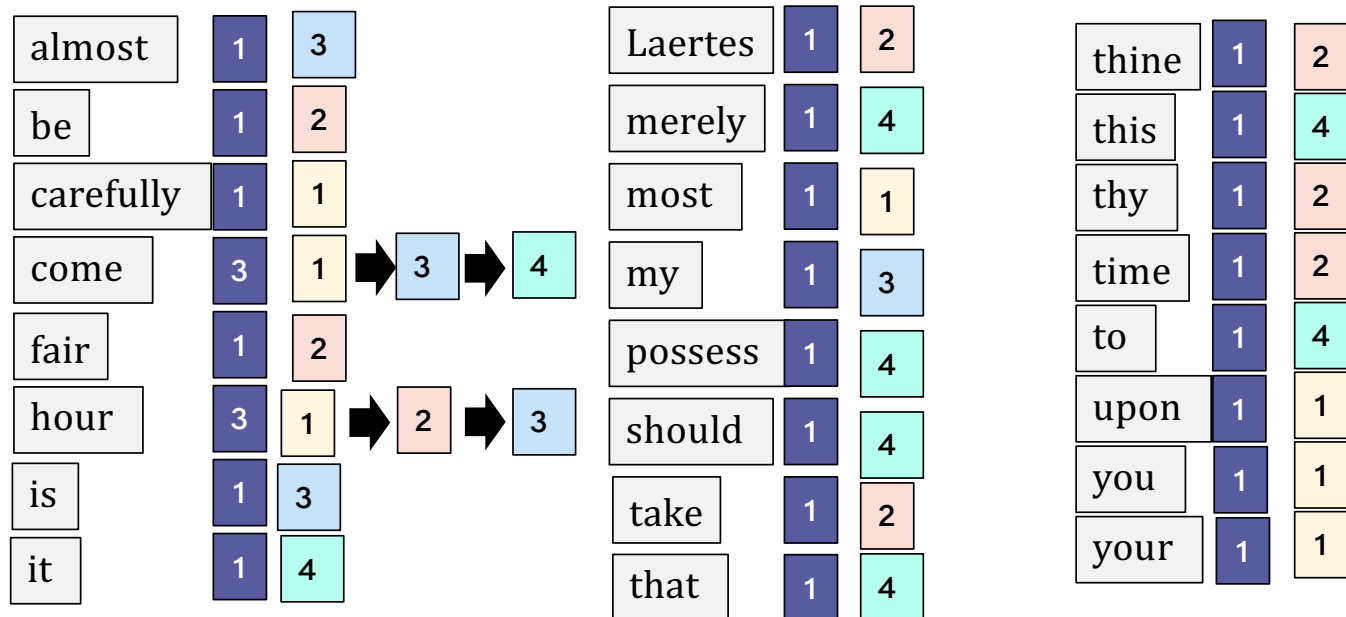
Document 2
Take thy fair **hour**, Laertes; time be thine,

Document 3
My **hour** is almost come,

Boolean (conjunctive) query

hour AND come

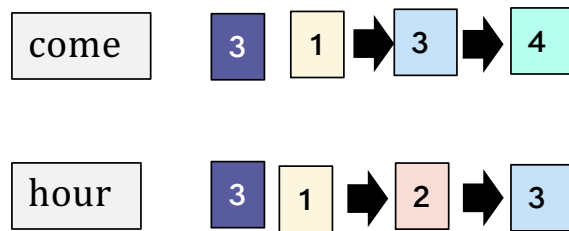
Single term query



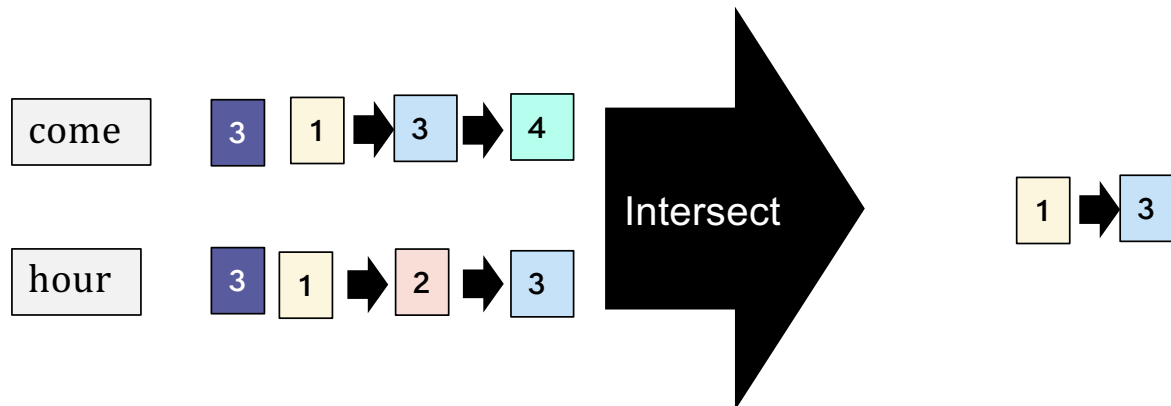
Single term query

almost	1	3		Laertes	1	2		thine	1	2
be	1	2		merely	1	4		this	1	4
carefully	1	1		most	1	1		thy	1	2
come	3	1	→ 3 → 4	my	1	3		time	1	2
fair	1	2		possess	1	4		to	1	4
hour	3	1	→ 2 → 3	should	1	4		upon	1	1
is	1	3		take	1	2		you	1	1
it	1	4		that	1	4		your	1	1

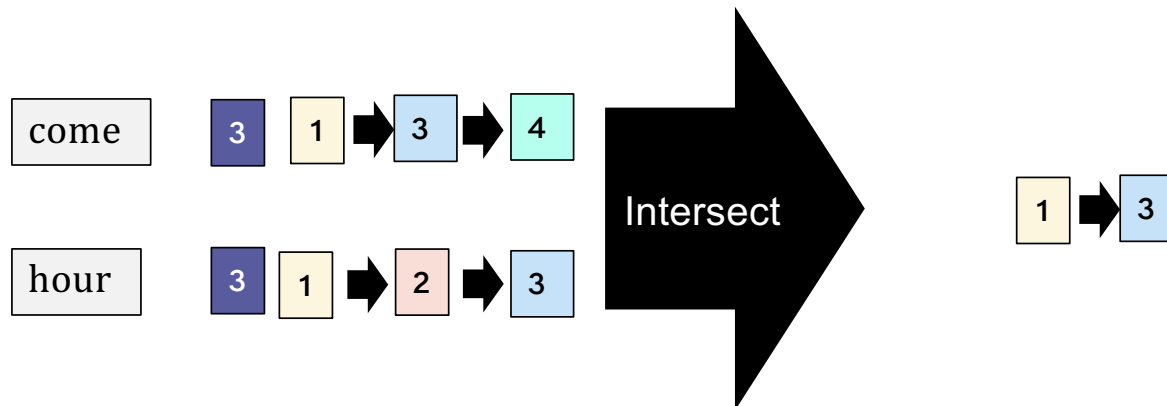
Single term query



Single term query



Single term query



Results

Document 1

You **come** most carefully upon your **hour**.

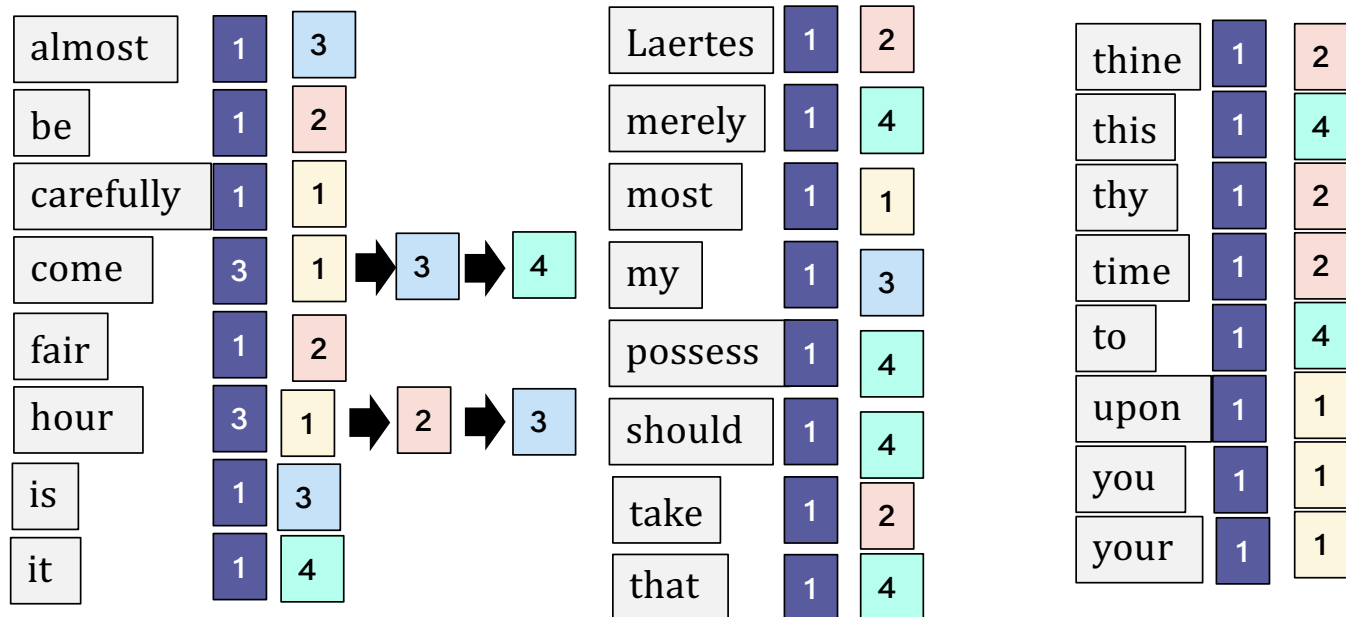
Document 3

My **hour** is almost **come**,

Boolean (conjunctive) query

hour OR this

Single term query

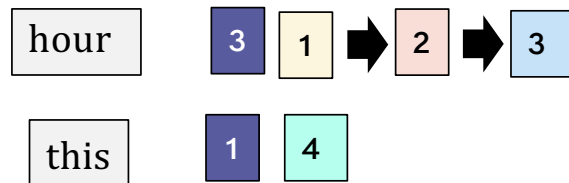


Single term query

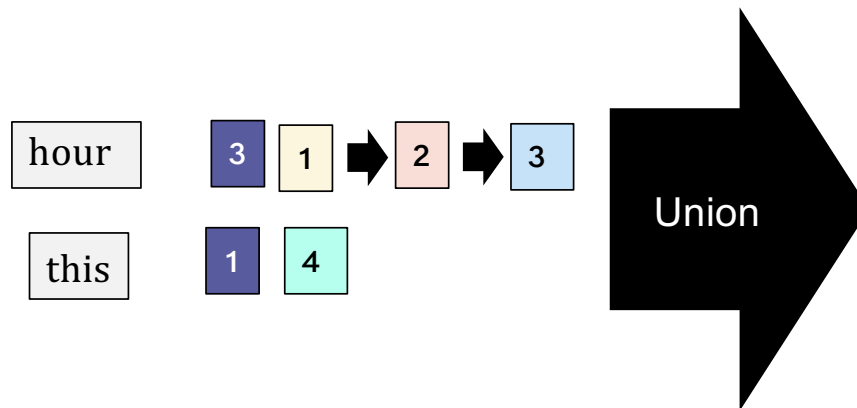
almost	1	3		Laertes	1	2	
be	1	2		merely	1	4	
carefully	1	1		most	1	1	
come	3	1	→ 3 → 4	my	1	3	
fair	1	2		possess	1	4	
hour	3	1	→ 2 → 3	should	1	4	
is	1	3		take	1	2	
it	1	4		that	1	4	

thine	1	2	
this	1	4	
thy	1	2	
time	1	2	
to	1	4	
upon	1	1	
you	1	1	
your	1	1	

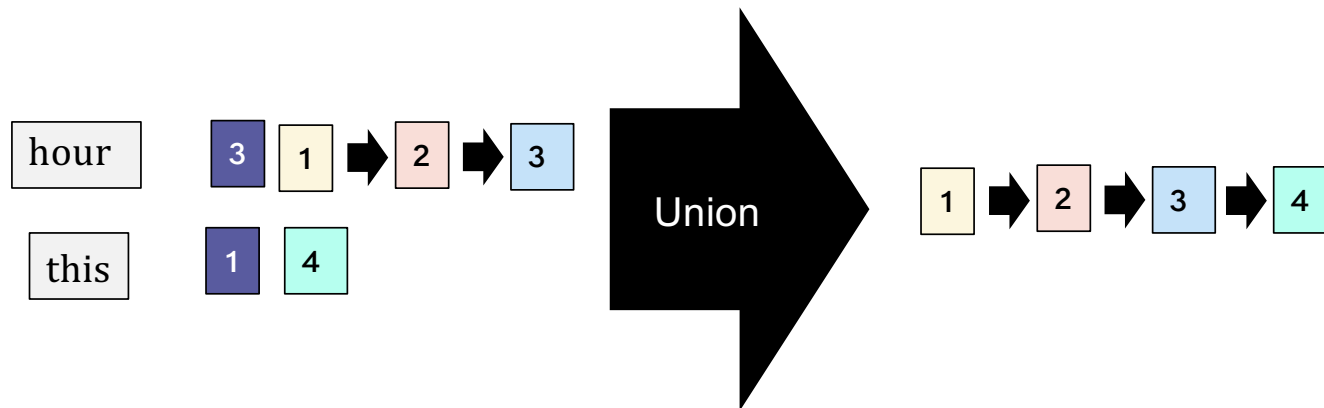
Single term query



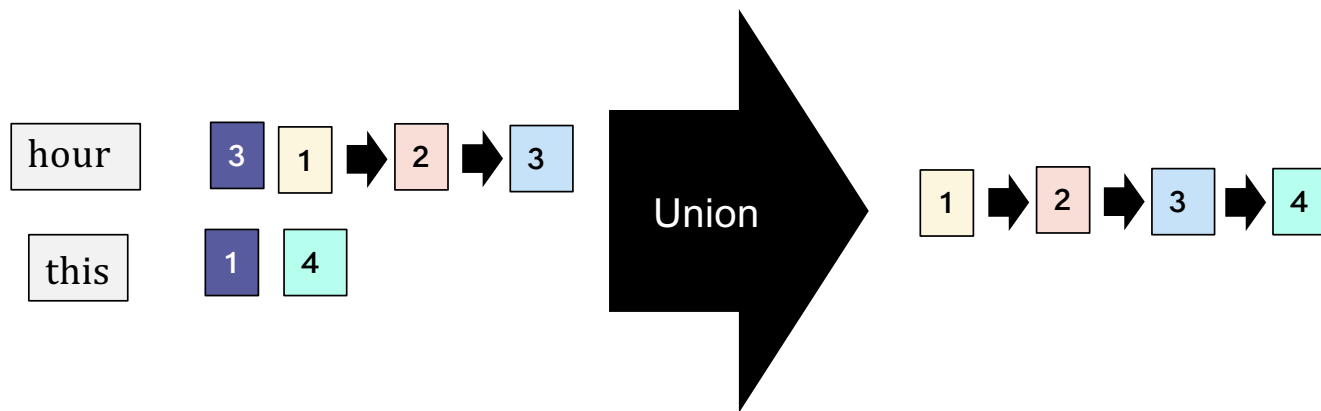
Single term query



Single term query



Single term query



Results

Document 1

You come most carefully upon your **hour**.

Document 2

Take thy fair **hour**, Laertes; time be thine,

Document 3

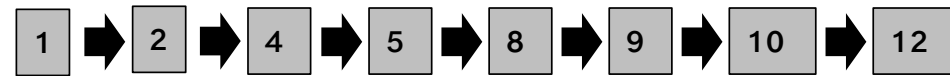
My **hour** is almost come,

Document 4

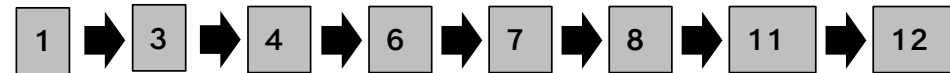
Possess it merely. That it should come to **this**!

Intersection algorithm

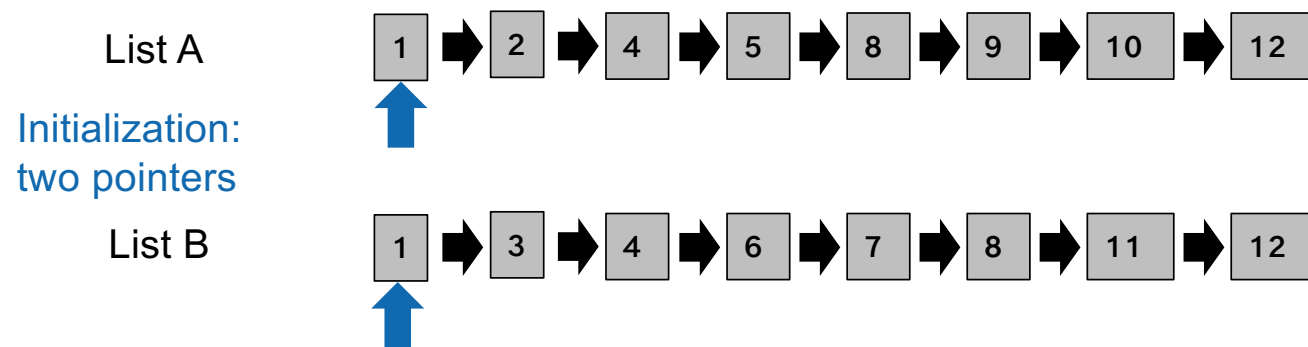
List A



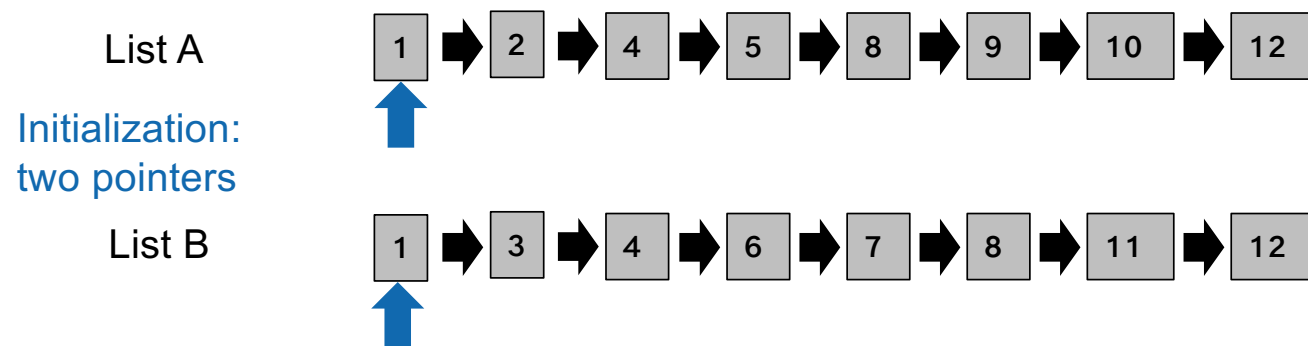
List B



Intersection algorithm

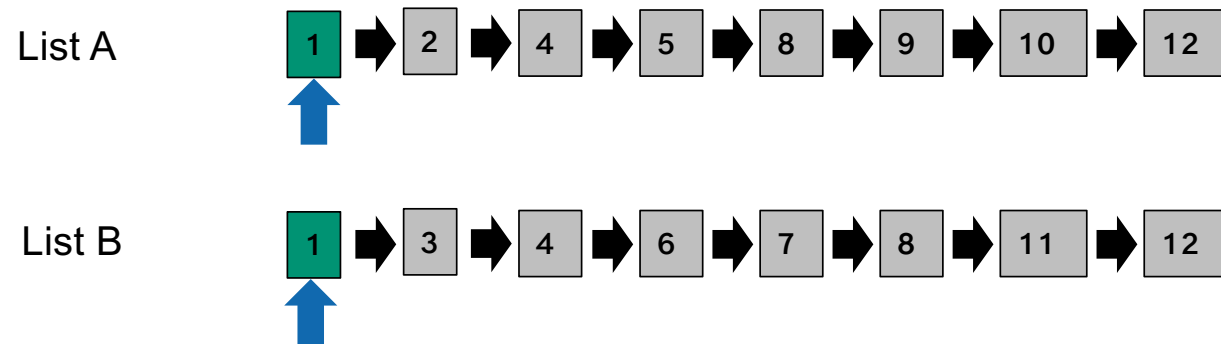


Intersection algorithm



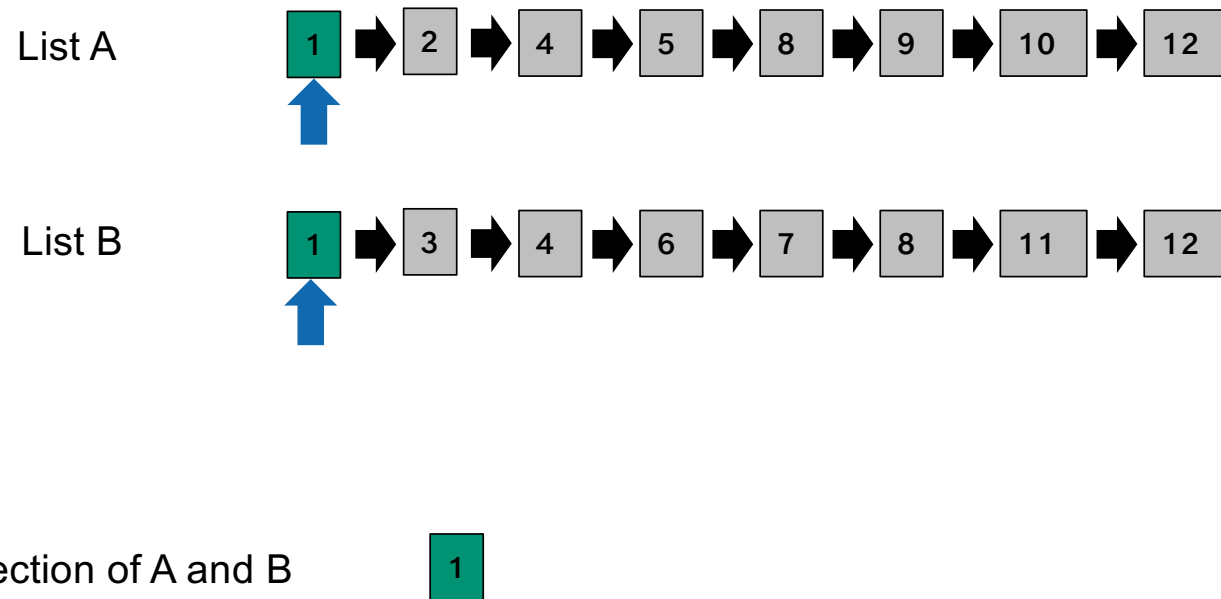
Intersection of A and B

Intersection algorithm

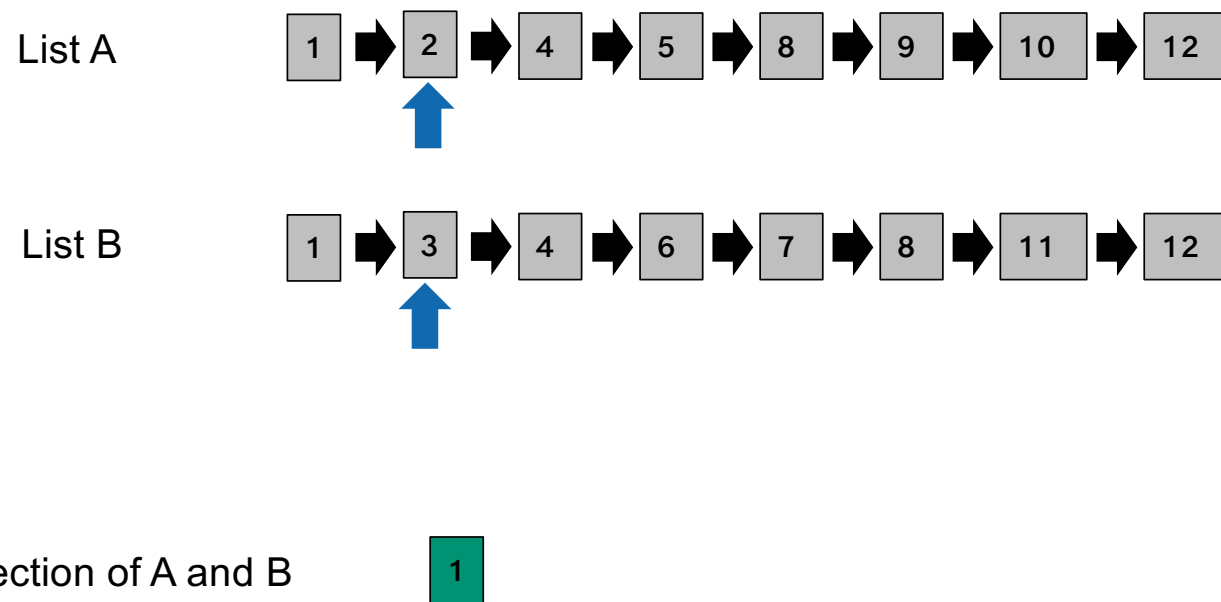


Intersection of A and B

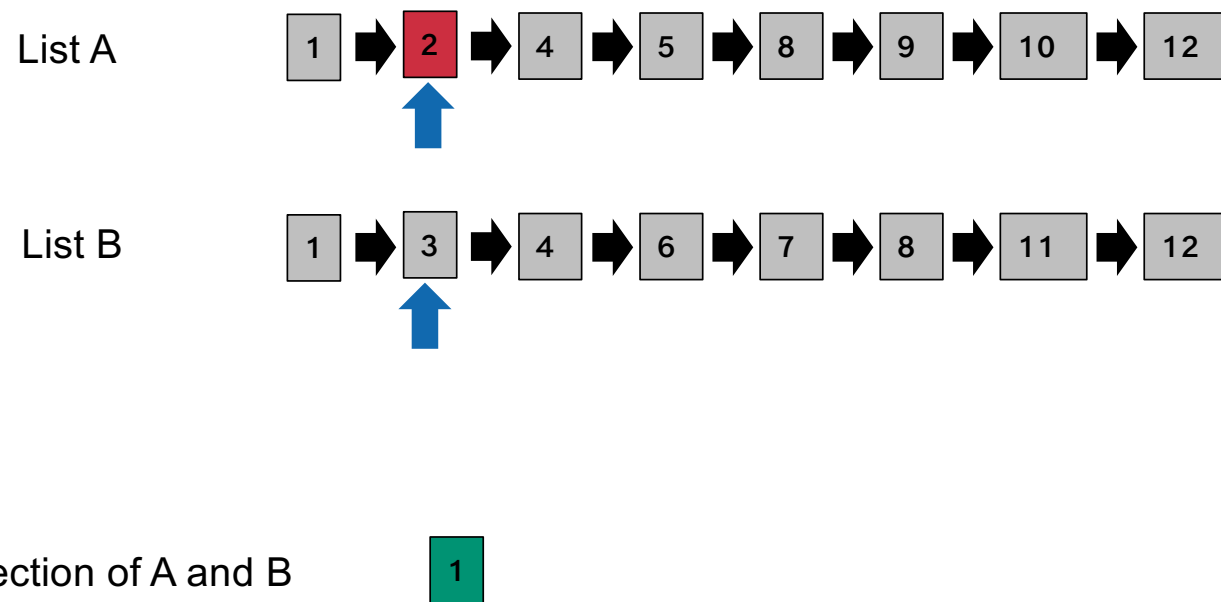
Intersection algorithm



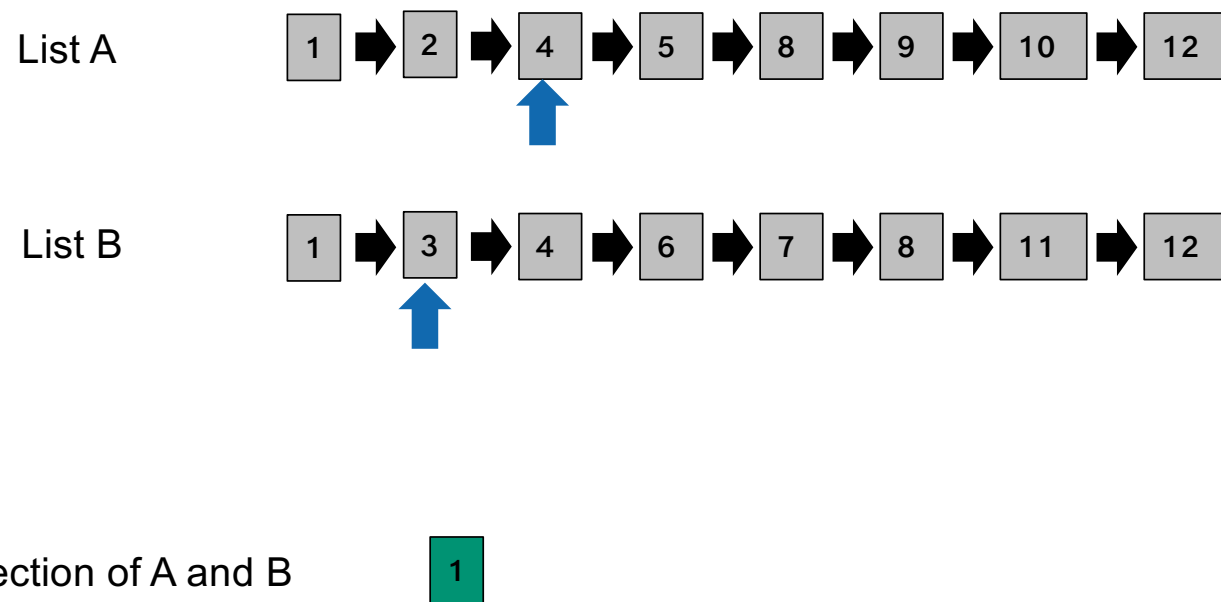
Intersection algorithm



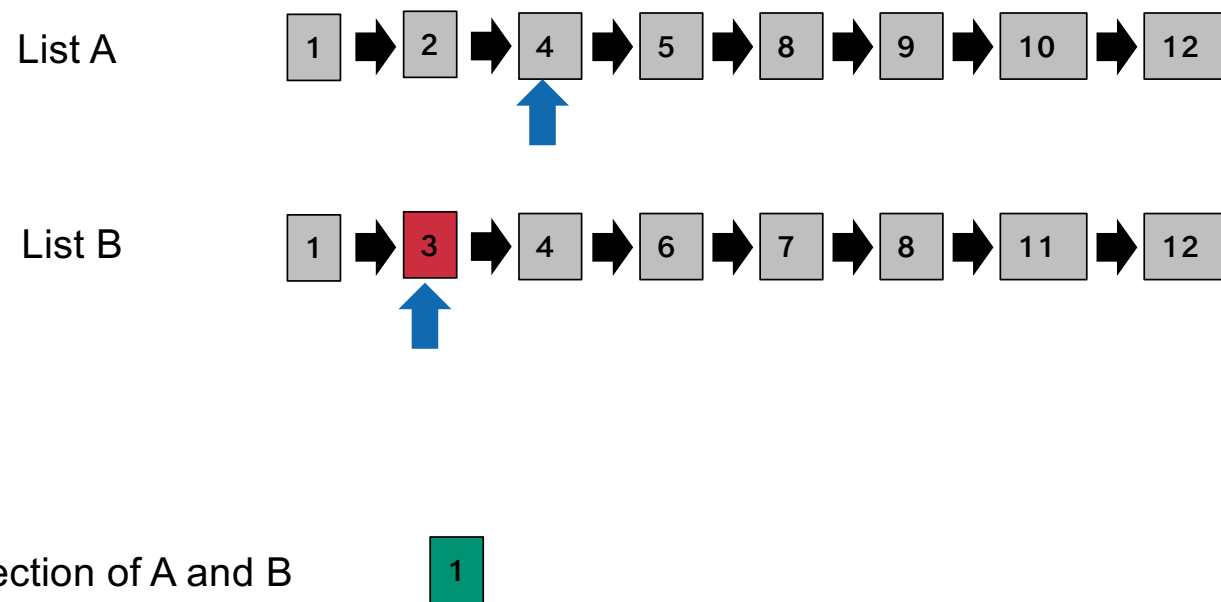
Intersection algorithm



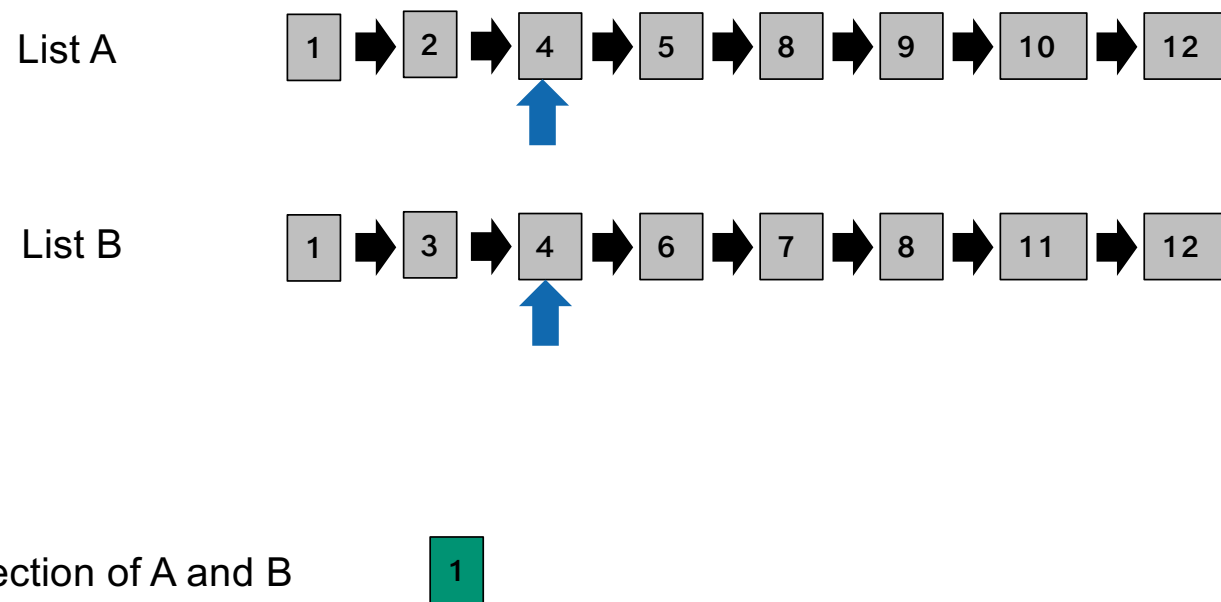
Intersection algorithm



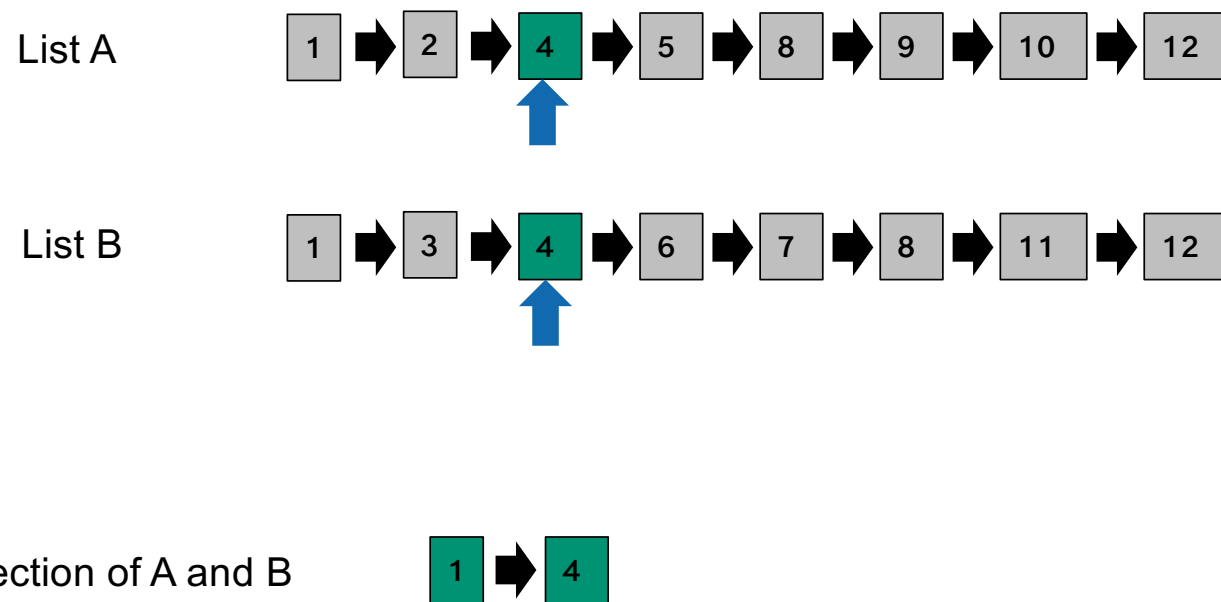
Intersection algorithm



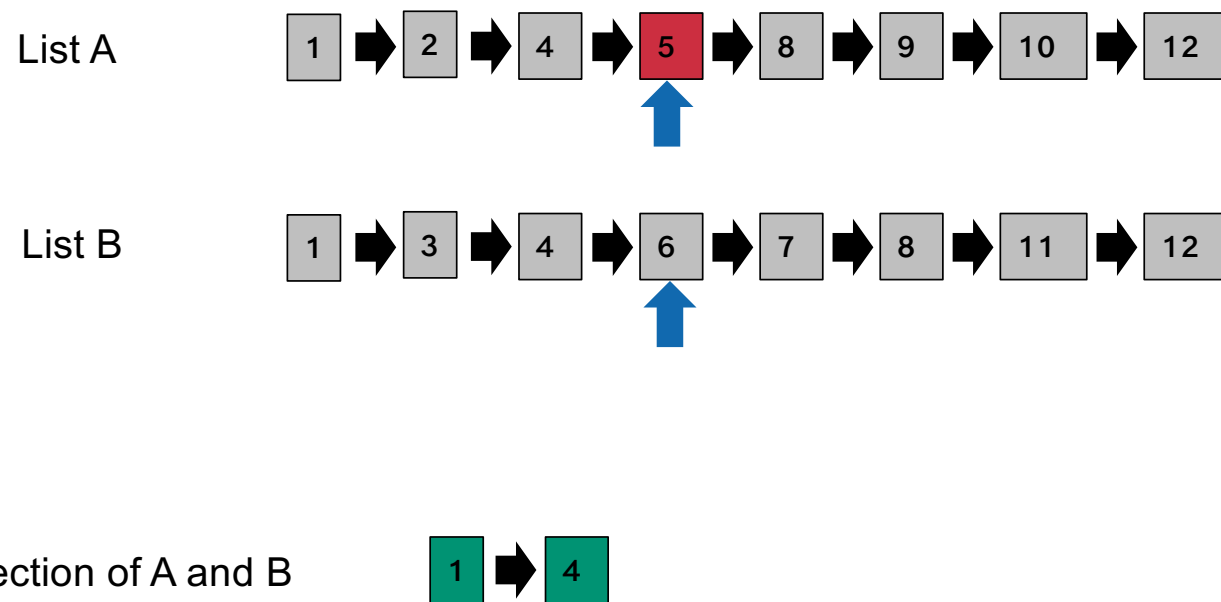
Intersection algorithm



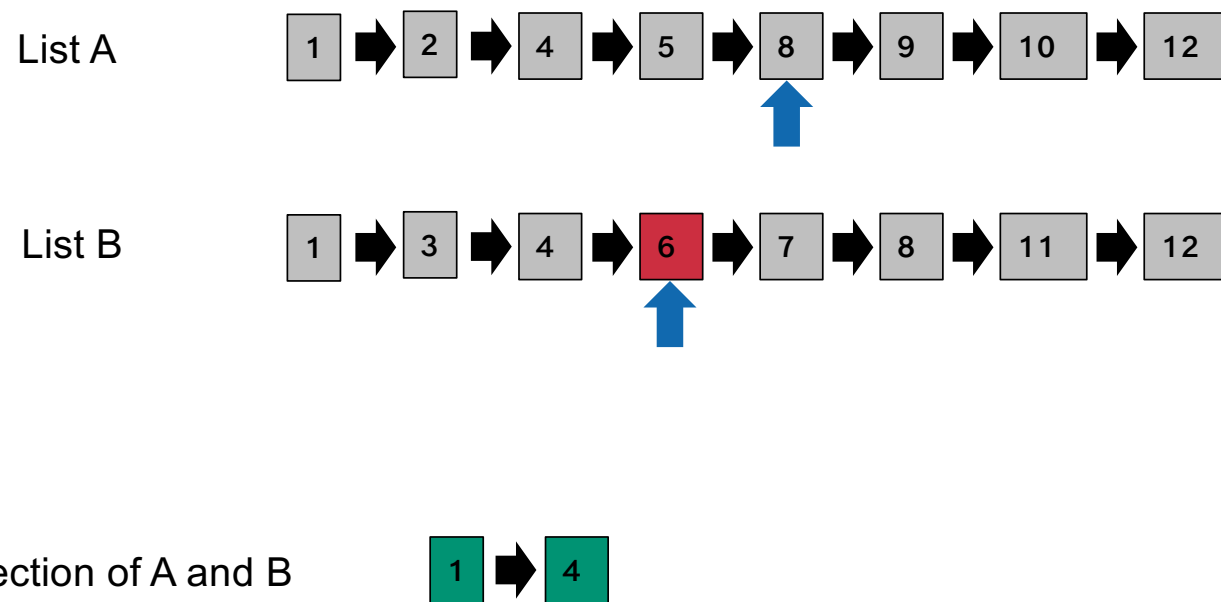
Intersection algorithm



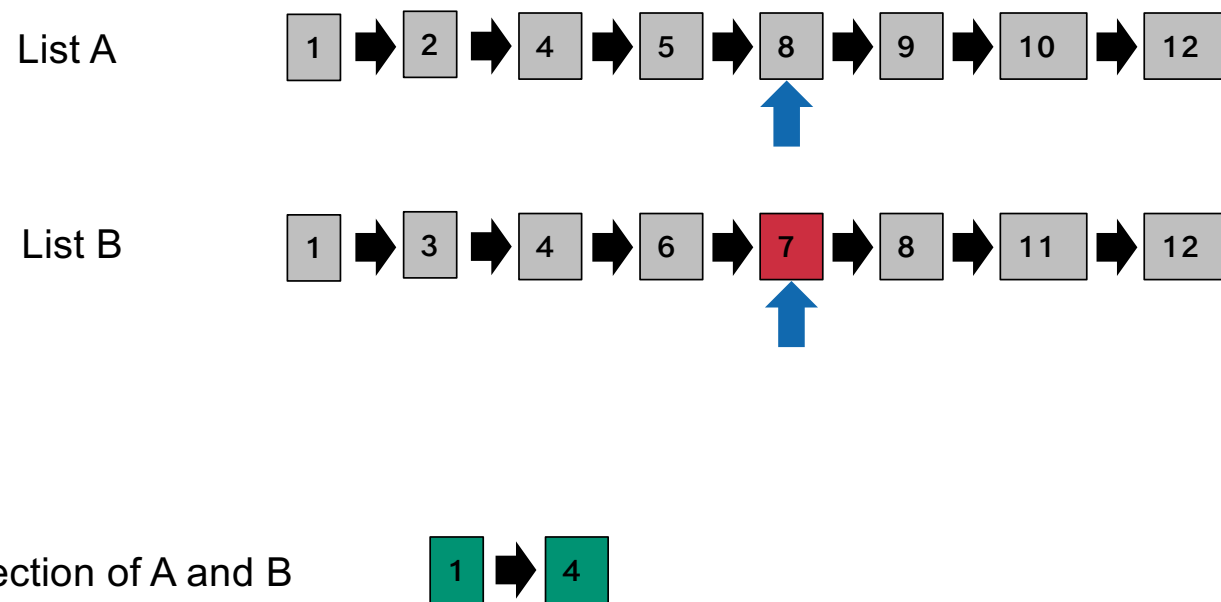
Intersection algorithm



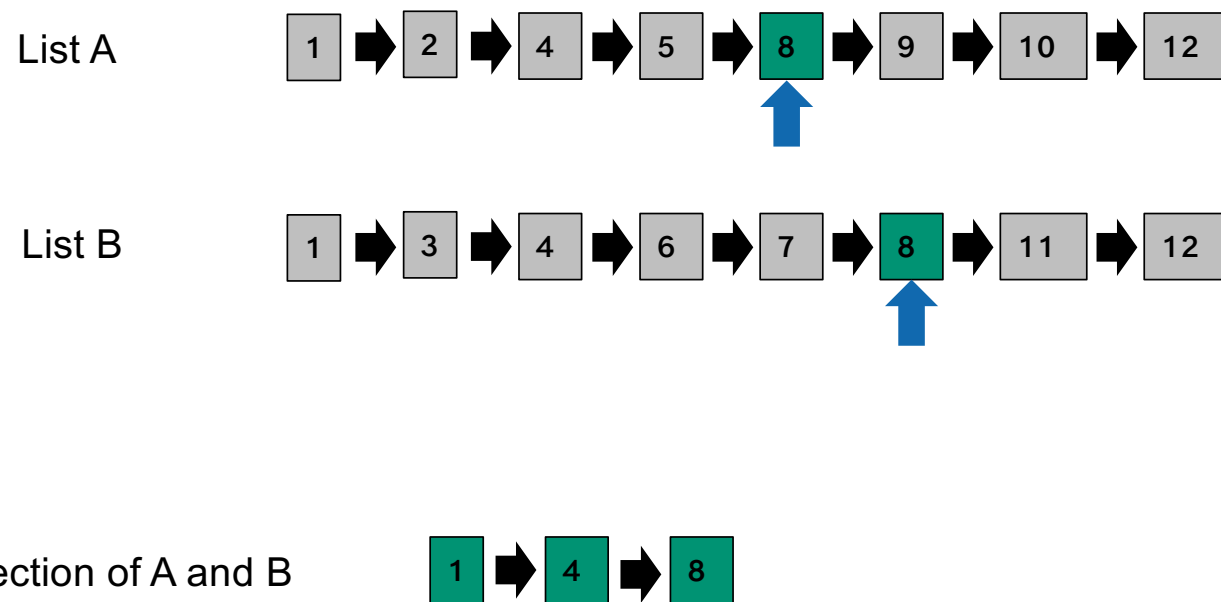
Intersection algorithm



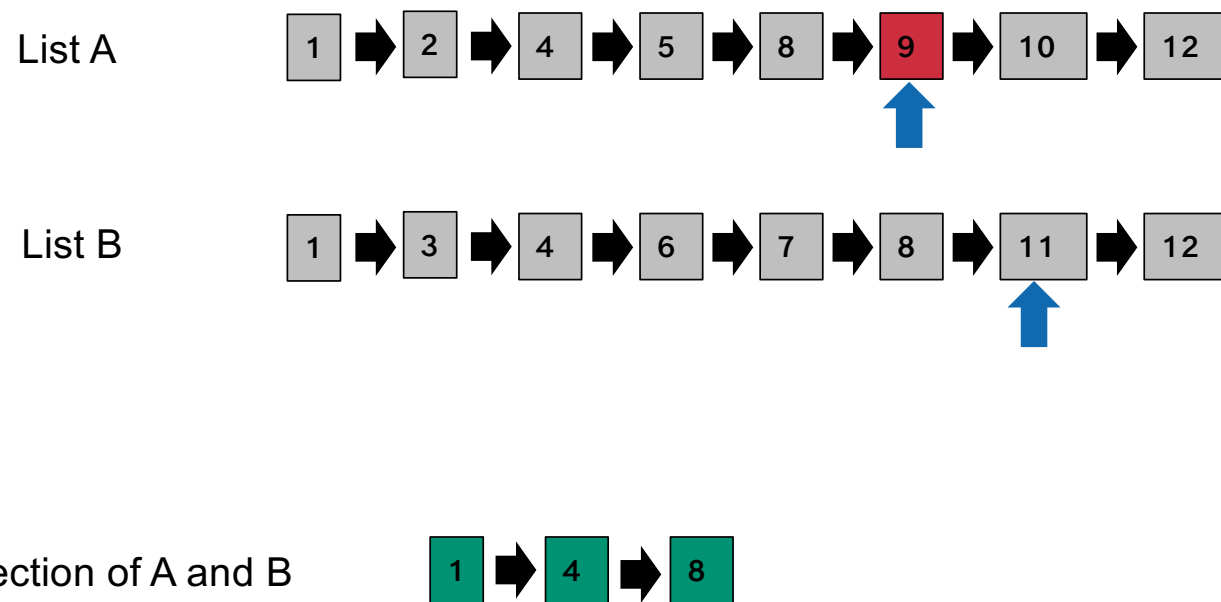
Intersection algorithm



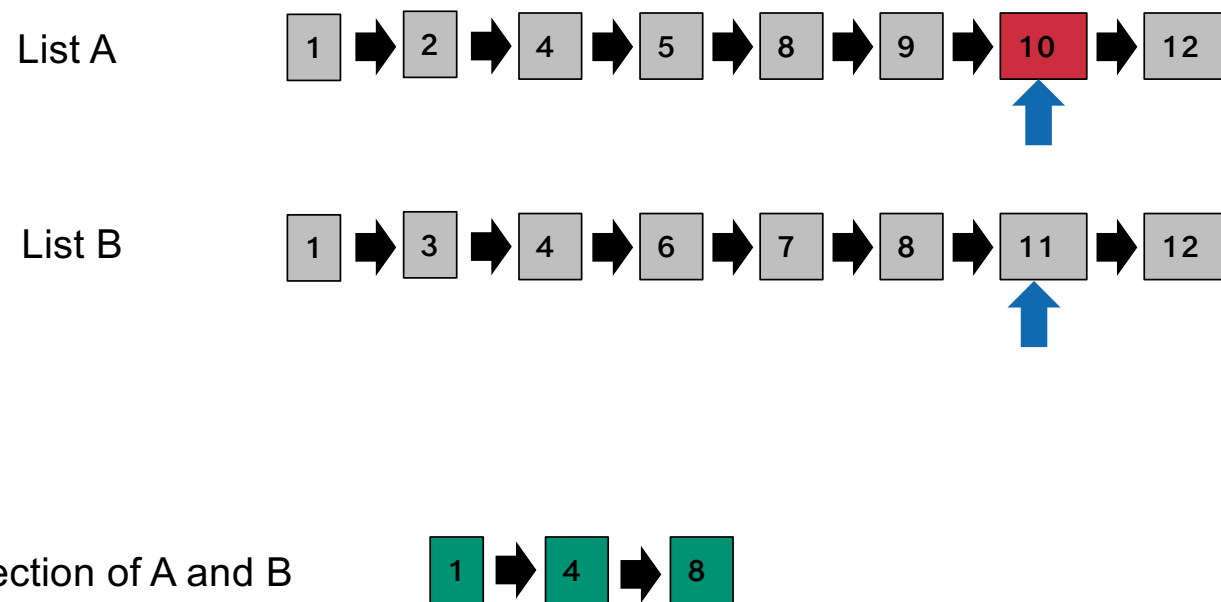
Intersection algorithm



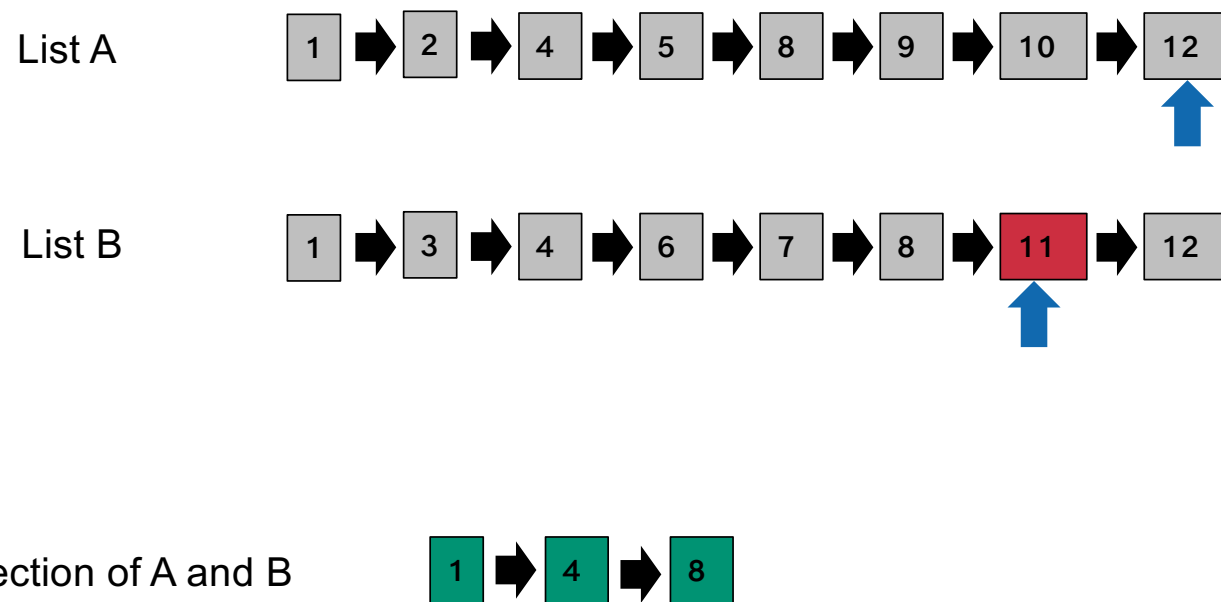
Intersection algorithm



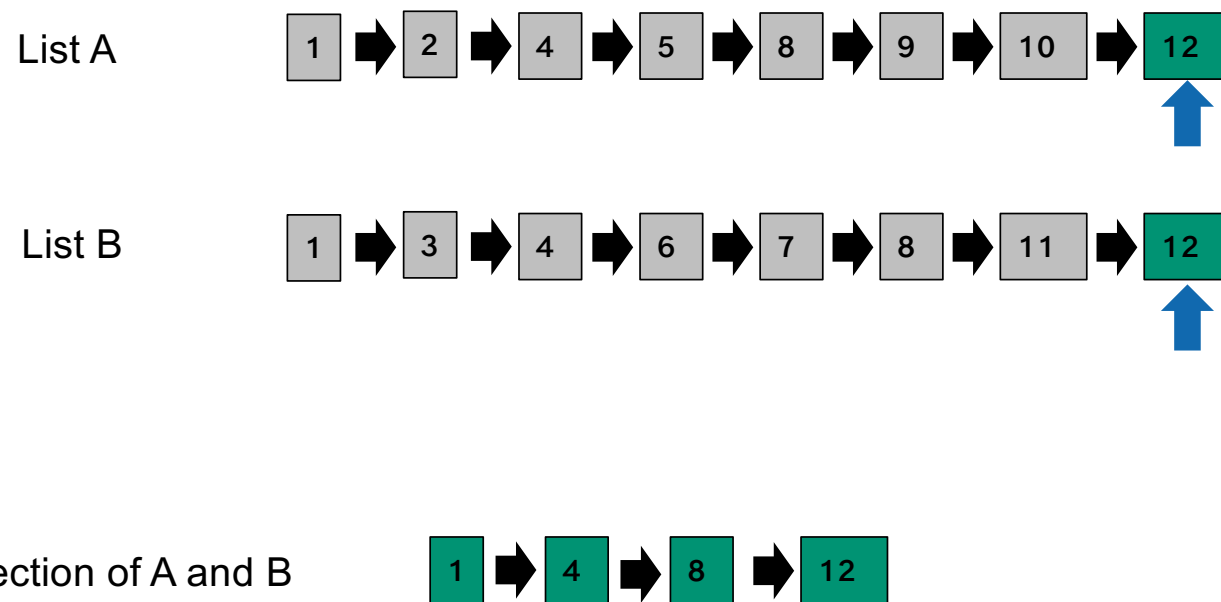
Intersection algorithm



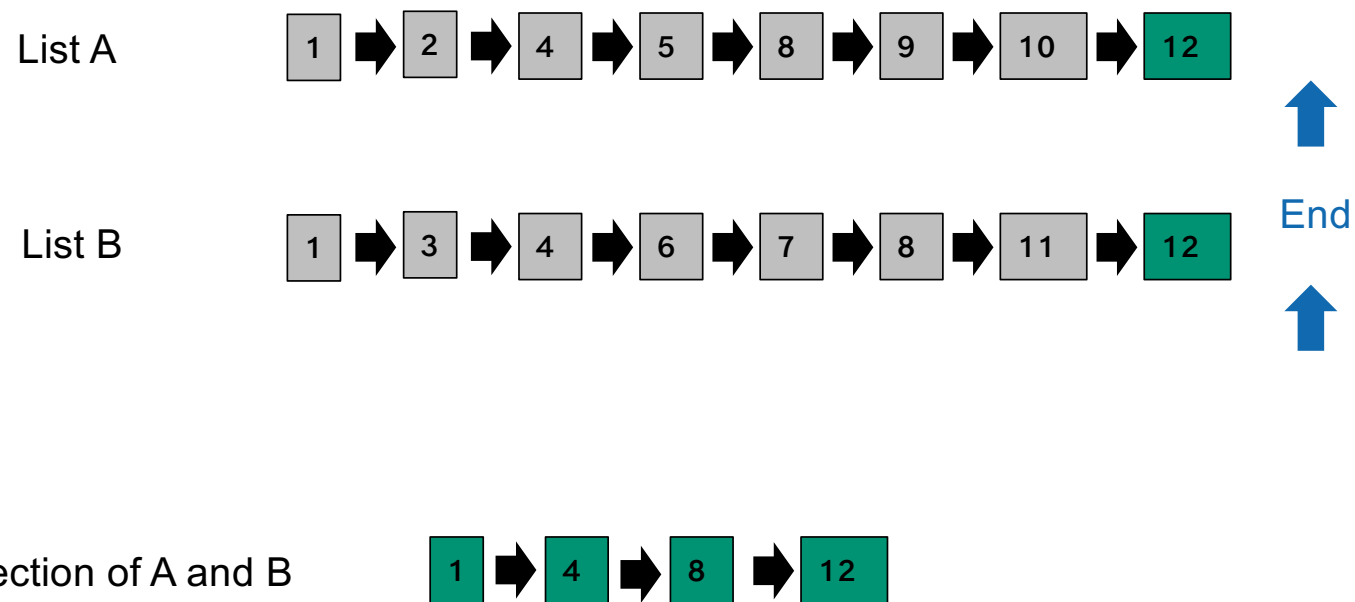
Intersection algorithm



Intersection algorithm

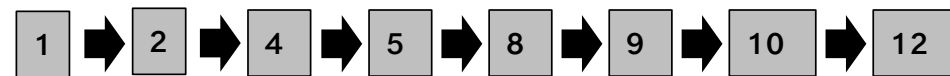


Intersection algorithm

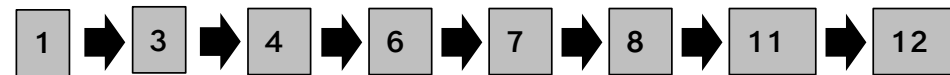


Union algorithm

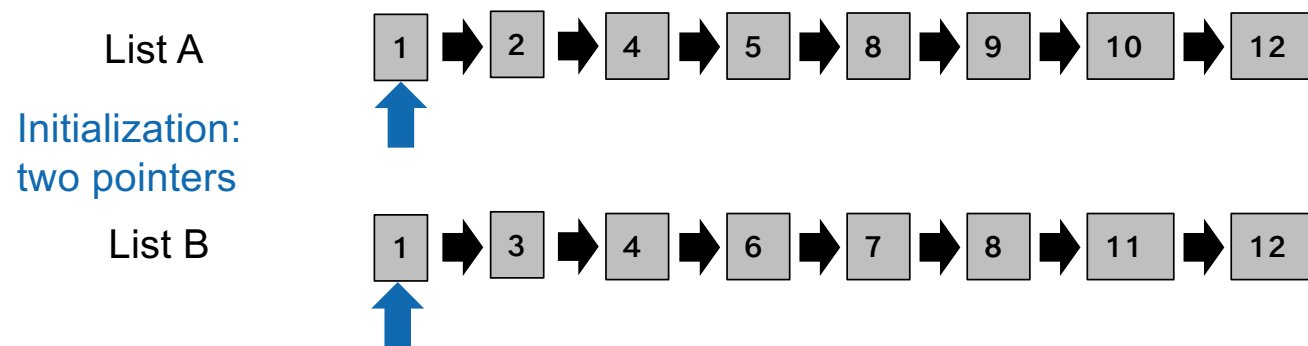
List A



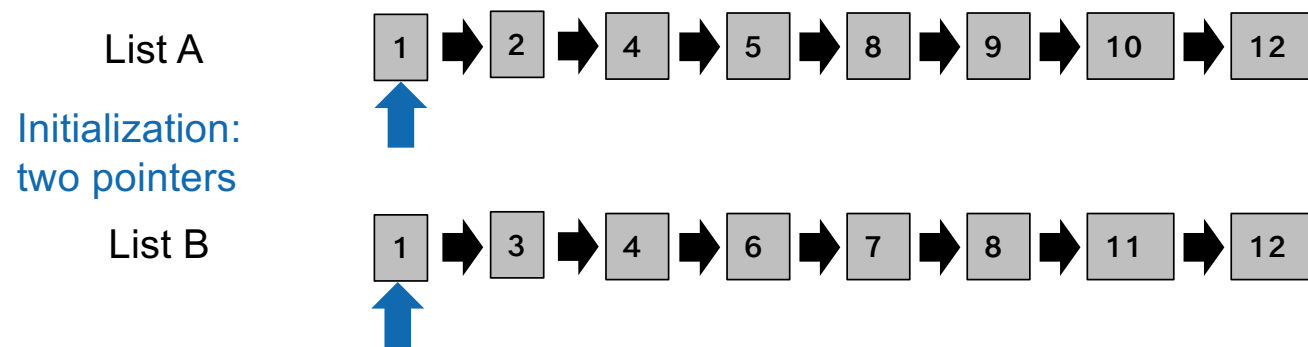
List B



Union algorithm

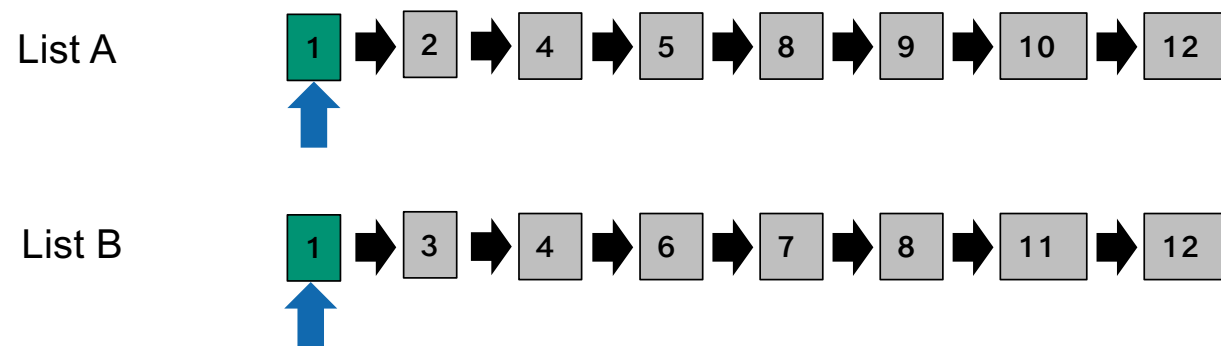


Union algorithm



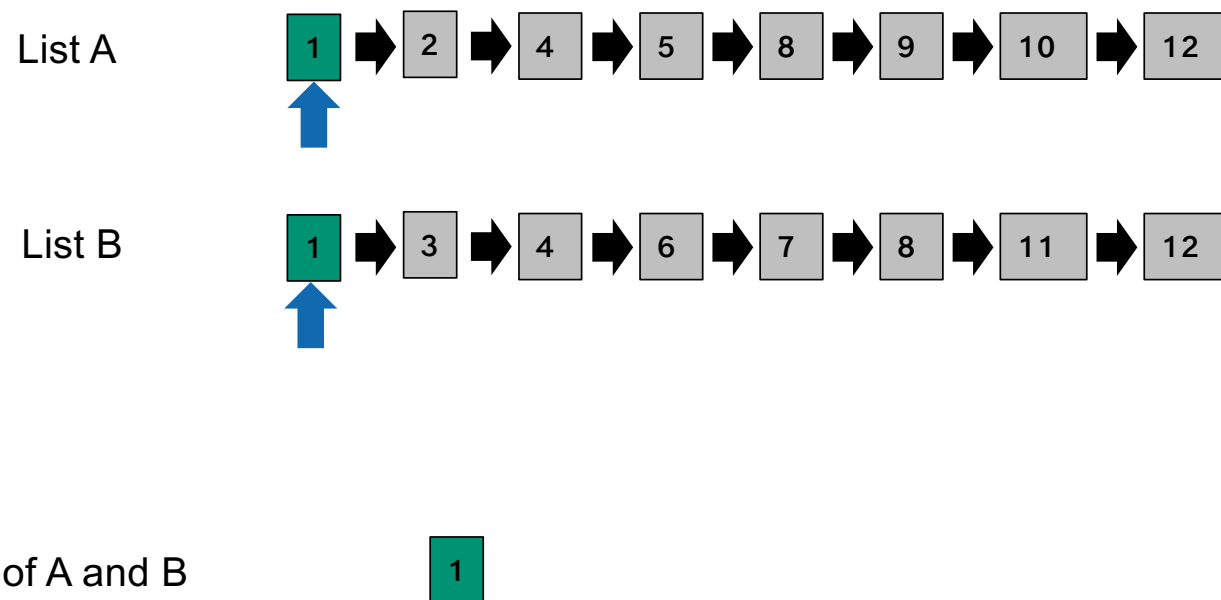
Union of A and B

Union algorithm



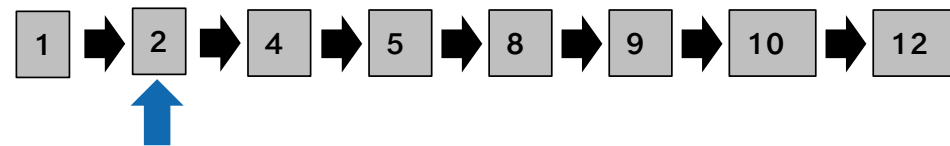
Union of A and B

Union algorithm

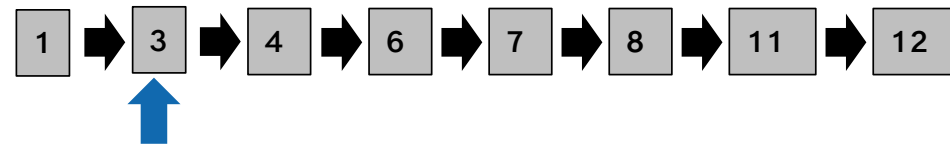


Union algorithm

List A



List B

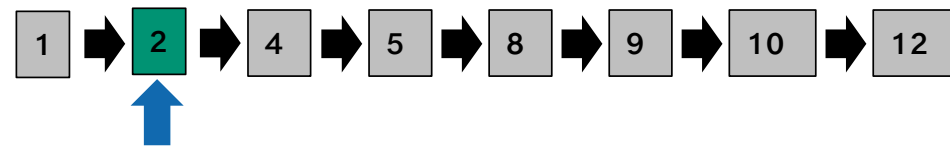


Union of A and B

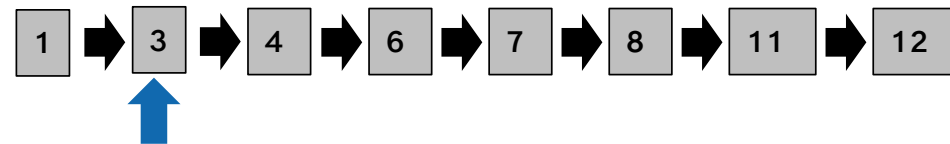


Union algorithm

List A



List B

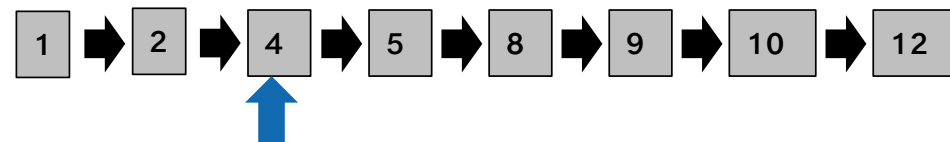


Union of A and B

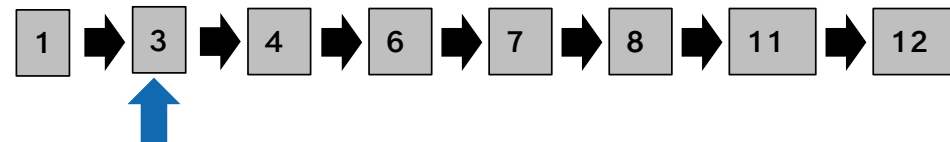


Union algorithm

List A



List B

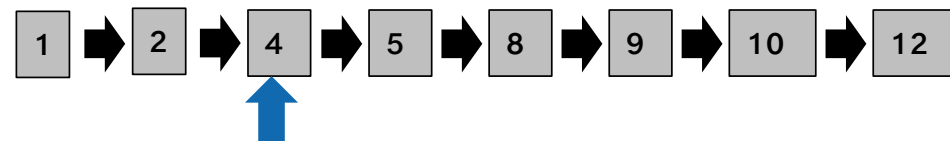


Union of A and B

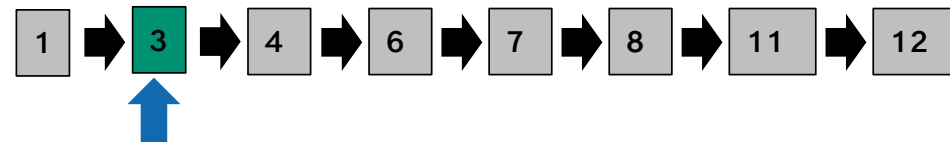


Union algorithm

List A



List B

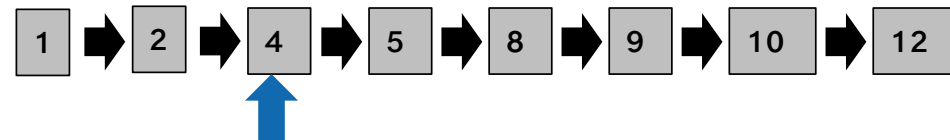


Union of A and B

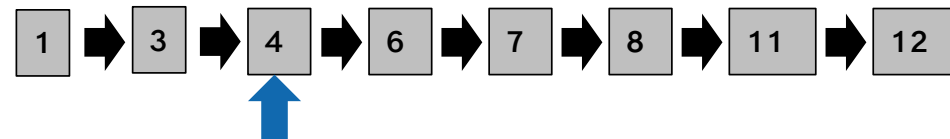


Union algorithm

List A



List B

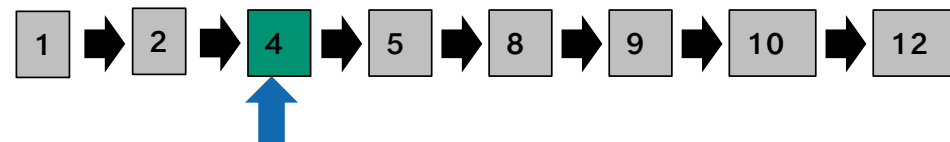


Union of A and B

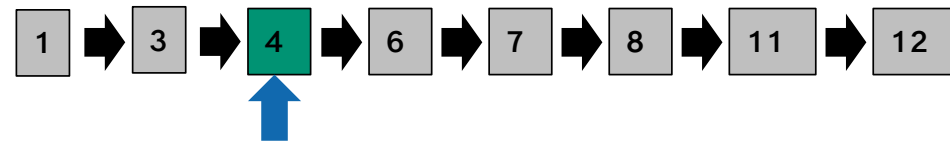


Union algorithm

List A



List B

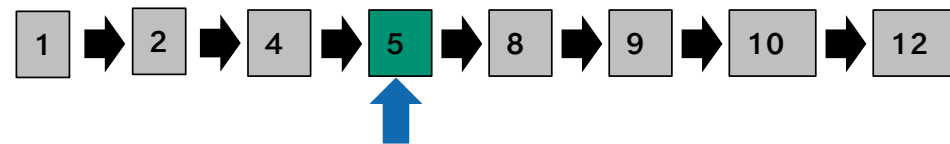


Union of A and B

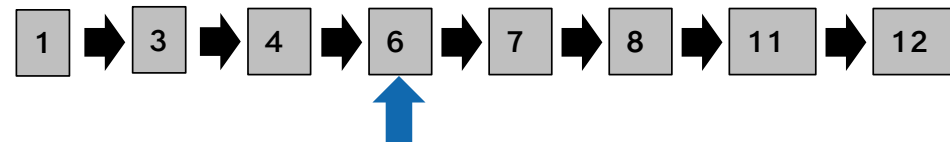


Union algorithm

List A



List B

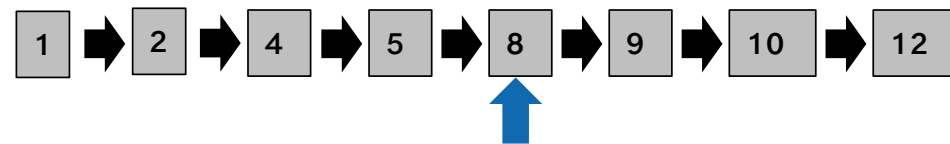


Union of A and B

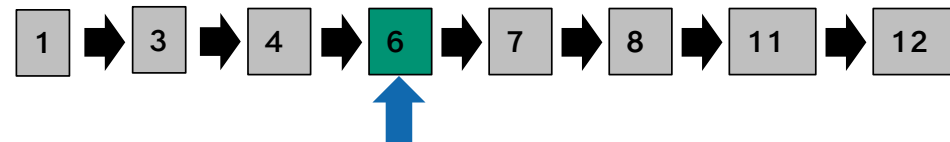


Union algorithm

List A



List B

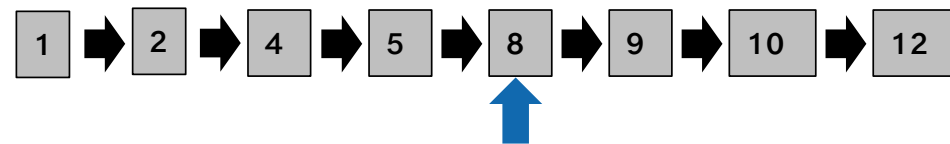


Union of A and B

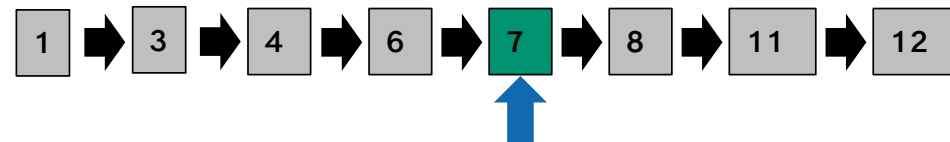


Union algorithm

List A



List B

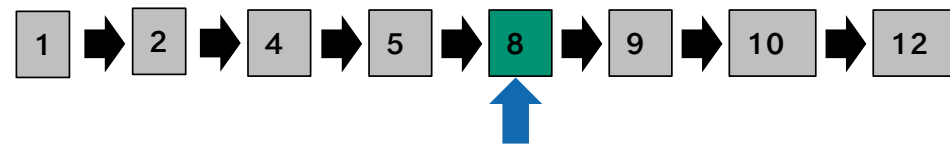


Union of A and B

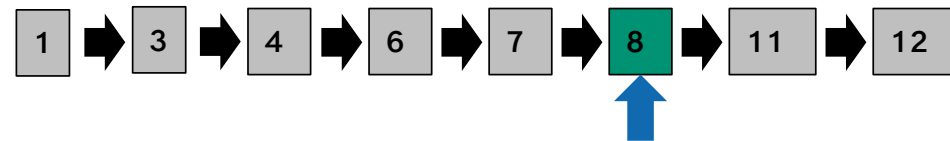


Union algorithm

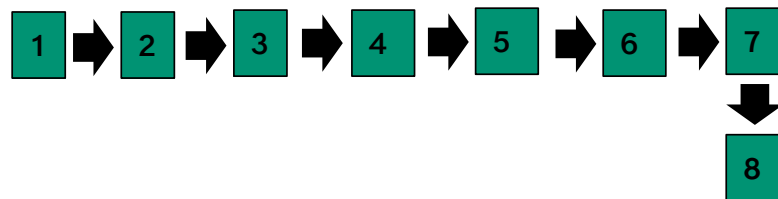
List A



List B

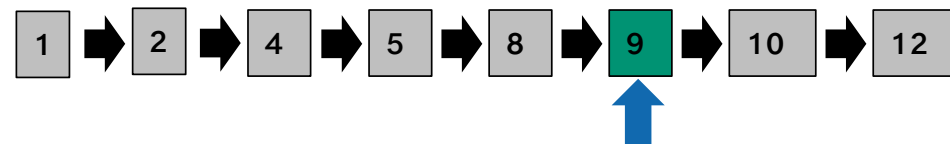


Union of A and B

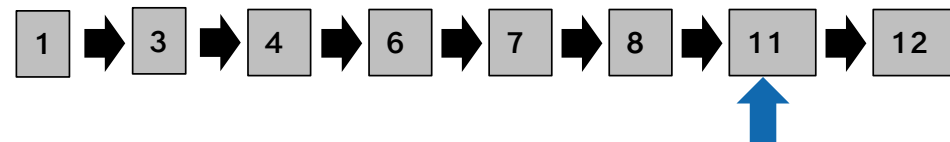


Union algorithm

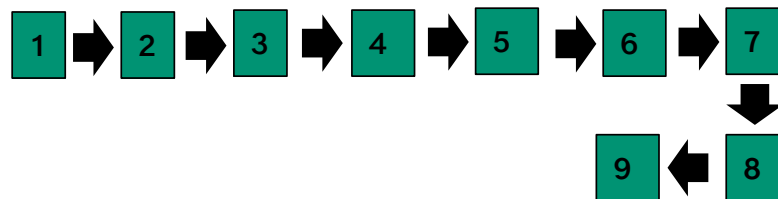
List A



List B

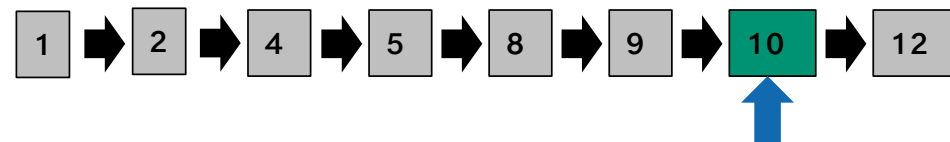


Union of A and B

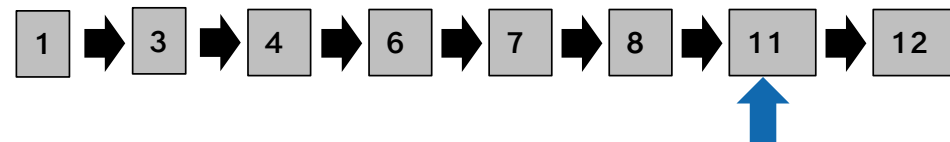


Union algorithm

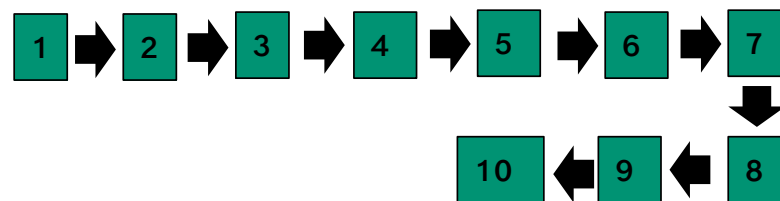
List A



List B

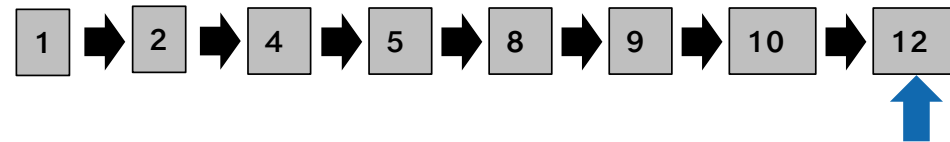


Union of A and B

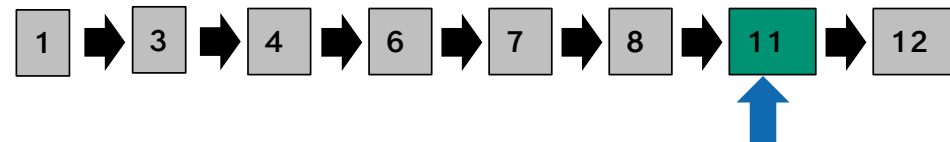


Union algorithm

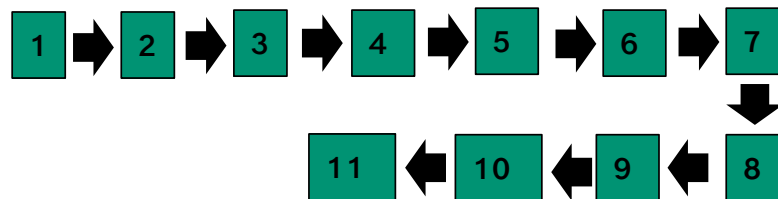
List A



List B

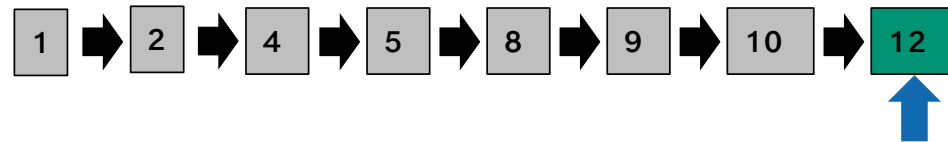


Union of A and B

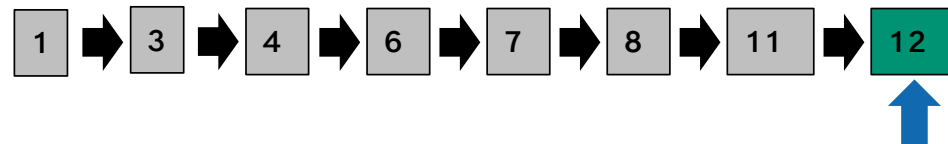


Union algorithm

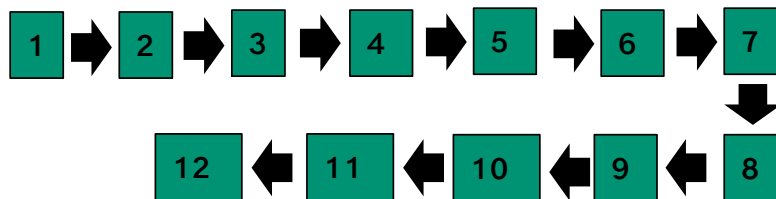
List A



List B

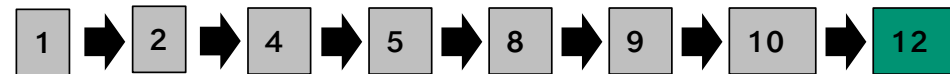


Union of A and B

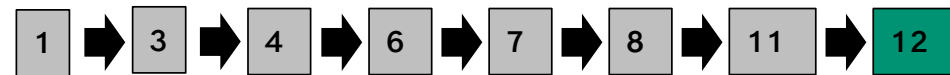


Union algorithm

List A



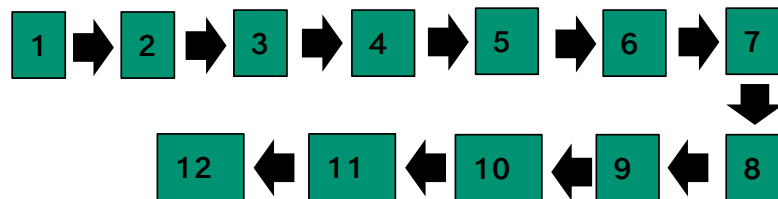
List B



End



Union of A and B



Optimizing

eth AND zurich AND bibliopole

The inverted index has the document frequencies!

Optimizing

eth AND zurich AND bibliopole

10,000 docs

10 docs

100,000 docs

Optimizing

bibliopole AND eth AND zurich

10 docs

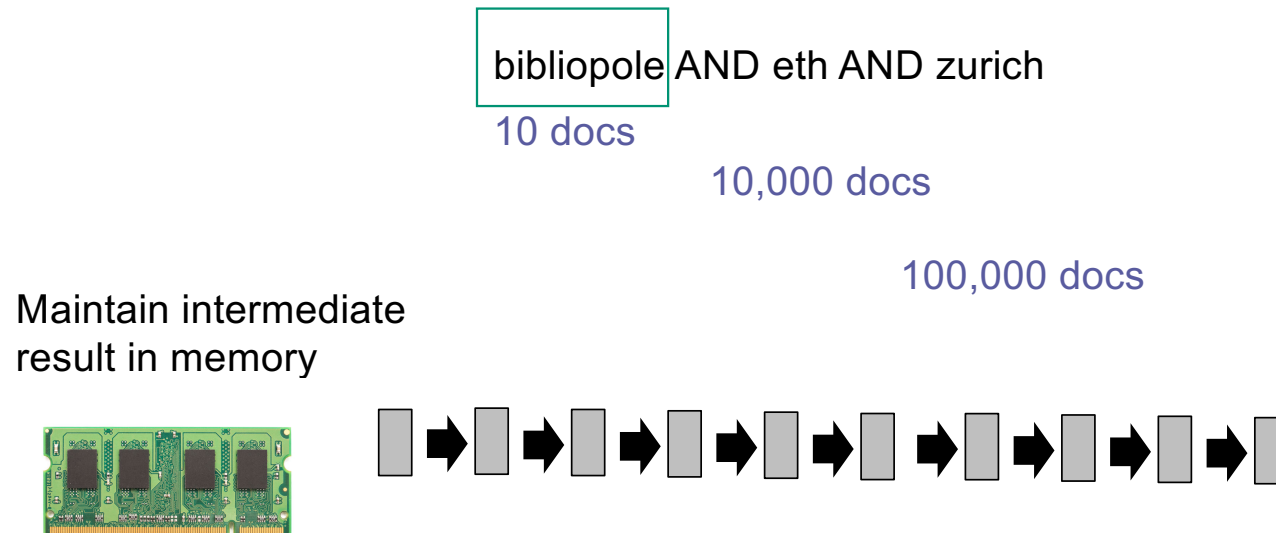
10,000 docs

100,000 docs



Sort query terms by increasing document frequency

Optimizing



Optimizing

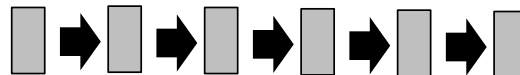
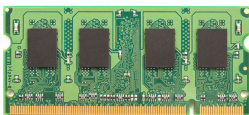
bibliopole AND eth AND zurich

10 docs

10,000 docs

100,000 docs

Maintain intermediate
result in memory



Optimizing

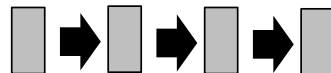
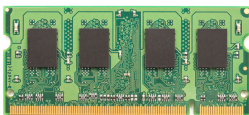
bibliopole AND eth AND zurich

10 docs

10,000 docs

100,000 docs

Maintain intermediate
result in memory



More complex queries

(epf OR eth) AND zurich AND bibliopole



We can approximate the (estimated) size of the postings list
by summing the two original lists

More complex queries

(epf OR eth) AND NOT zurich AND bibliopole



We can adapt the intersection algorithm
to "virtually" walk down the negated list

Simple Boolean language (EBNF)

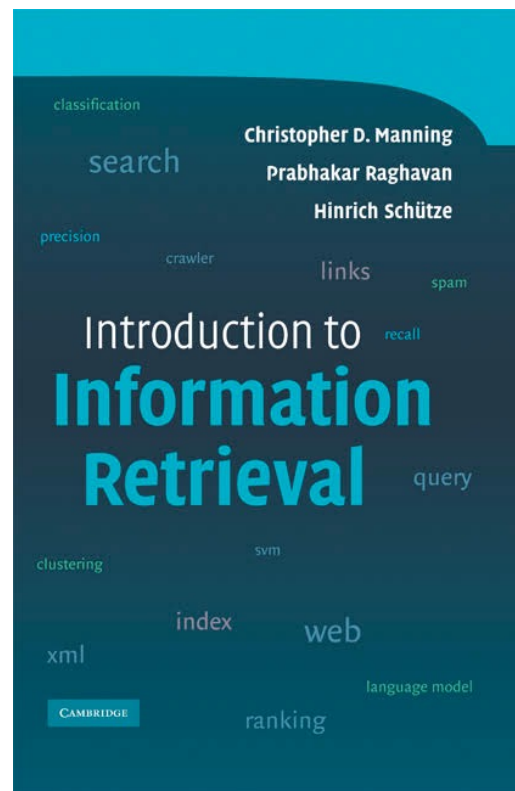
`PrimaryExpr ::= Term | "(" Expr ")"`

`NotExpr ::= "NOT"? PrimaryExpr`

`AndExpr ::= NotExpr ("AND" NotExpr)*`

`Expr ::= NotExpr ("OR" AndExpr)*`

This week's reading



Chapter 1

Boolean Retrieval

Poll

Go *now* to:

<https://eduapp-app1.ethz.ch/>

